

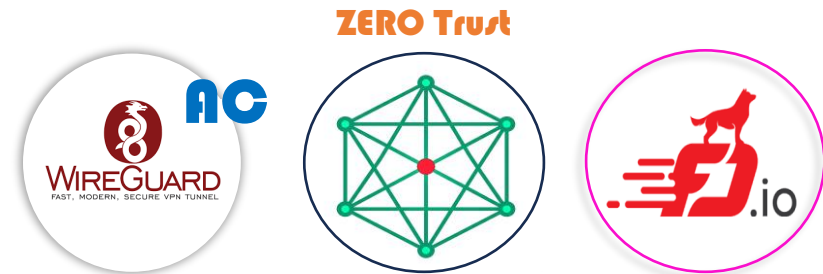
MightyConnect™ Security Platform

- Mesh P2P VPN
- Powered by WireGuard & FD.io VPP

MightyLink Lab.
(www.mightyconn.com)
Doc. Revision: 1.01

Table of Contents

1. Zero Trust P2P Security Platform
2. WireGuard
3. FD.io VPP
4. MightyConnect™ Security Box
5. MightyConnect™ Devices
6. References



(1) "WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld
(2) FD.io VPP and FD.io logo are registered trademarks of The Fast Data Project, a series of LF Projects, LLC

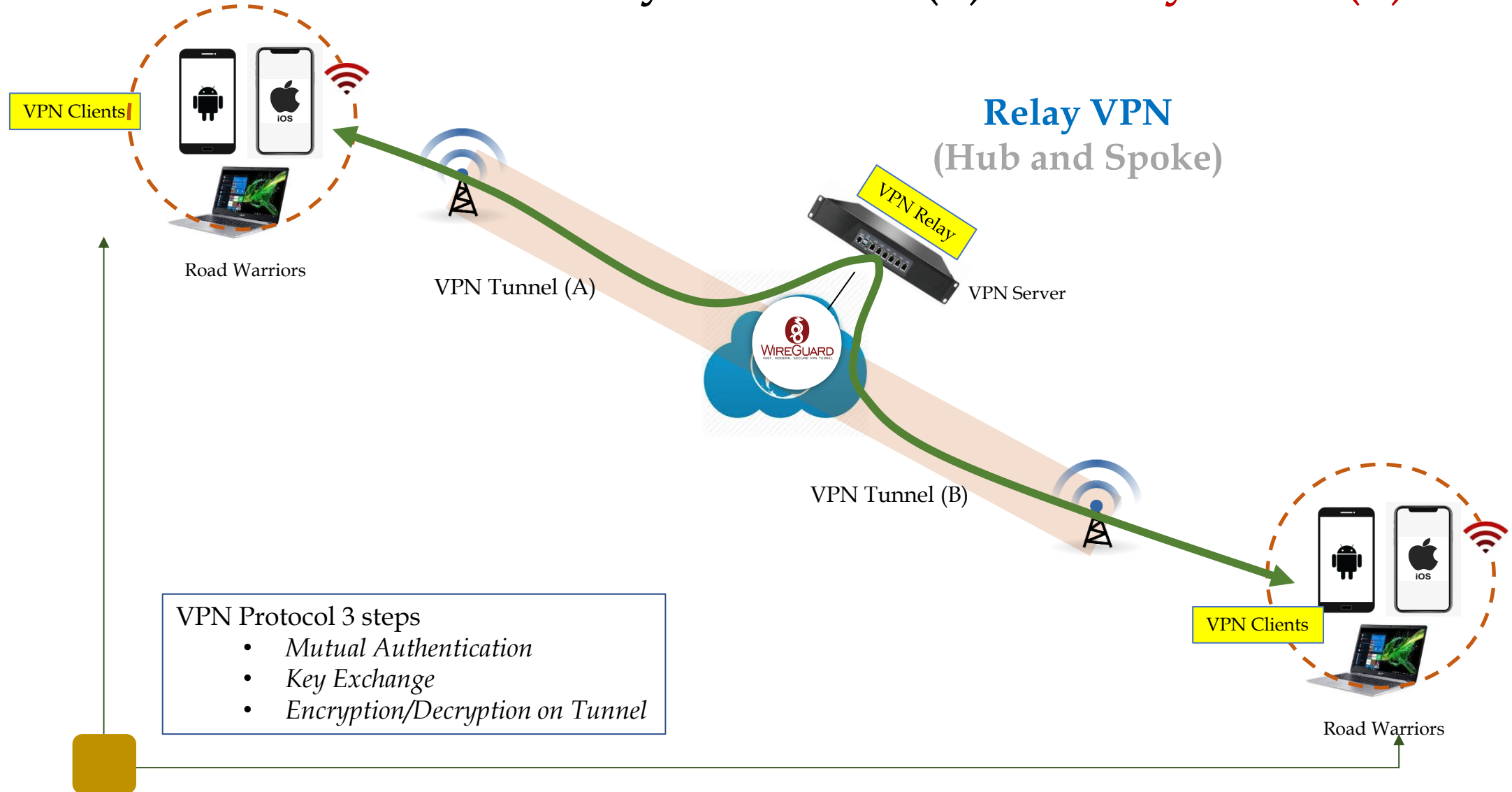
1. Zero Trust P2P Security Platform

Zero Trust P2P Security Platform

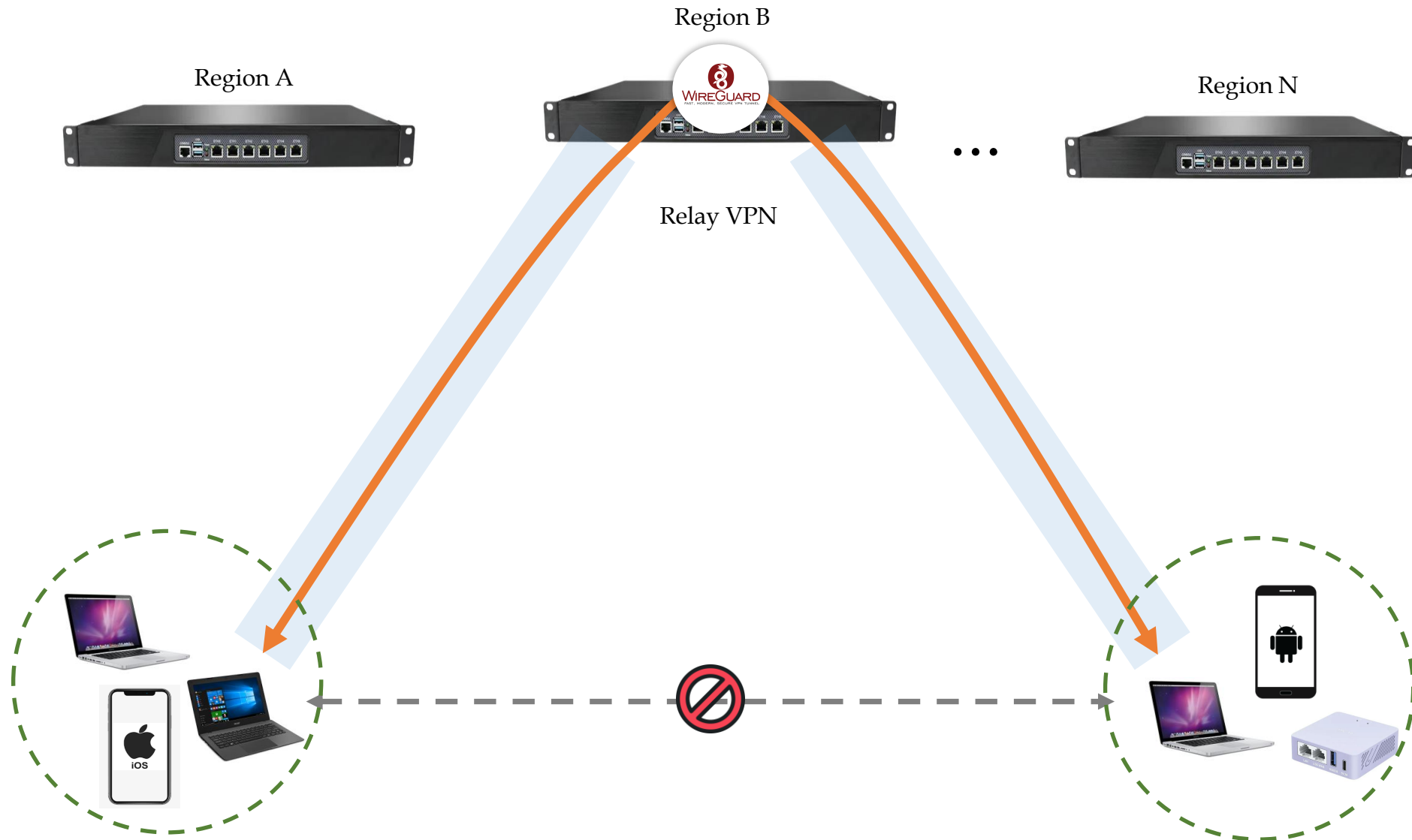
- ZTNA/Secure Network
- No modifications for any applications
- Auto Connect, Direct Connect and Relay Connect



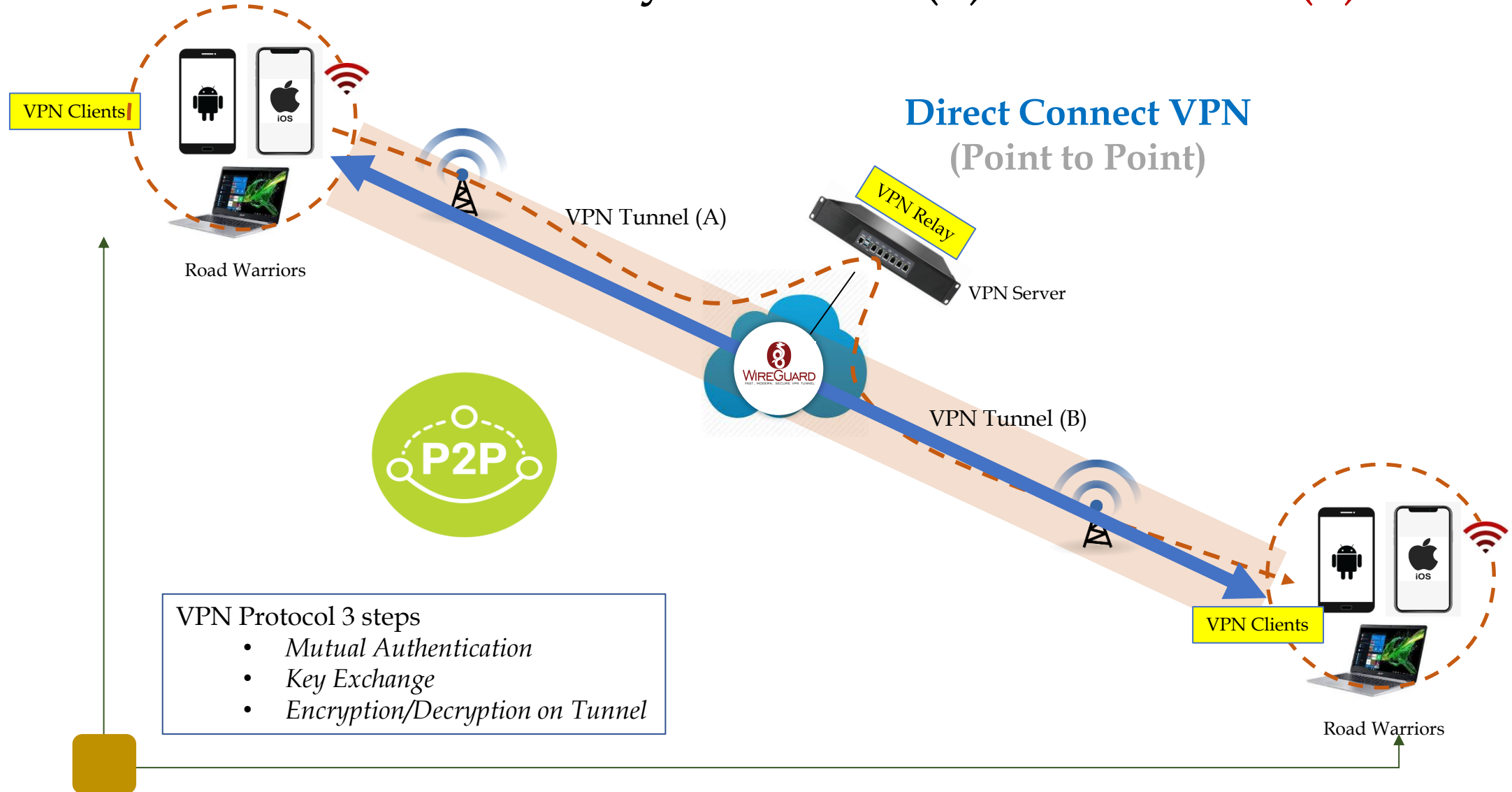
1. Zero Trust P2P Security Platform(1) – Relay VPN(1)



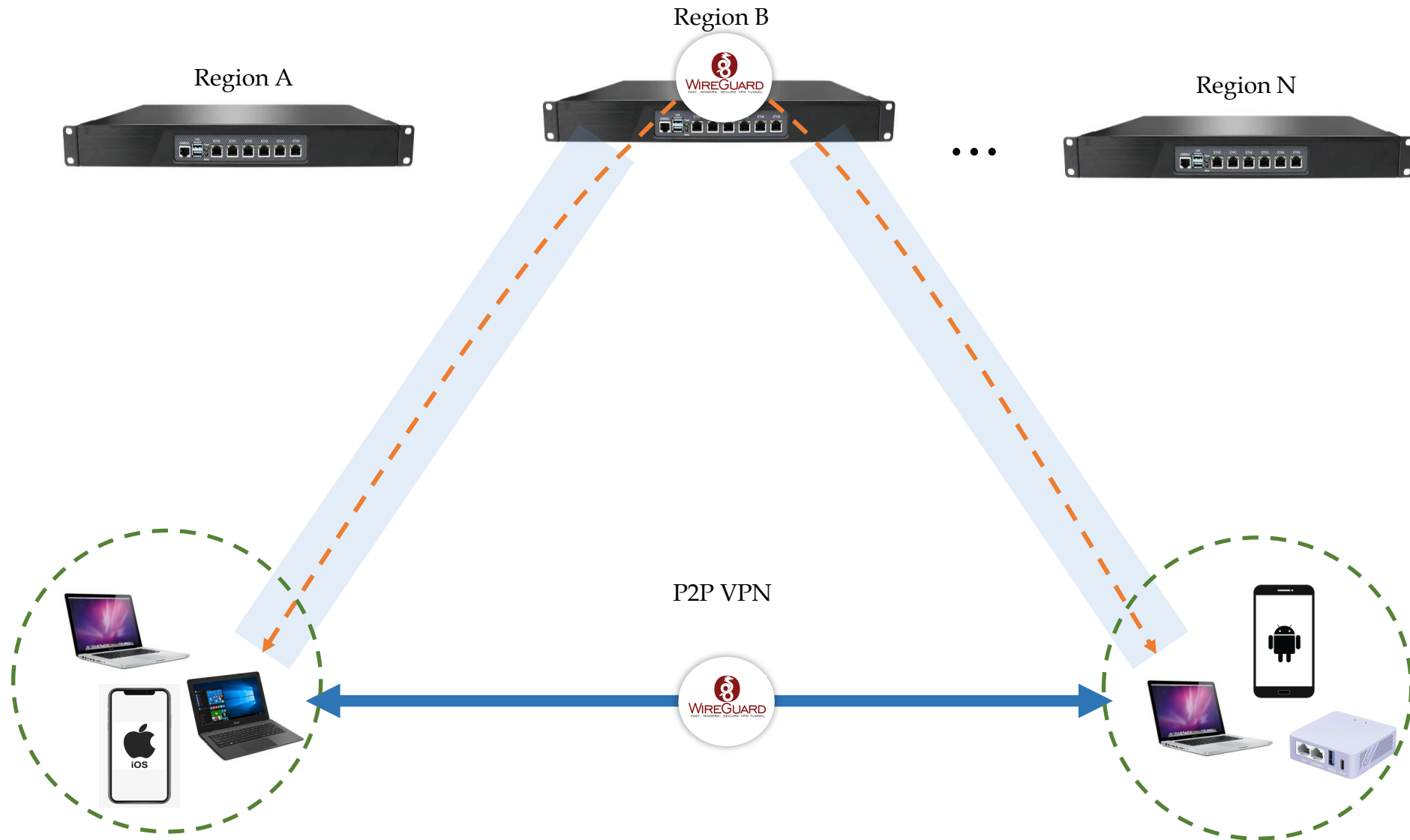
1. Zero Trust P2P Security Platform(1) – Relay VPN(2)



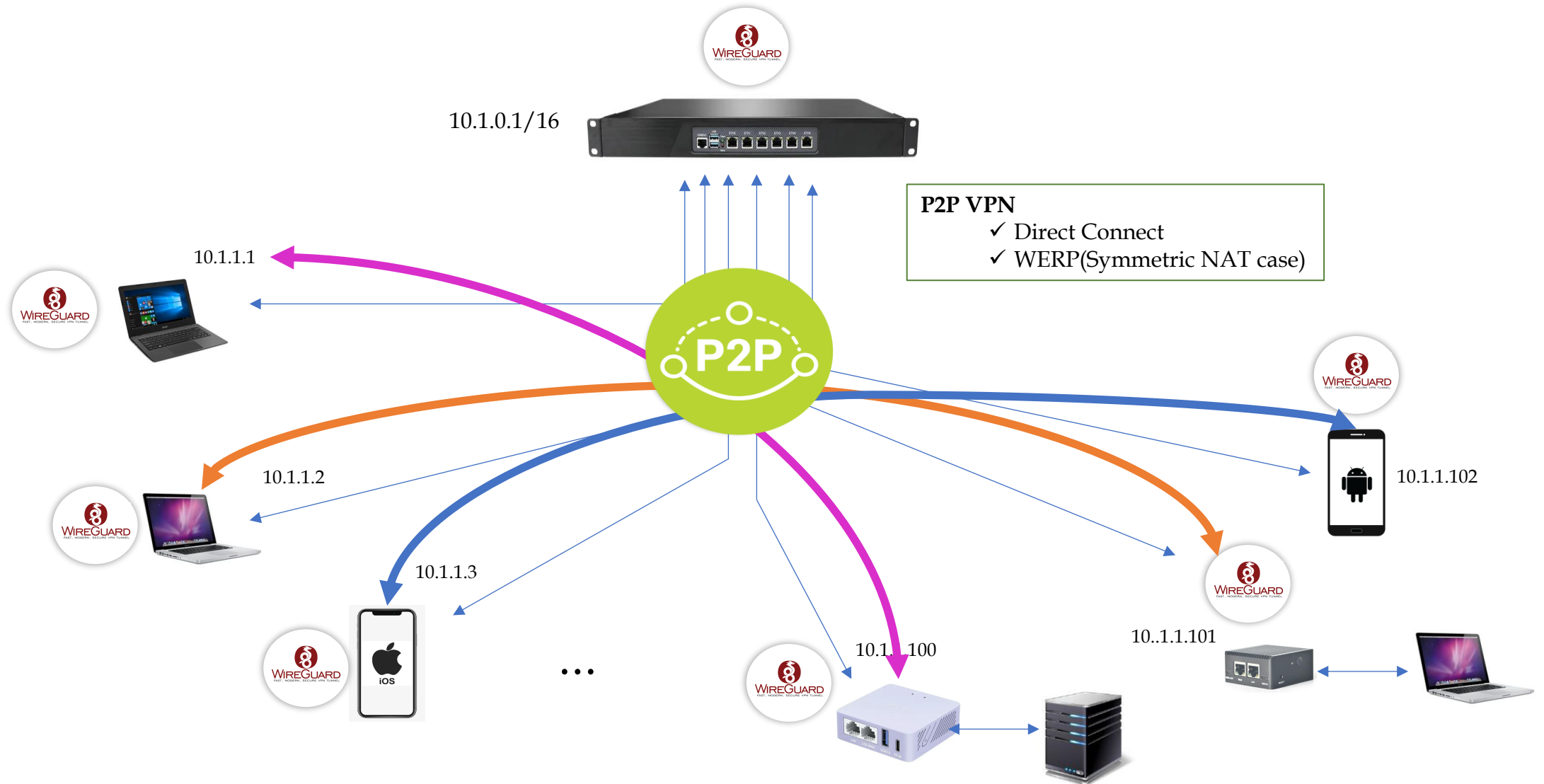
1. Zero Trust P2P Security Platform(2) – P2P VPN(1)



1. Zero Trust P2P Security Platform(2) – P2P VPN(2)



1. Zero Trust P2P Security Platform(2) – P2P VPN(3)



1. Zero Trust P2P Security Platform(3) – ZTNA(1-1)

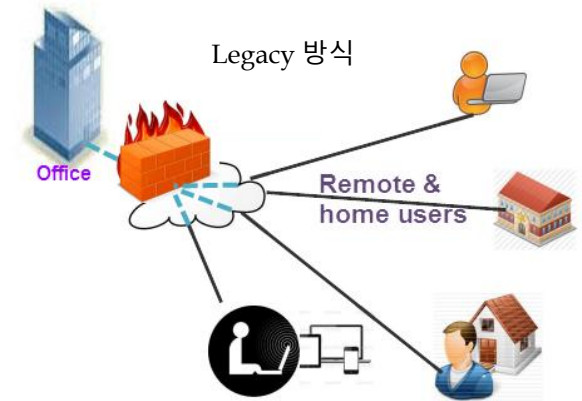
- Legacy 방식(Firewall/VPN/IPS/UTM)

- Internal network(Trusted Area)과 External network(Untrusted Area) 구분
- Internal network <=> Perimeter Security 제품 ⇔ External network : 안전한 내부망과 불안정한 외부망 경계 (Perimeter)에 보안 제품 설치
- Static Configuration, IP/Port 기반의 통제

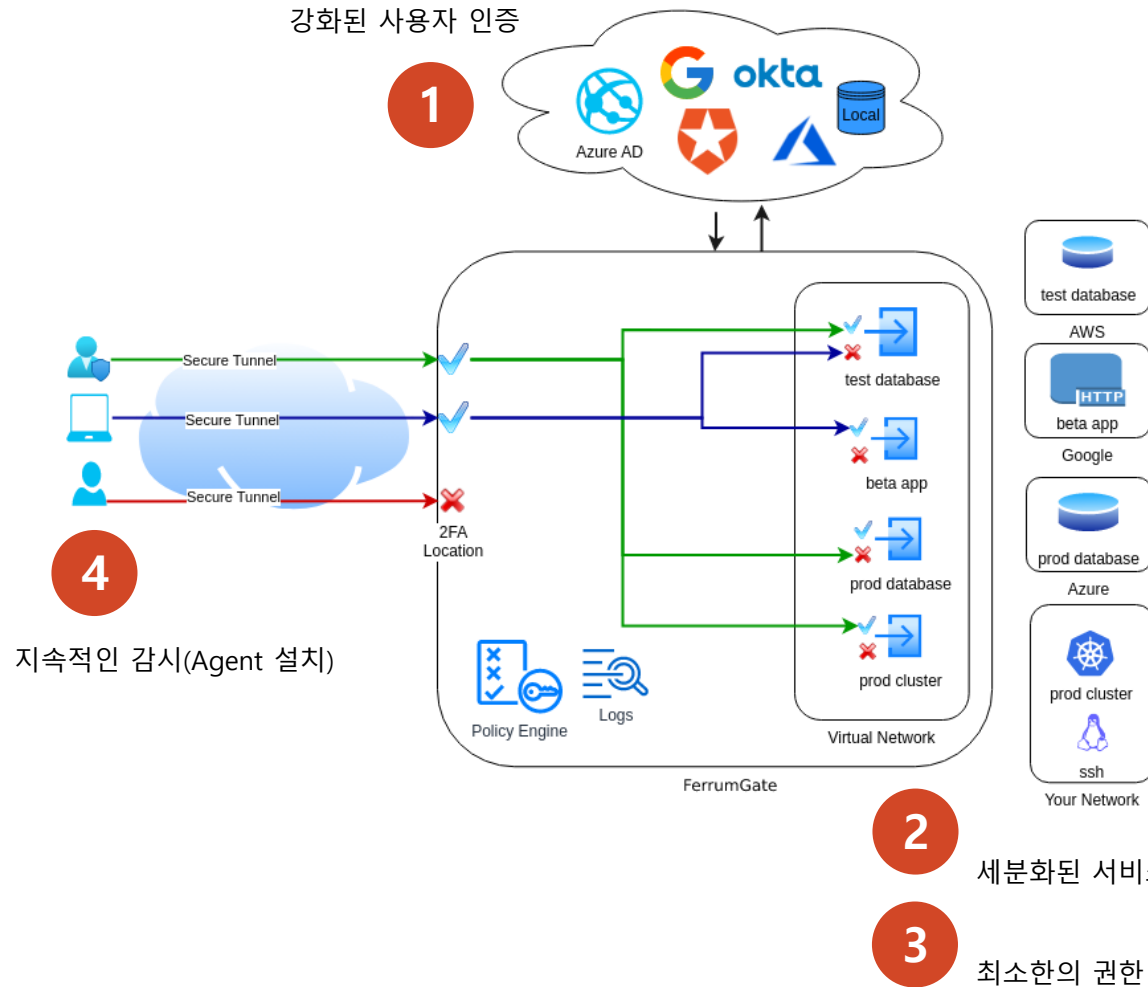
VPN is Dead

- ZTNA or SDP(Software Defined Perimeter) 방식

- 환경의 변화: Cloud, Mobile, BYOD
 - *이제는 중요한 것들이 Perimeter 바깥에 있는 세상이 되었다.*
- Zero Trust: "Internal, External network 모두 안전하지 않다"에서 출발
- S/W 방식의 중앙 Controller에 의한 접근 통제
 - *실제 network에 접근하기 전에, 인증(Authentication) & 접근 통제(Authorization)을 수행하는 개념*
 - *내부 Resource는 Appliance(Gateway)에 의해 모두 Hidden 상태, 인증을 통과한 후에 외부에 Open되는 구조. Dark Network(외부에 보이지 않는 network) 개념 출현*
- Dynamic Control, IP 기반이 아니라, 사용자 기반의 통제



1. Zero Trust P2P Security Platform(3) – ZTNA(1-2)

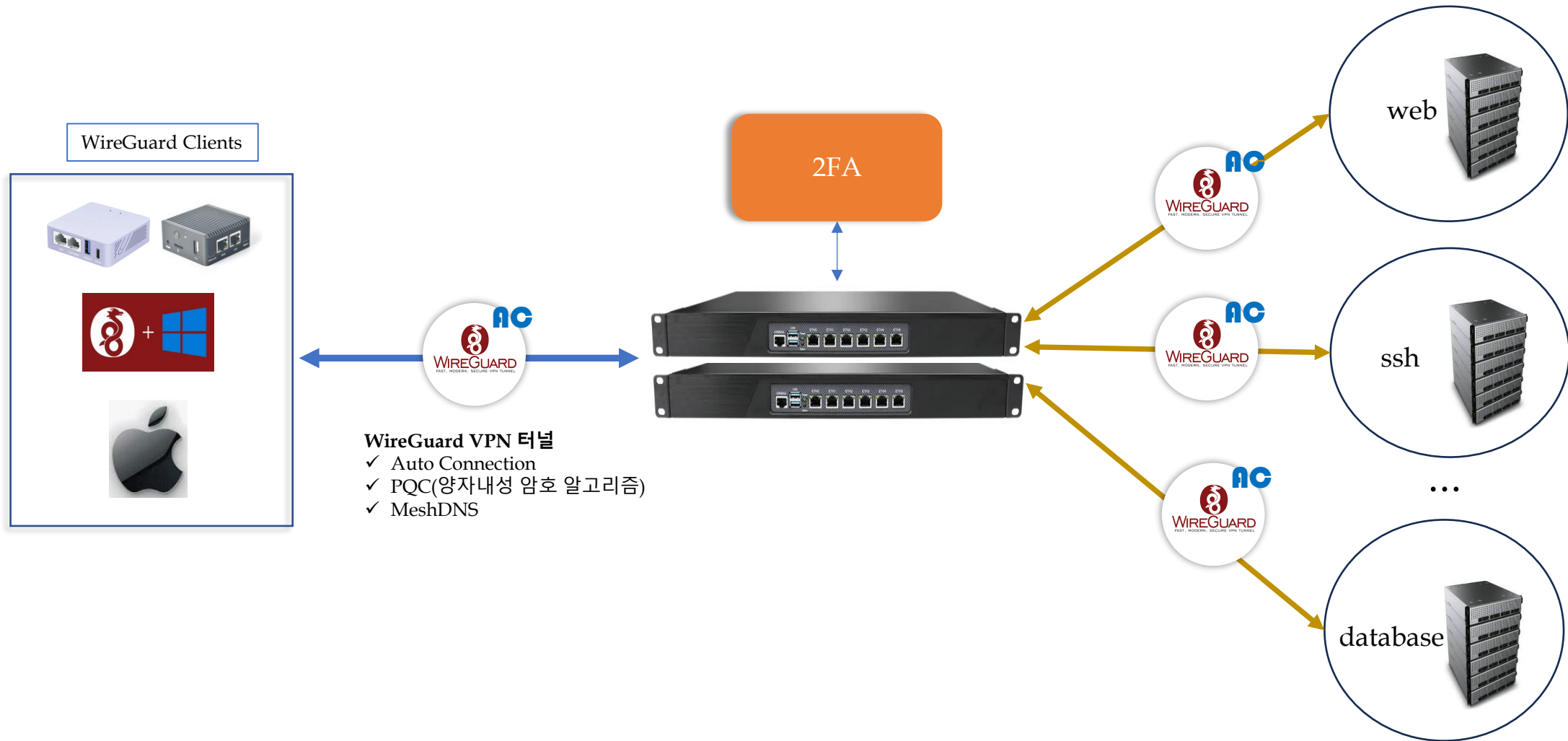


ZTNA의 요건

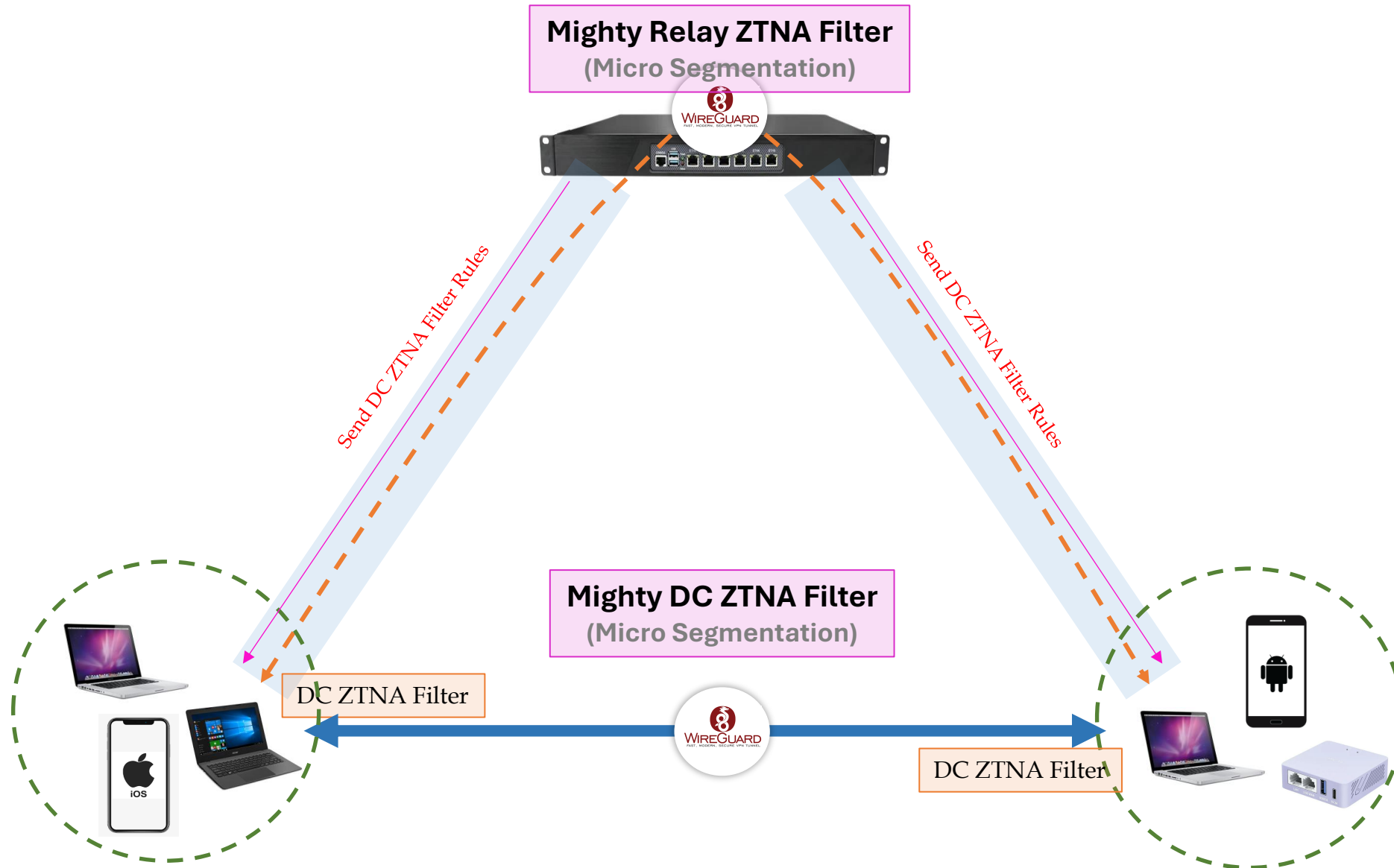
- More Authentication Options
 - MFA: SAML, OAuth2, LDAP, FIDO2
 - Location Check, Time Check, Device Posture Check
- Micro-segmentation
 - Virtual Application Networks
- Least privilege access
 - RBAC(Role Based Access Control)
- Continuous monitoring
 - Every packet, every flow, every connection

[출처] <https://ferrumgate.com/>

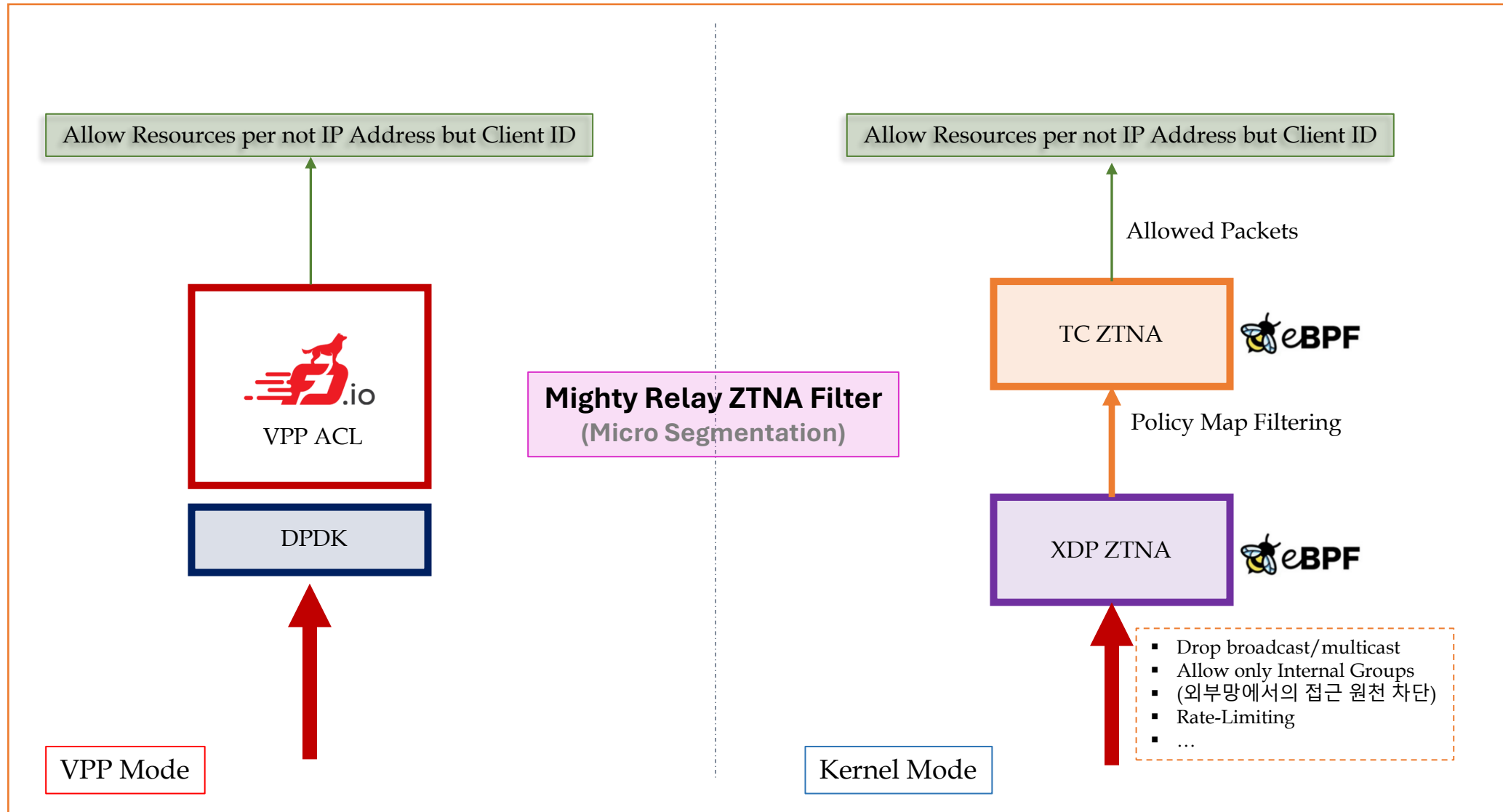
1. Zero Trust P2P Security Platform(3) – ZTNA(2)



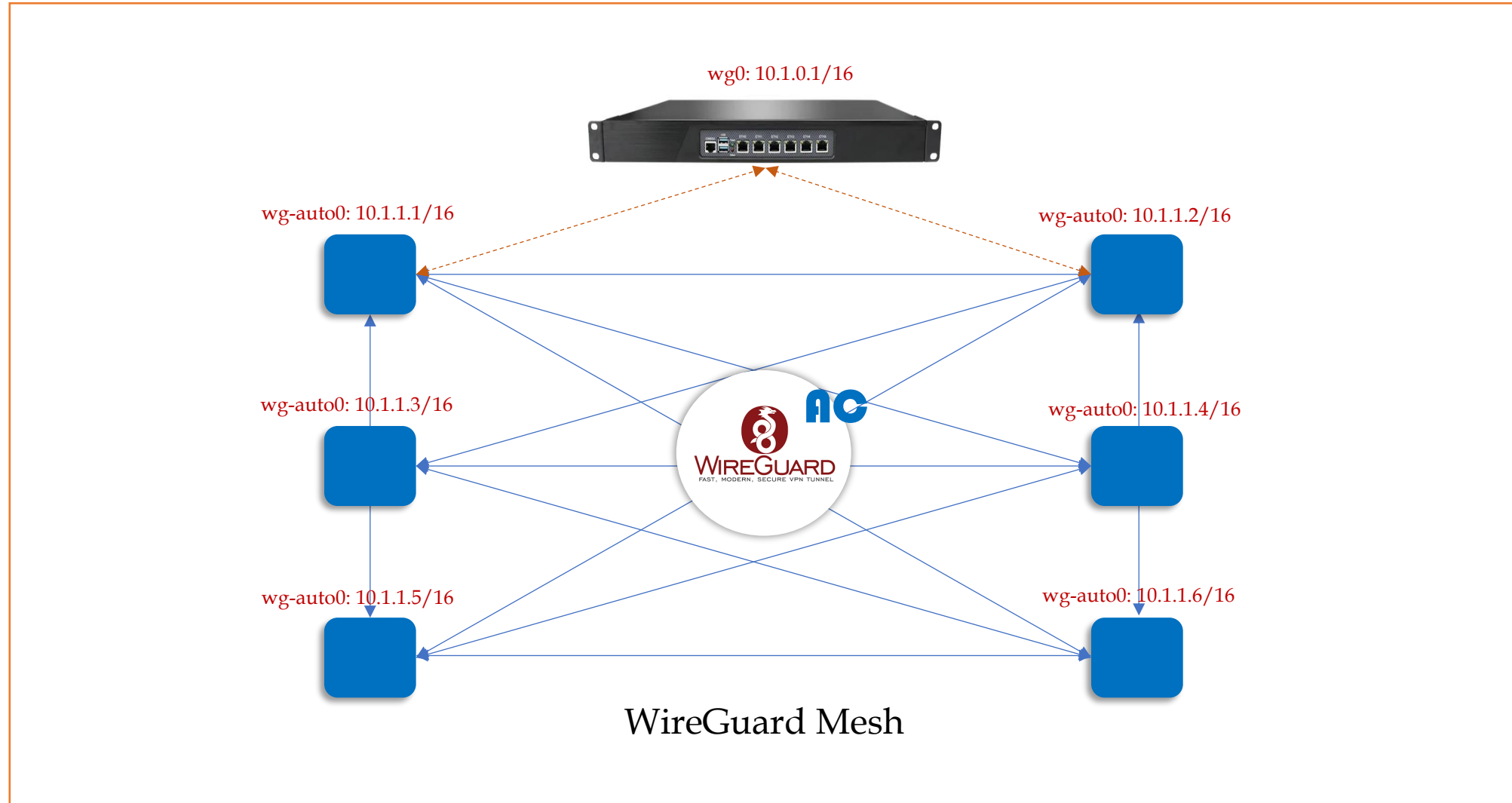
1. Zero Trust P2P Security Platform(3) – ZTNA(3-1)



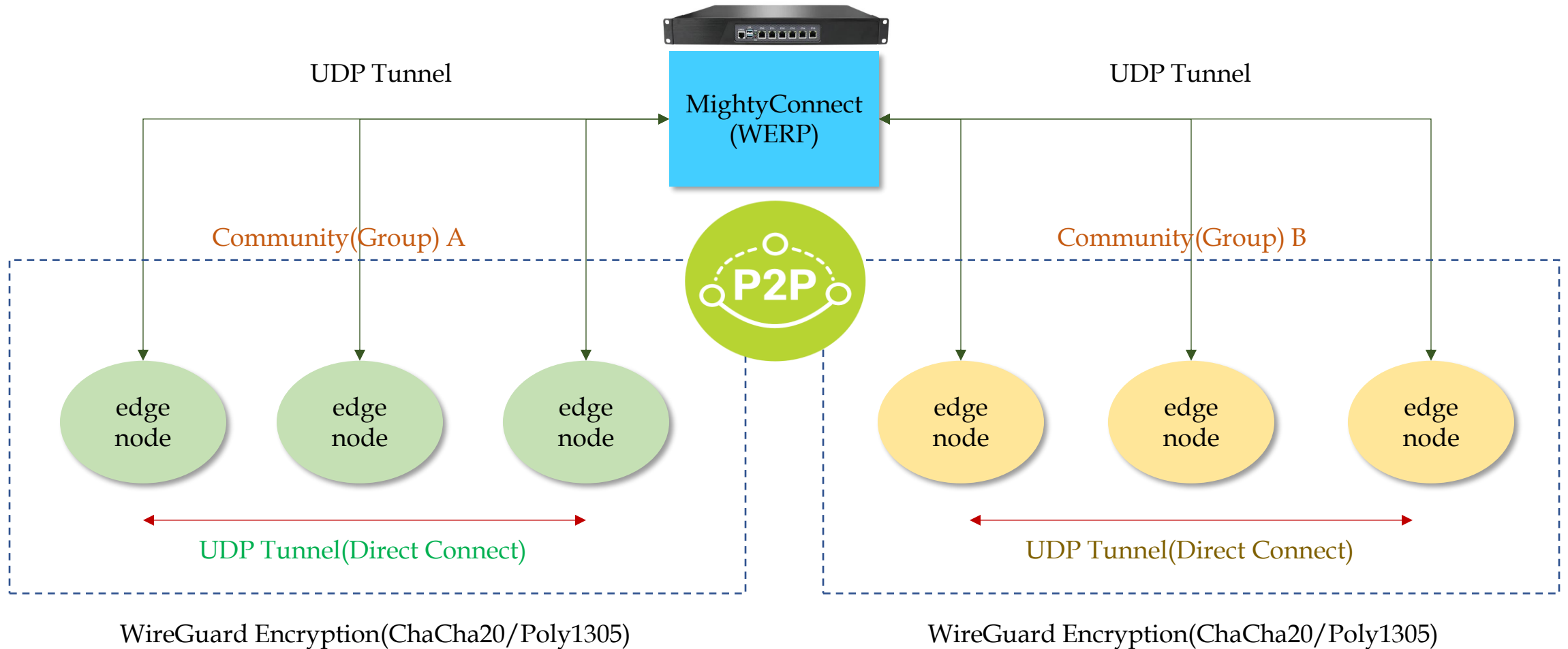
1. Zero Trust P2P Security Platform(3) – ZTNA(3-2)



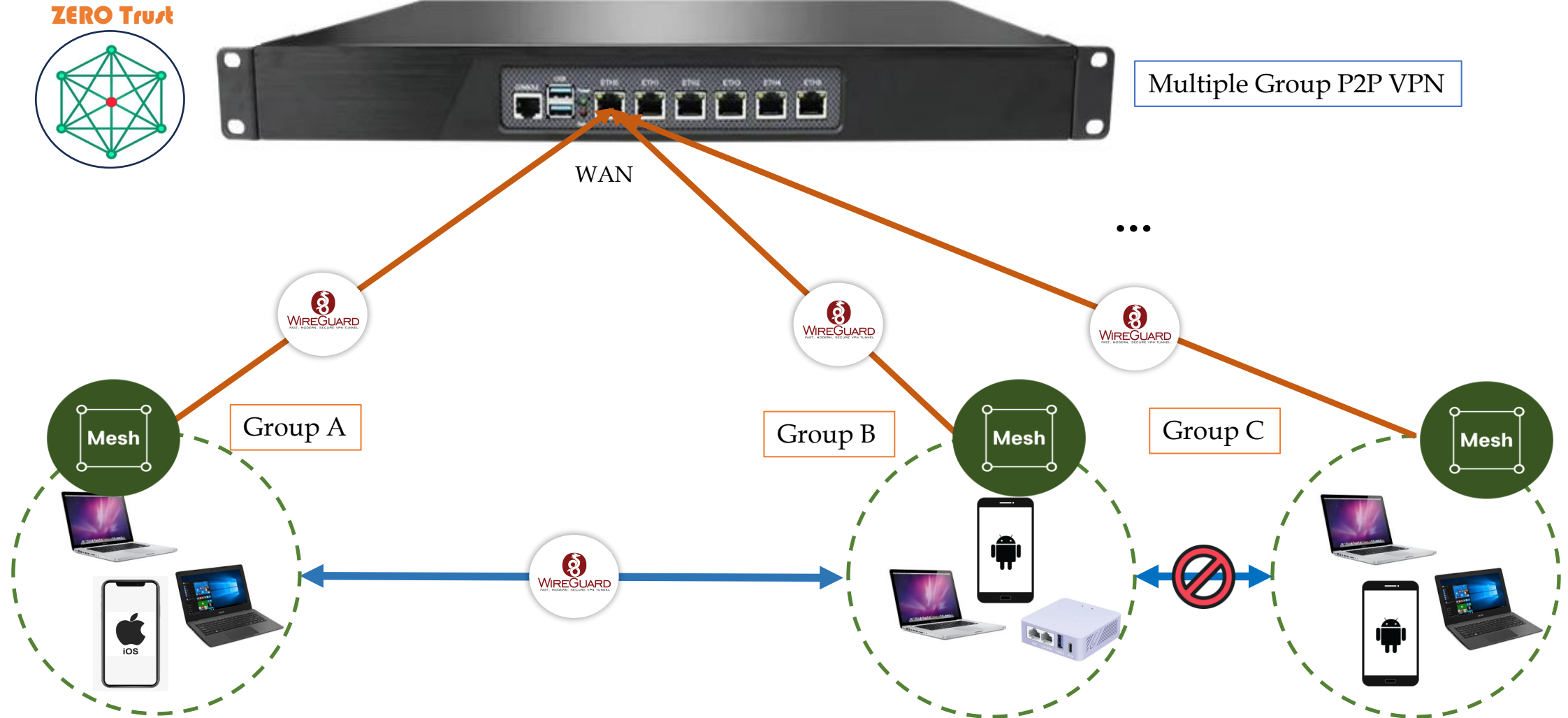
1. Zero Trust P2P Security Platform(4) - Mesh(1)



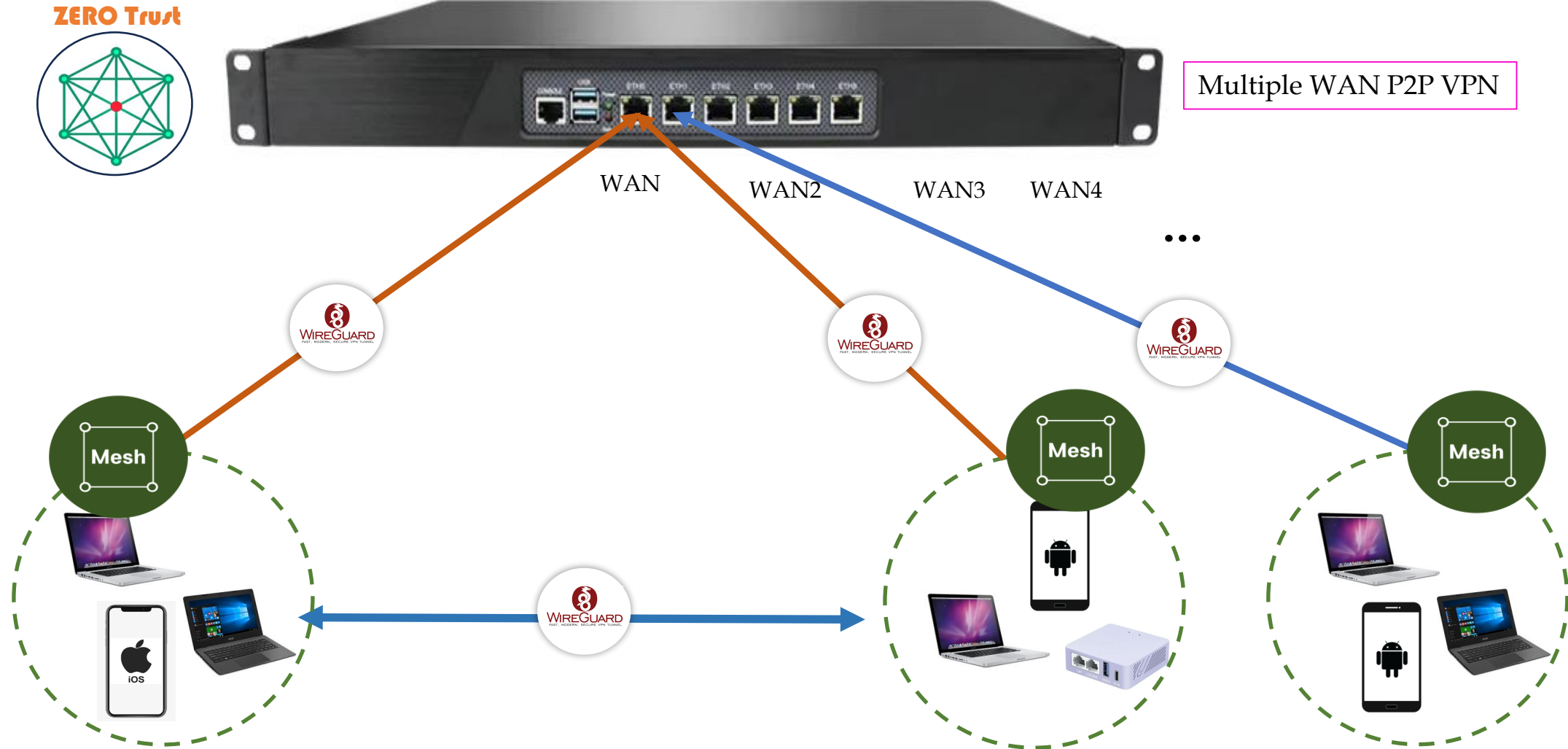
1. Zero Trust P2P Security Platform(4) – Mesh(2)



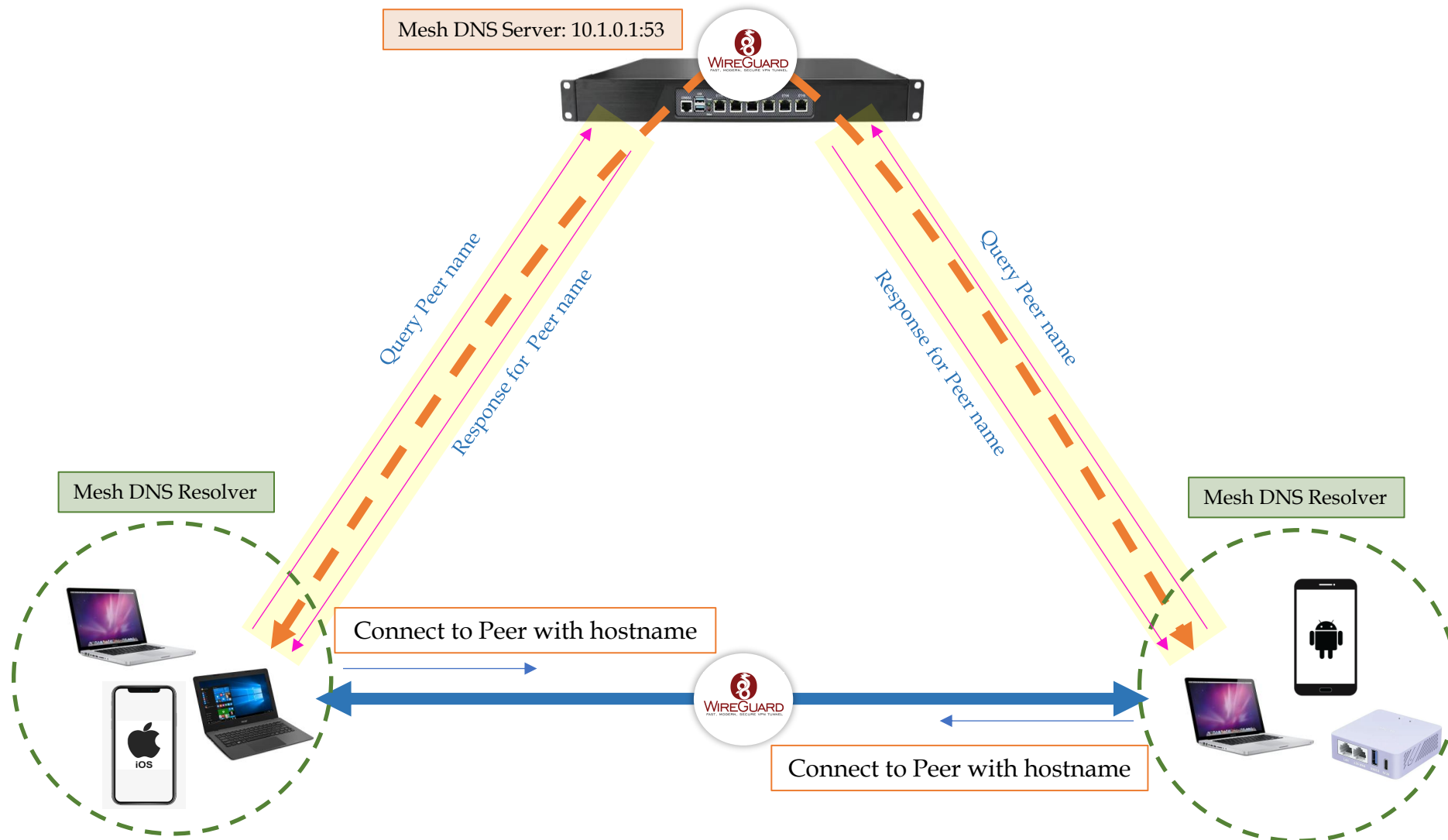
1. Zero Trust P2P Security Platform(4) – Mesh(3)



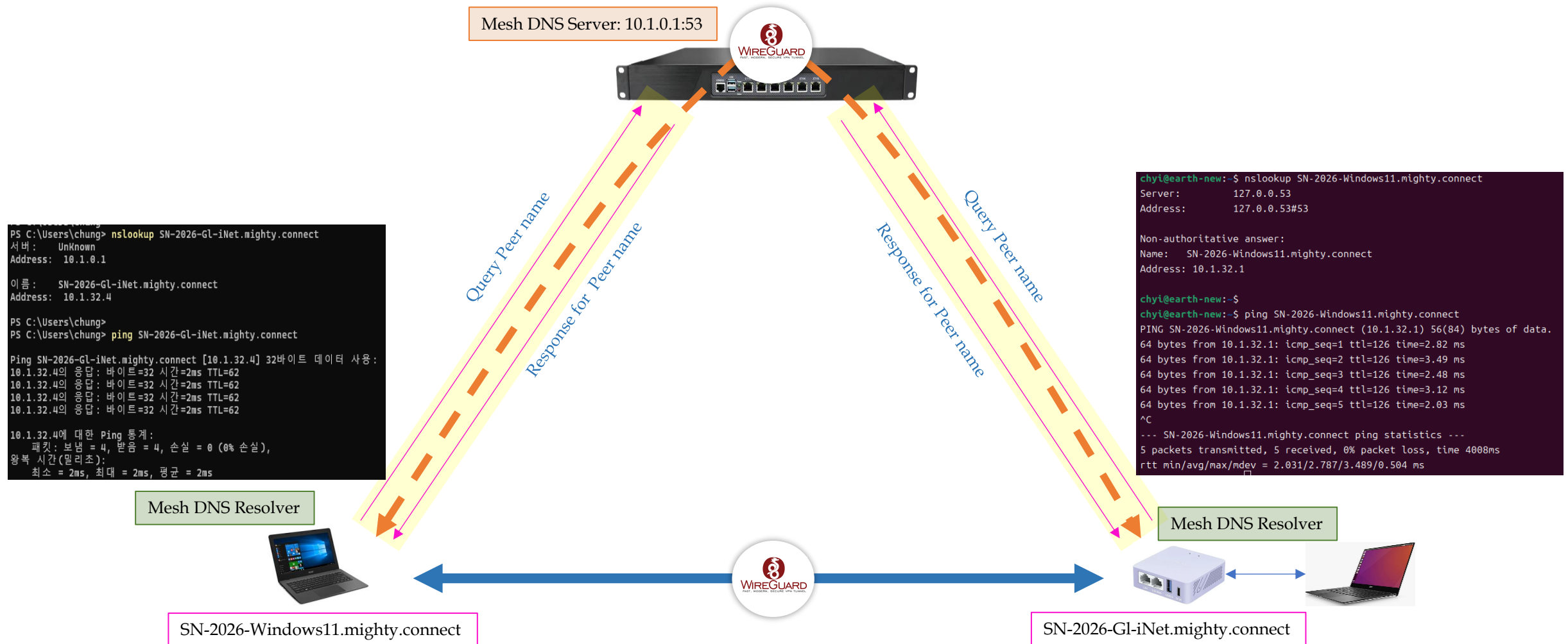
1. Zero Trust P2P Security Platform(4) - Mesh(4)



1. Zero Trust P2P Security Platform(5) – Mesh DNS(1)



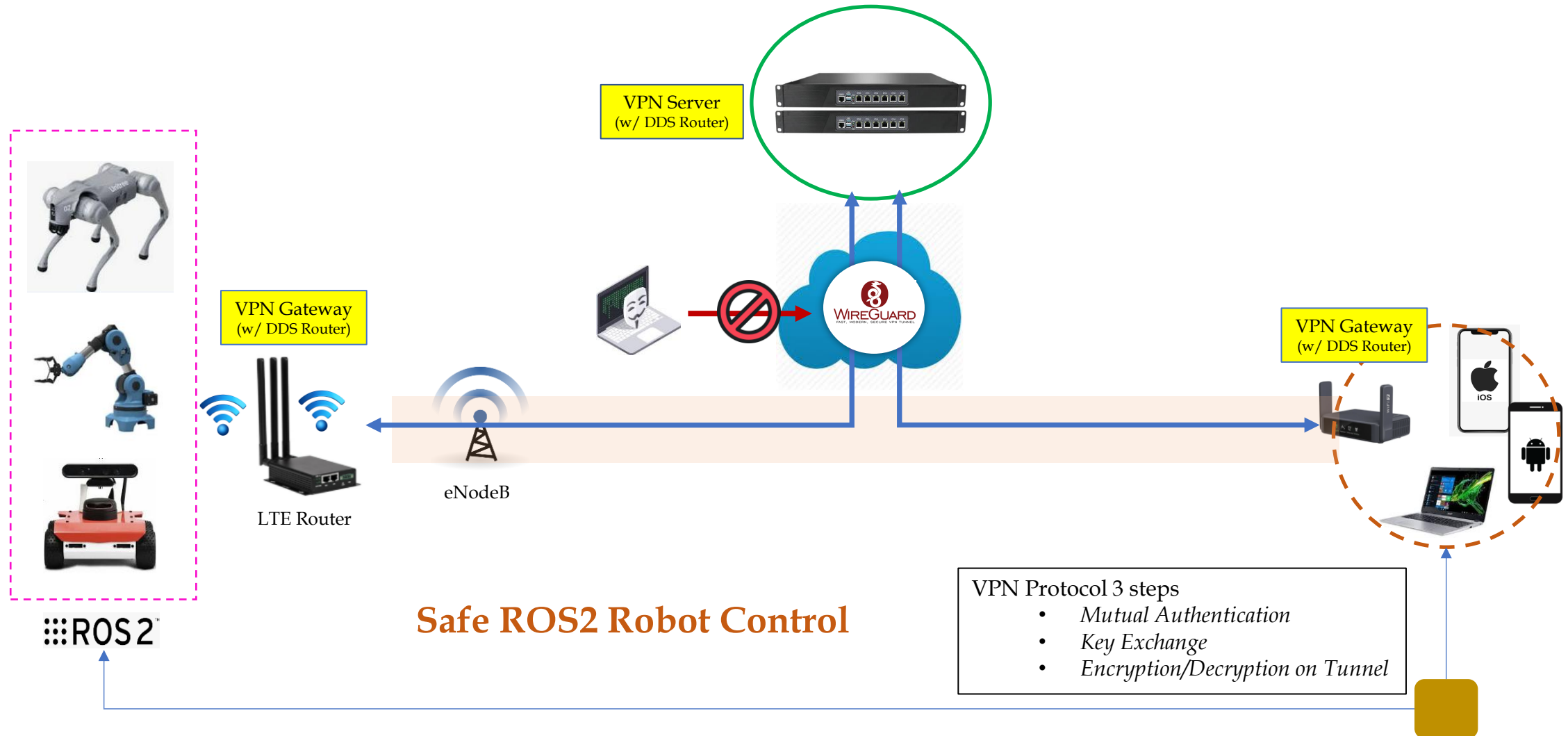
1. Zero Trust P2P Security Platform(5) – Mesh DNS(2)



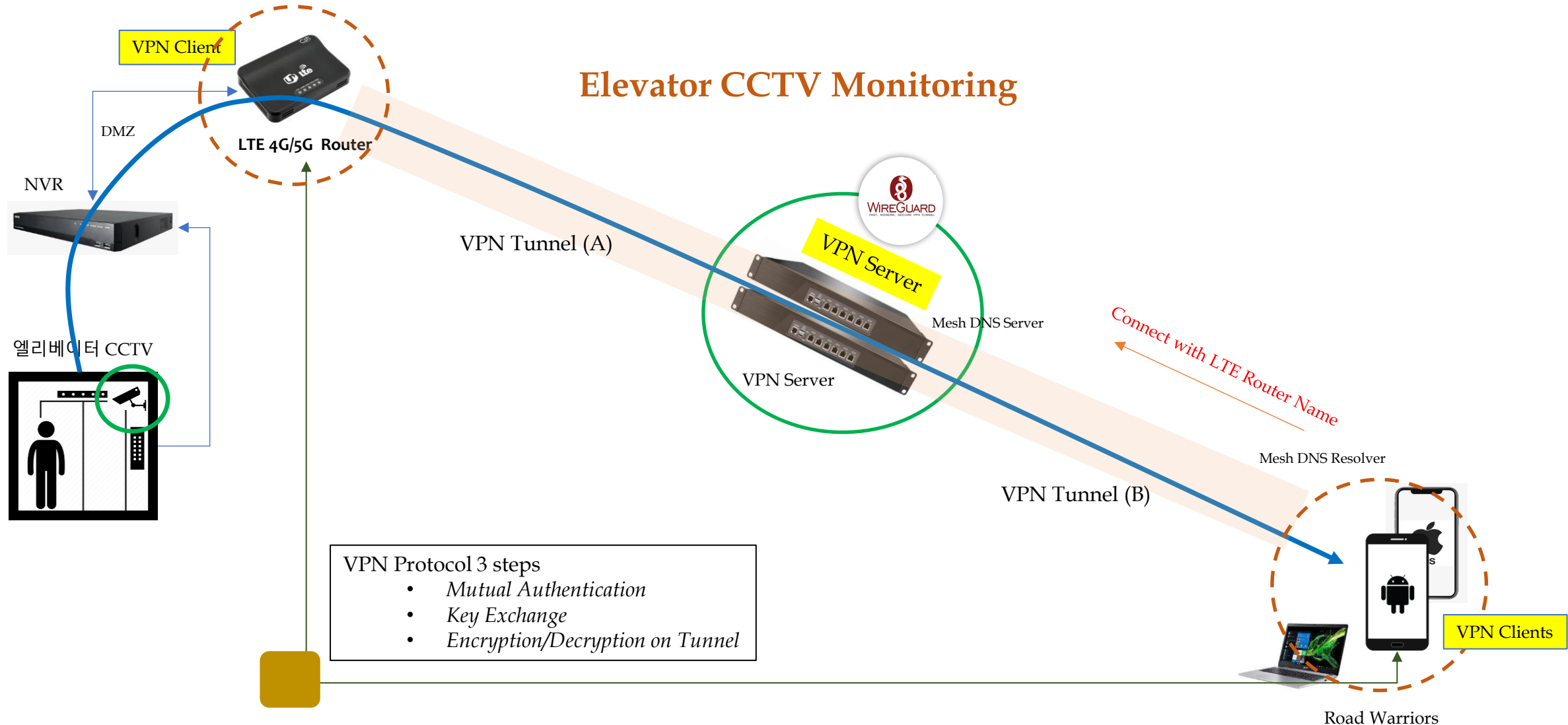
1. Zero Trust P2P Security Platform(6) – Use Cases(1)



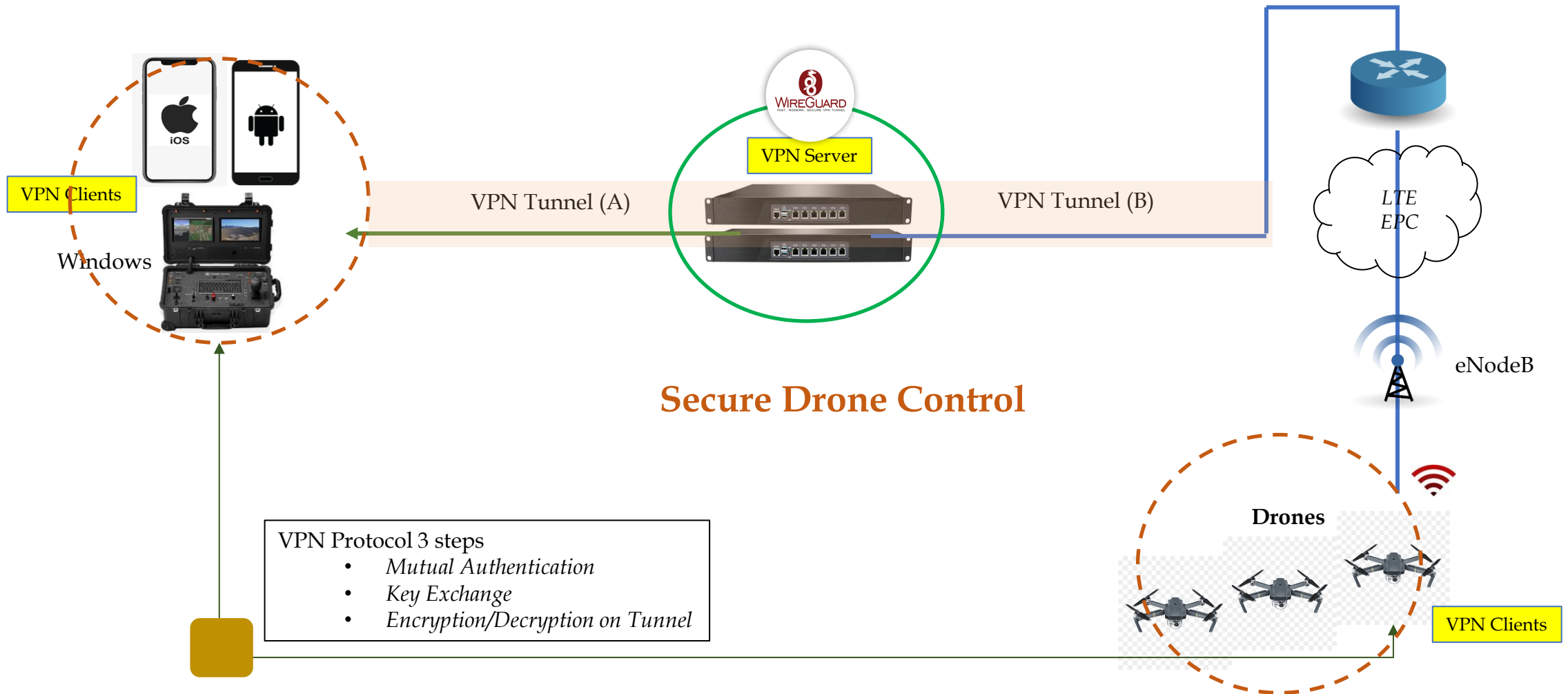
1. Zero Trust P2P Security Platform(6) - Use Cases(2)



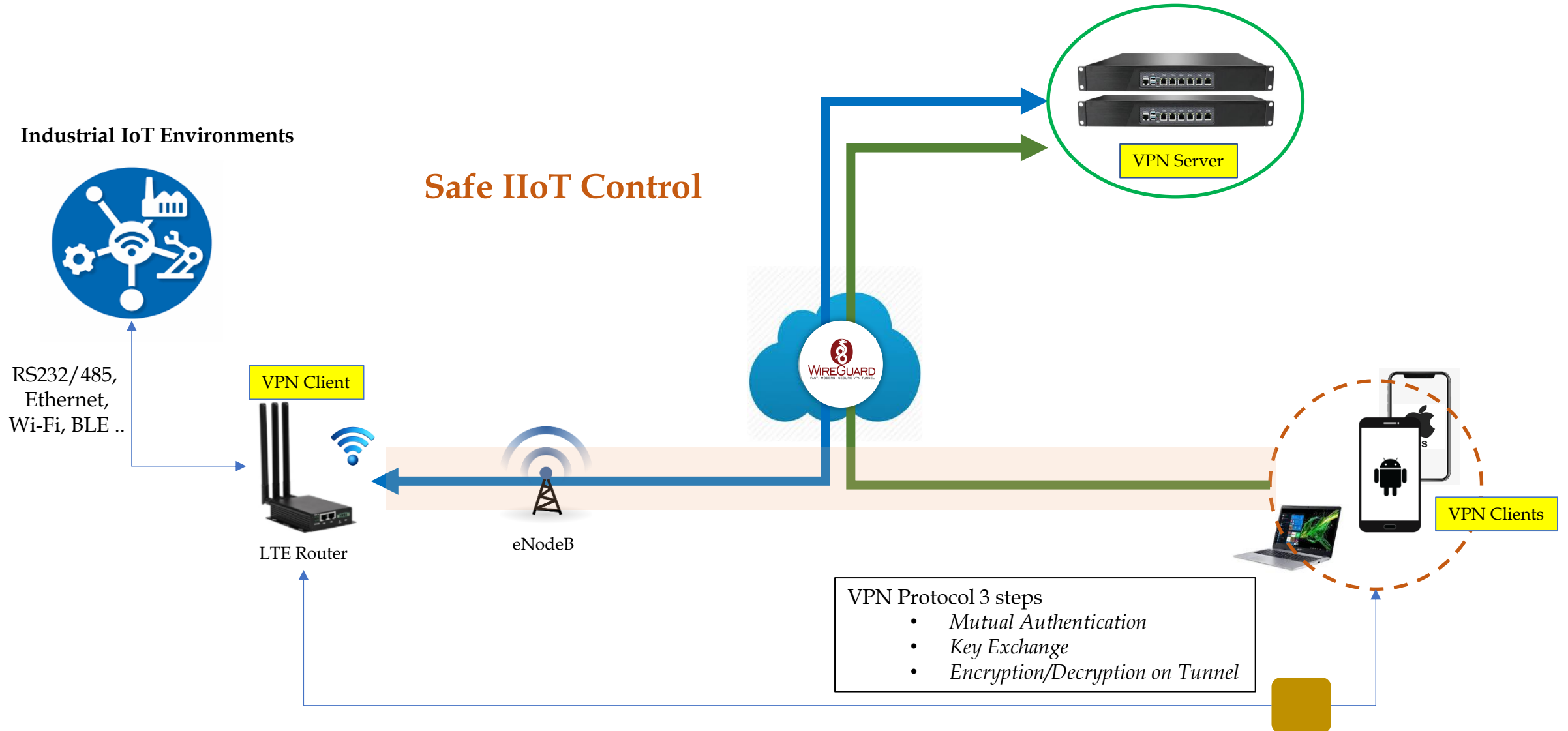
1. Zero Trust P2P Security Platform(6) - Use Cases(3)



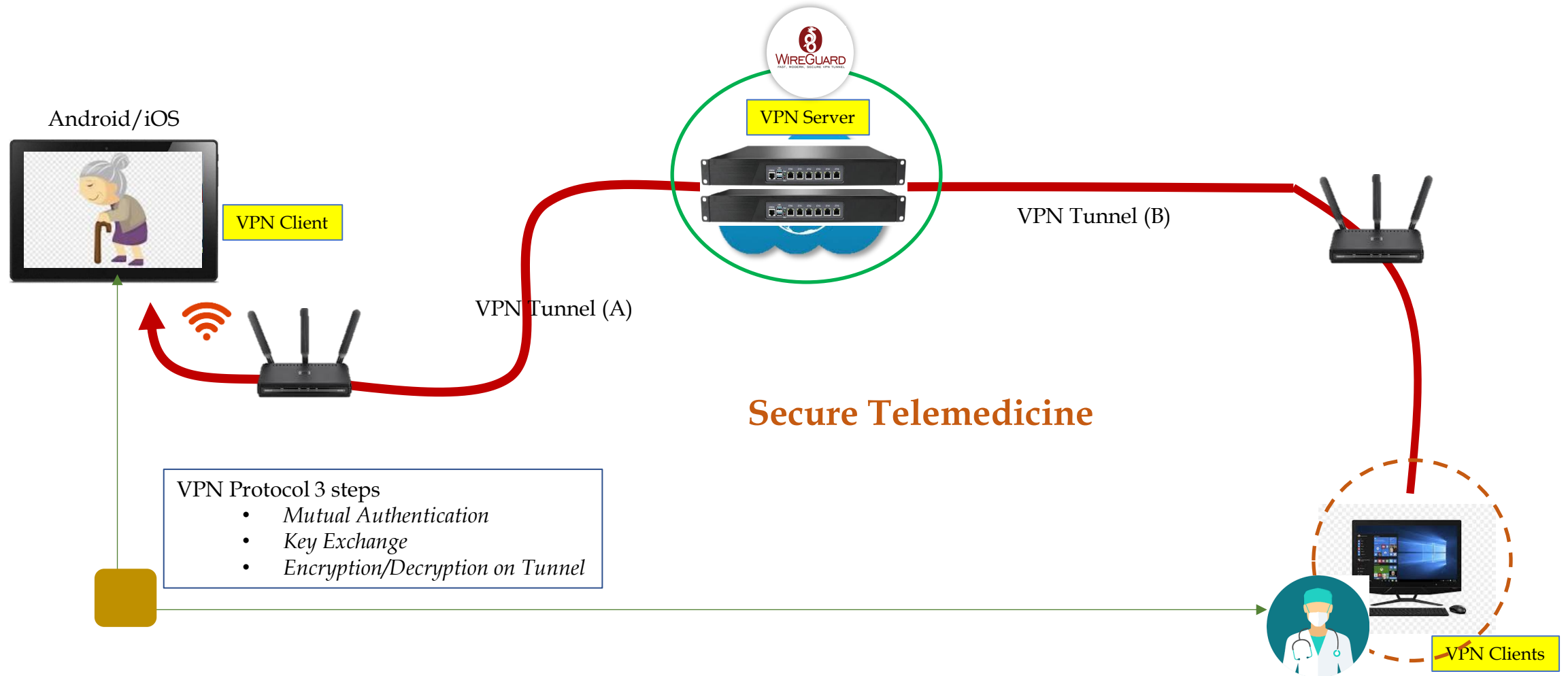
1. Zero Trust P2P Security Platform(6) - Use Cases(4)



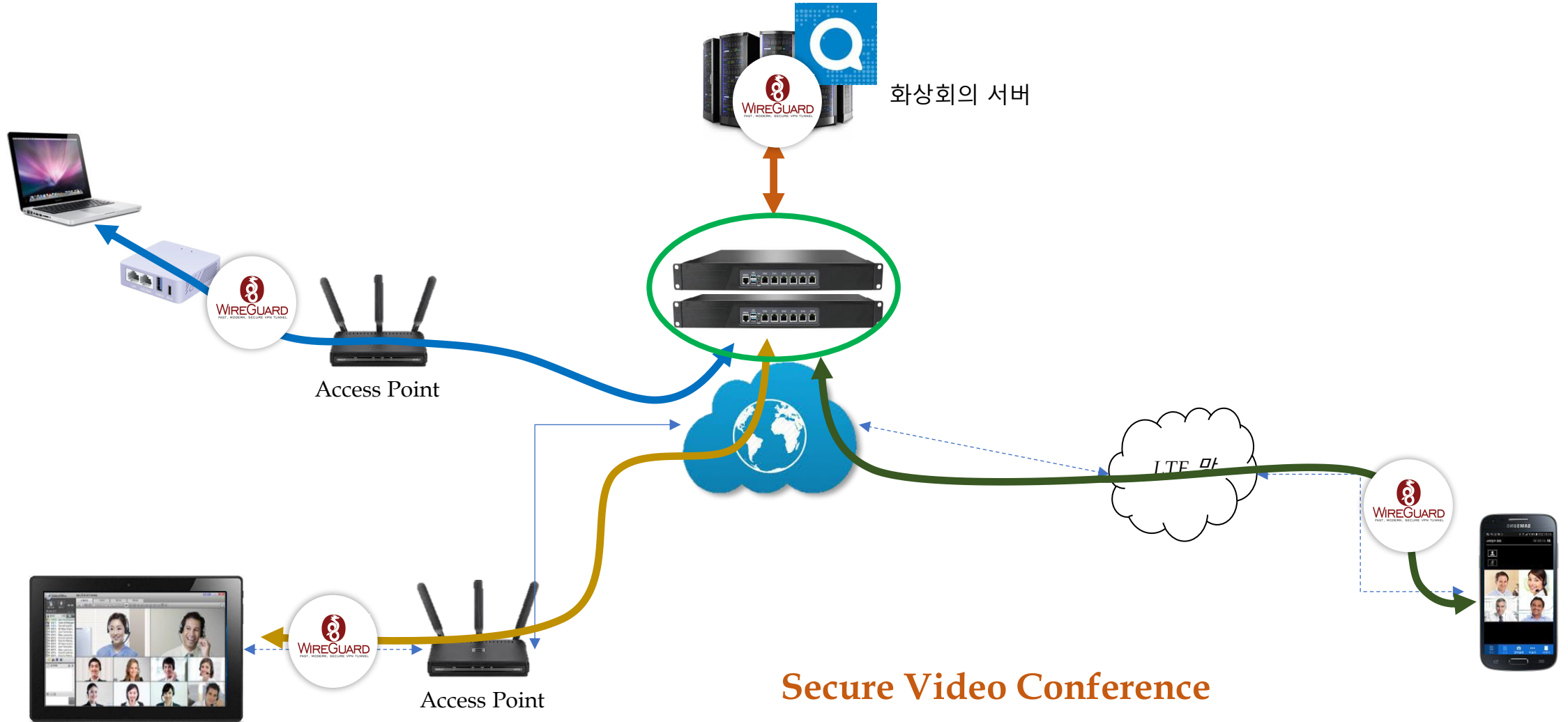
1. Zero Trust P2P Security Platform(6) - Use Cases(5)



1. Zero Trust P2P Security Platform(6) - Use Cases(6)



1. Zero Trust P2P Security Platform(6) - Use Cases(7)



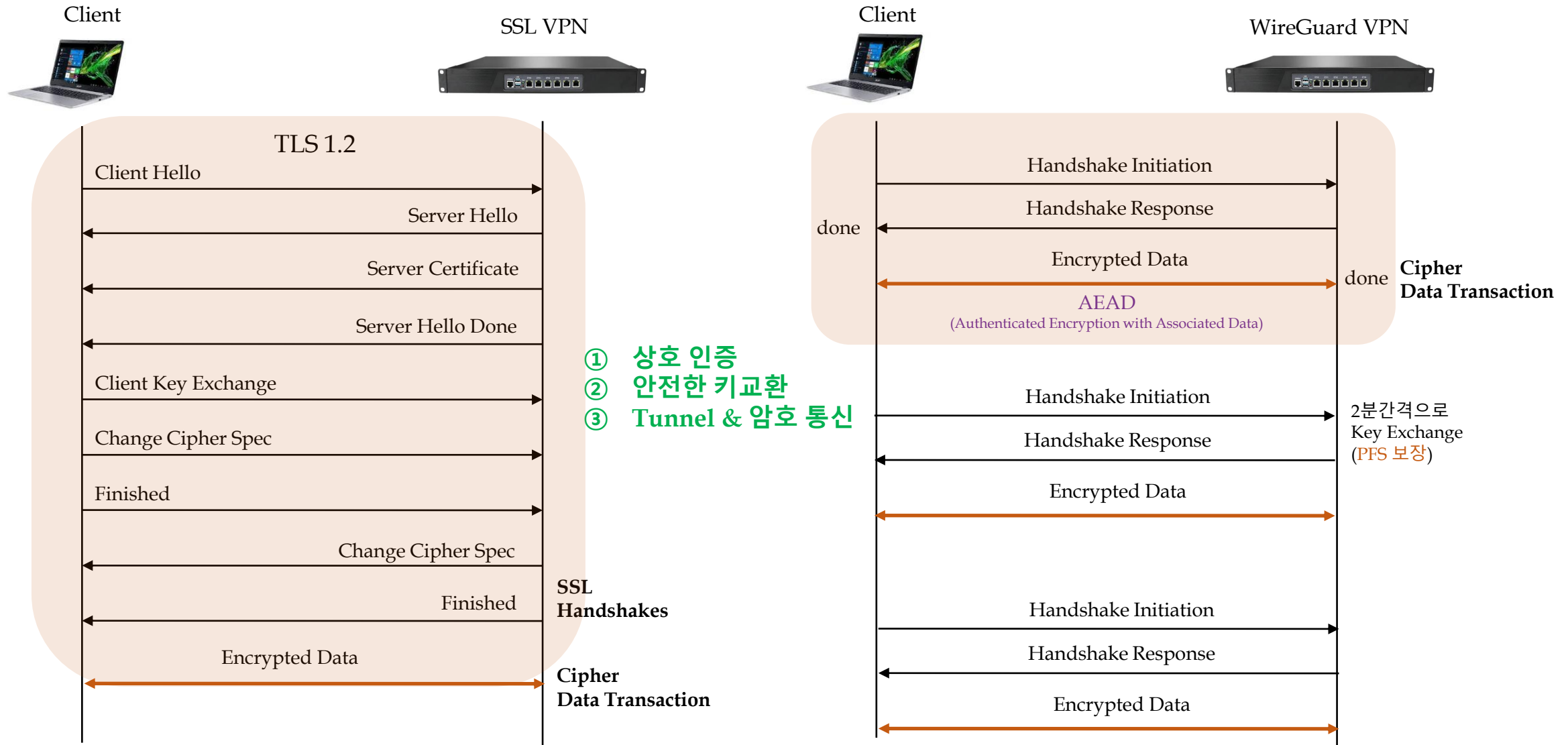
2. WireGuard

New Generation VPN Protocol



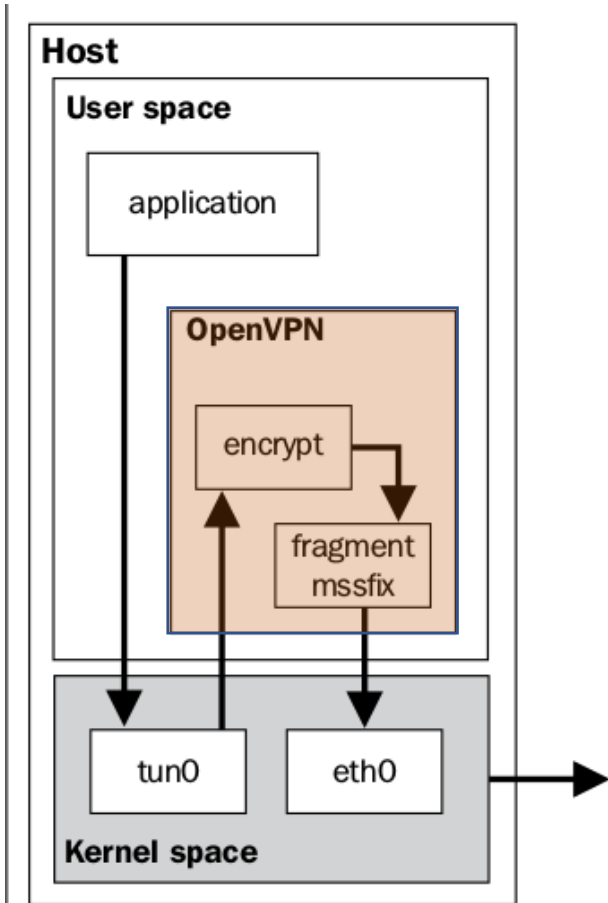
"WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld

2. WireGuard(1) - Architecture(1)



2. WireGuard(1) - Architecture(1)

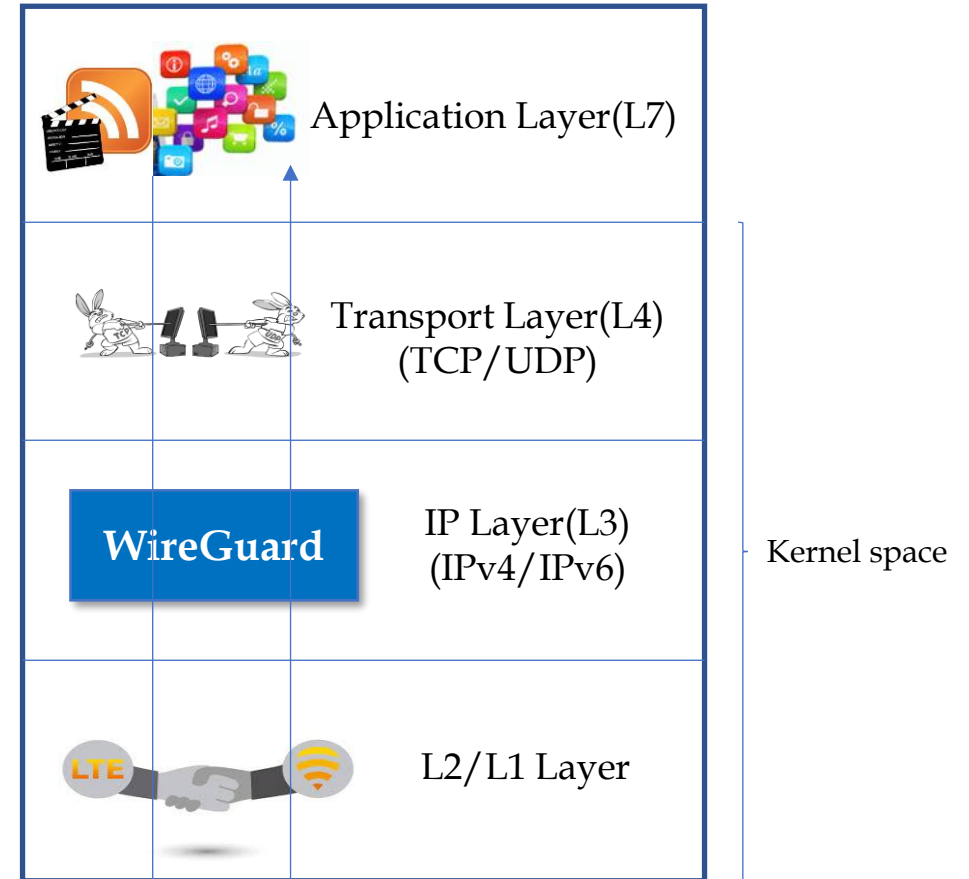
Limitations of OpenVPN



SSL VPN Tunnel(Before DCO)

VS

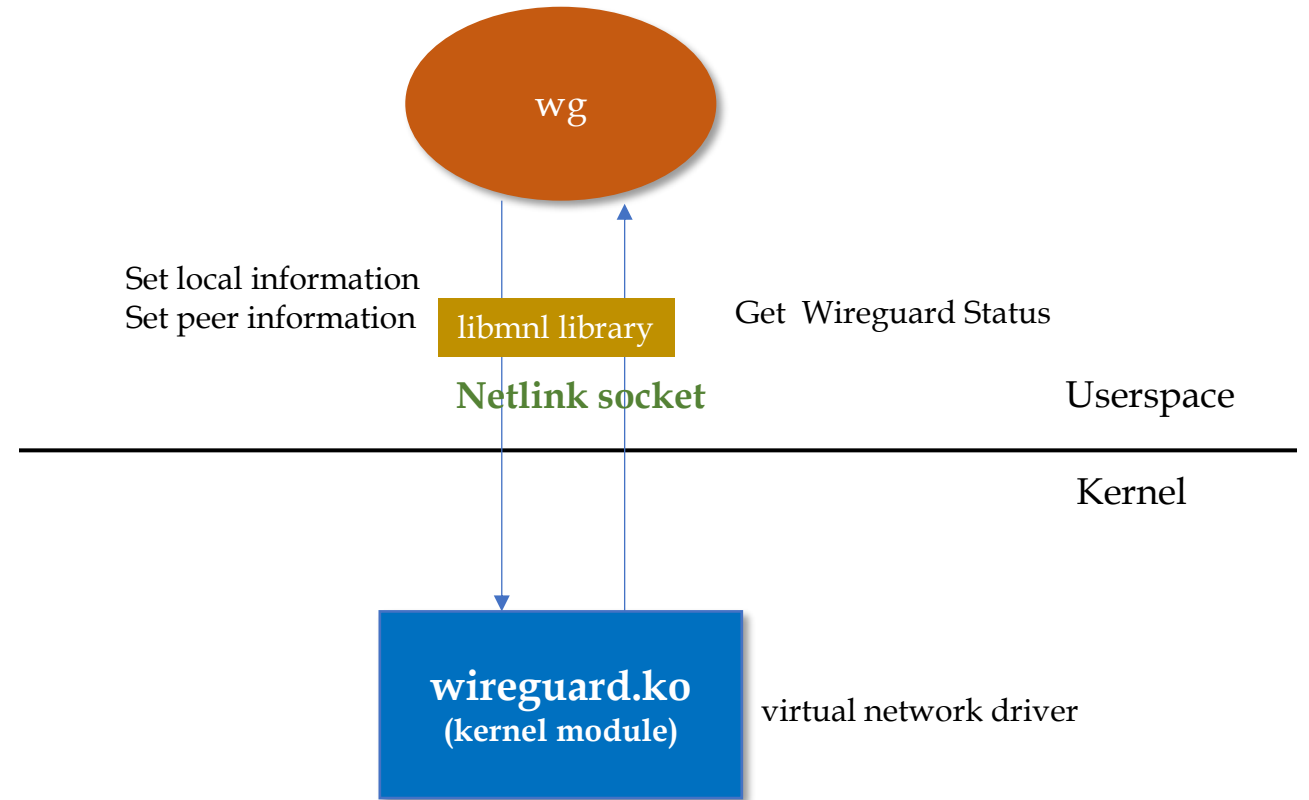
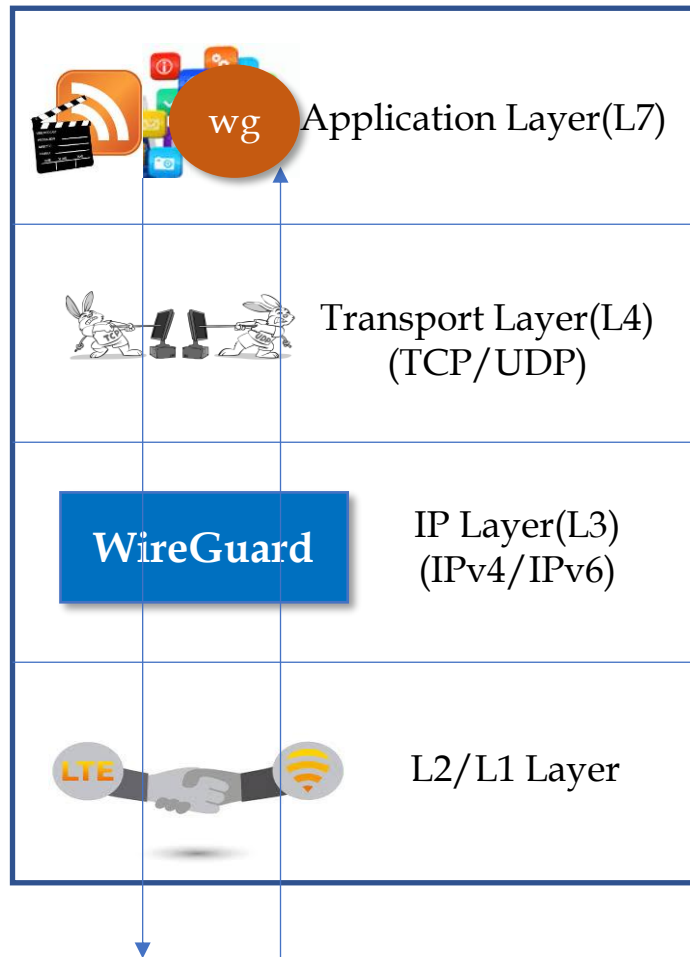
No Limitation. Cryptokey Routing



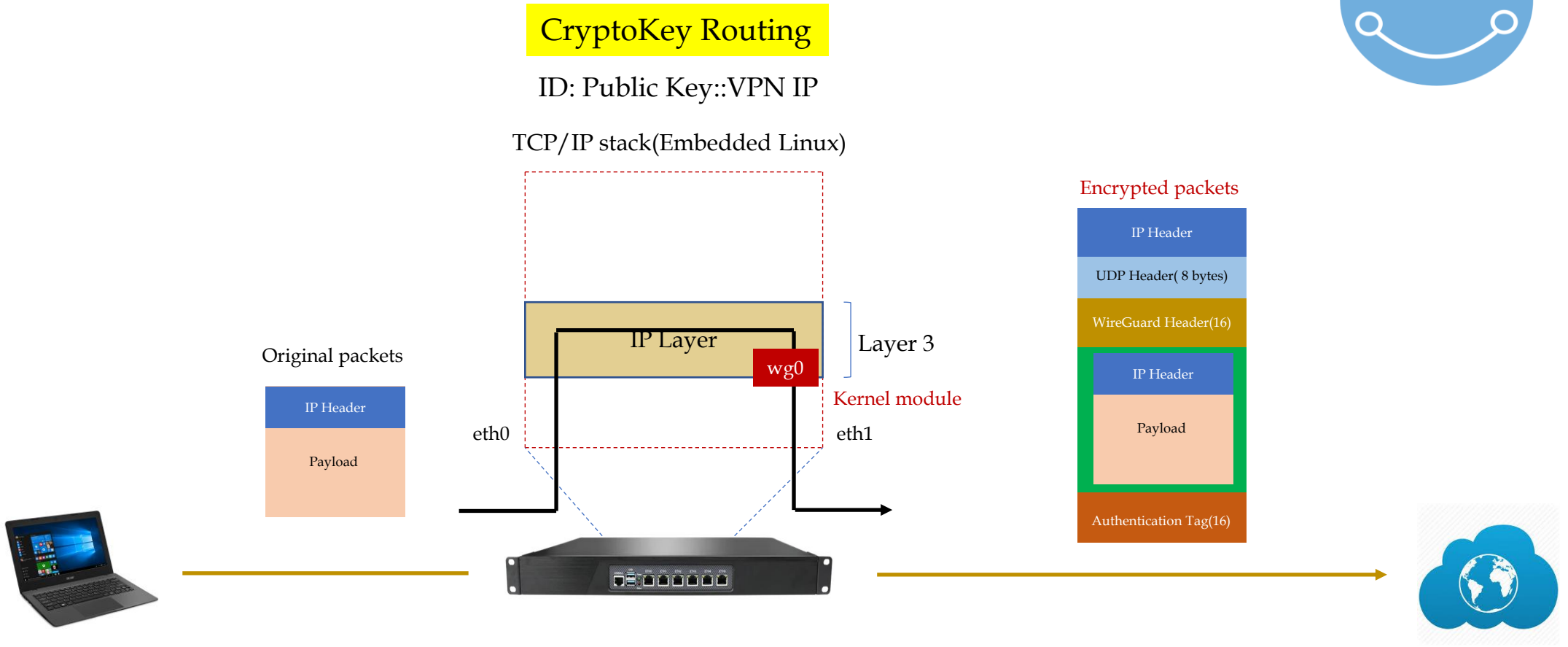
L3 VPN Tunnel

Why is WireGuard faster than OpenVPN?

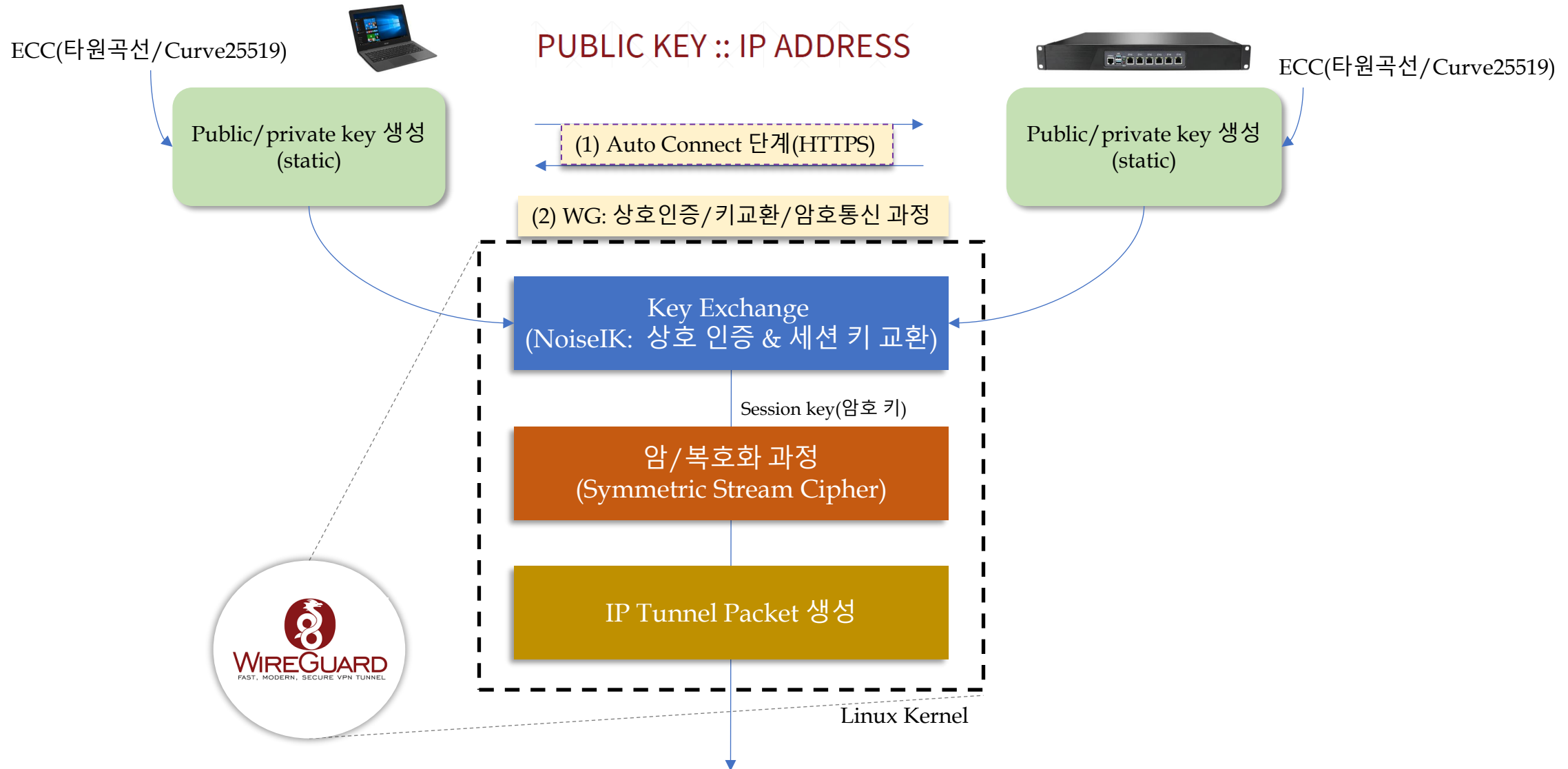
2. WireGuard(1) - Architecture(2)



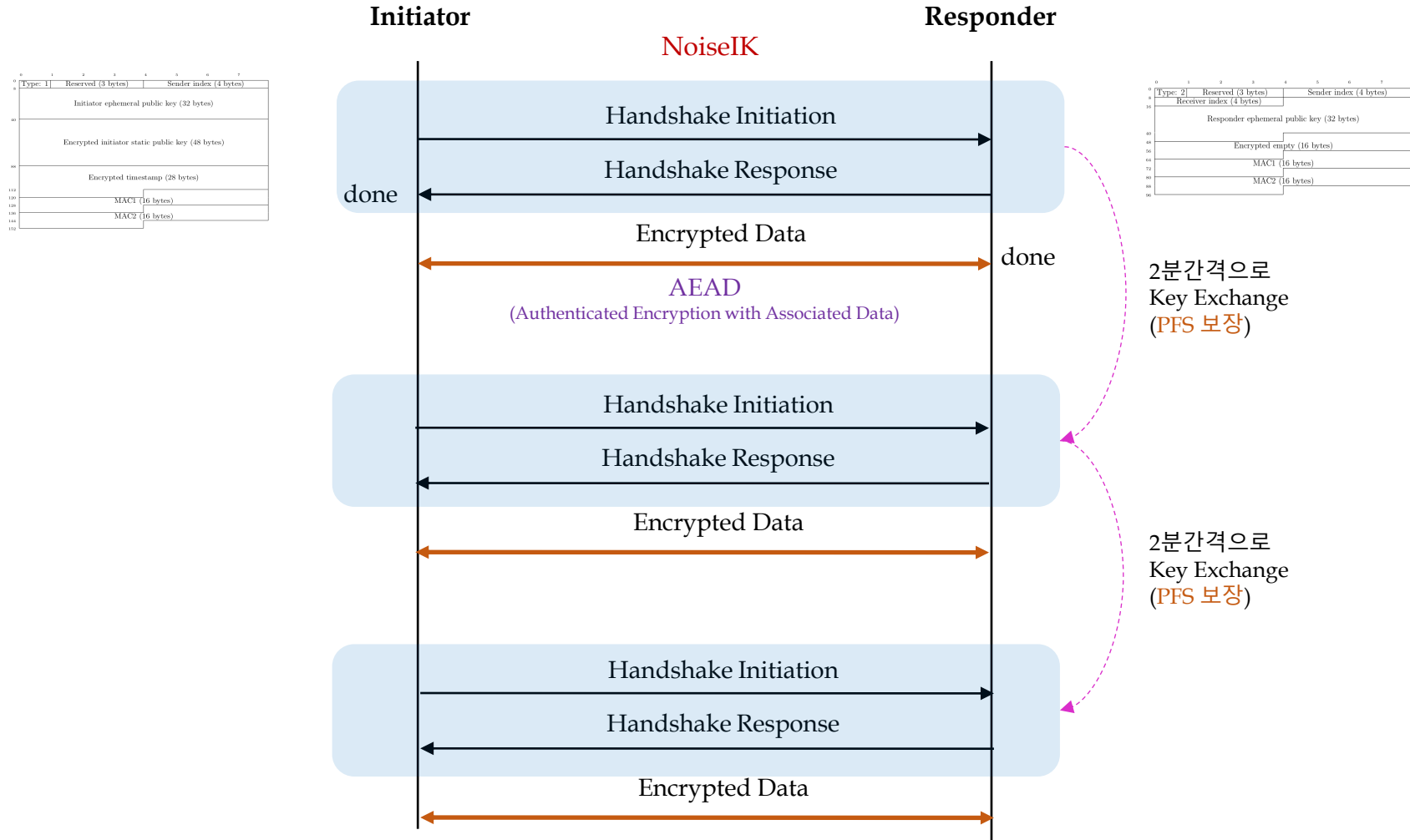
2. WireGuard(1) - Architecture(3)



2. WireGuard(2) - Protocol(1)

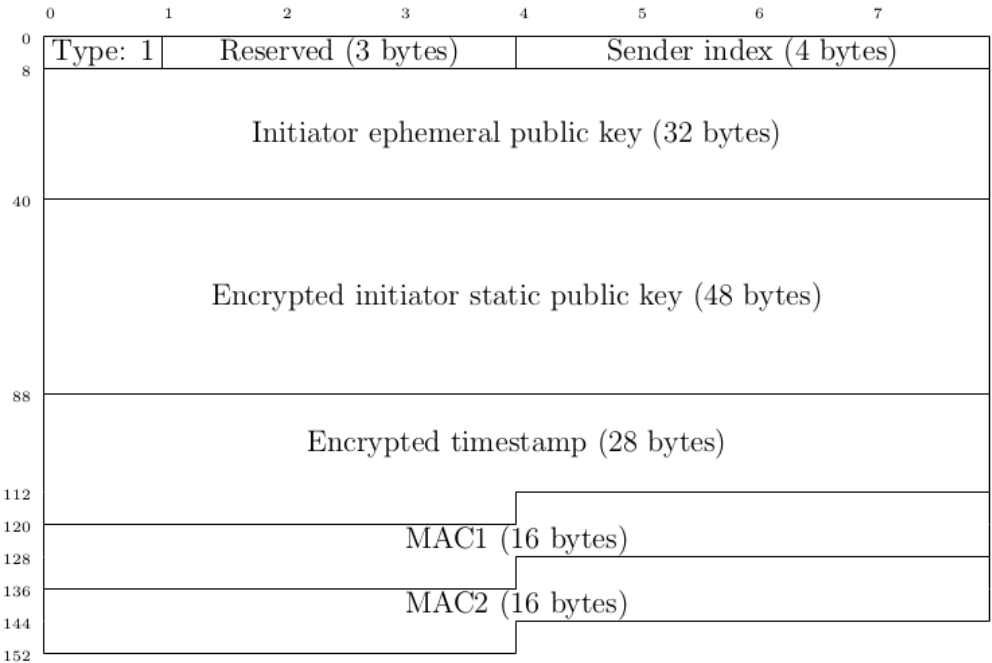


2. WireGuard(2) - Protocol(2)

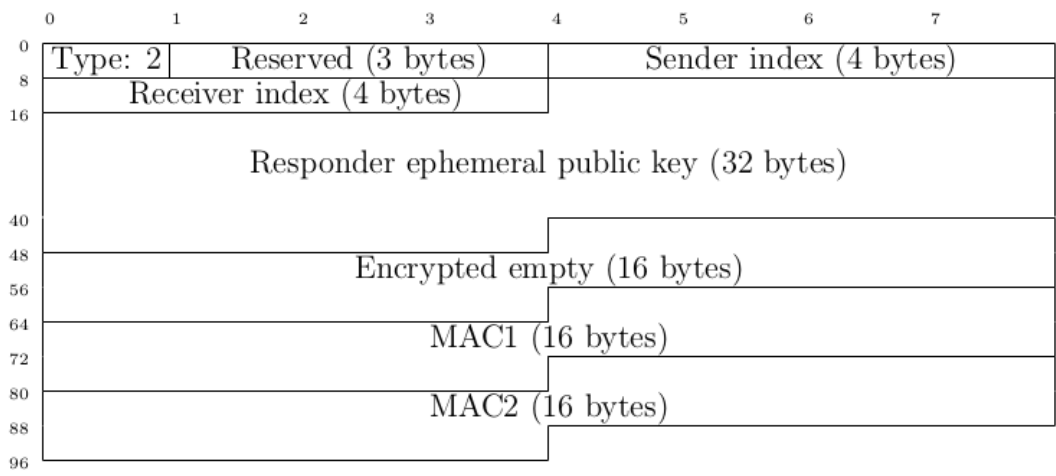


2. WireGuard(2) - Protocol(3)

Initiation Packet Format



Response Packet Format



Type	Message Name	Length
1	Handshake Initiation	148 bytes
2	Handshake Response	92 bytes
3	Cookie Reply	64 bytes
4	Transport Data	≥ 32 bytes

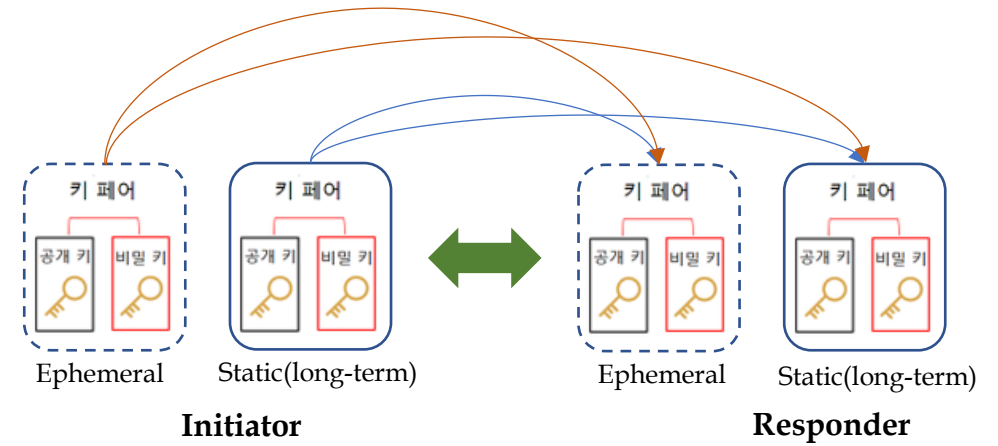
2. WireGuard(2) - Protocol(4)

Session keys = Noise(
ECDH(ephemeral, static),
ECDH(static, ephemeral),
ECDH(ephemeral, ephemeral),
ECDH(static, static)
)

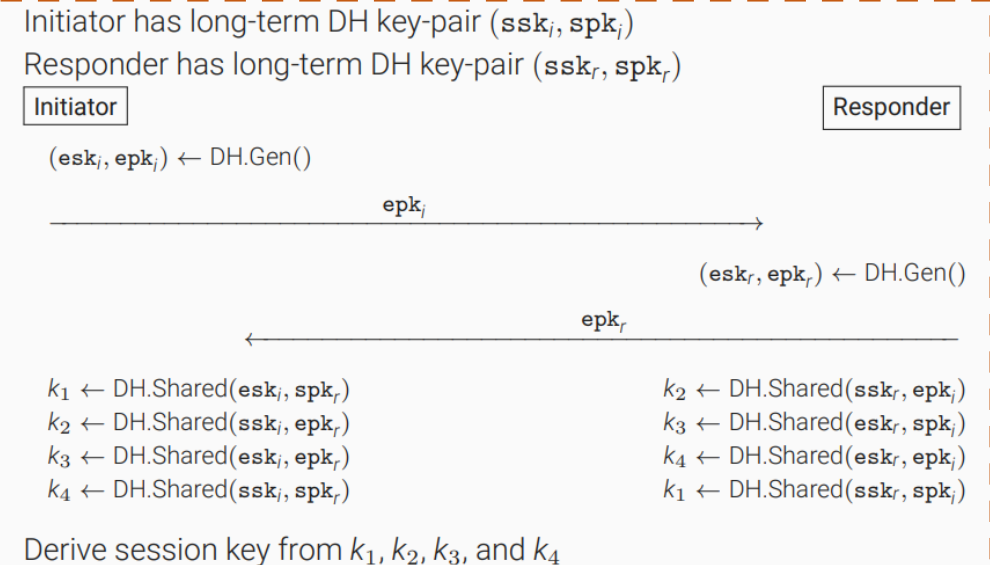
256bits(32bytes)



AEAD



상호 인증
및
키 교환



2. WireGuard(3) – Algorithms(1)

Layer 3(IPv4, IPv6 지원) VPN Protocol

handshake pattern IKpsk2 (Noise Protocol Framework 기반)

DH function X25519

cipher function ChaCha20-Poly1305

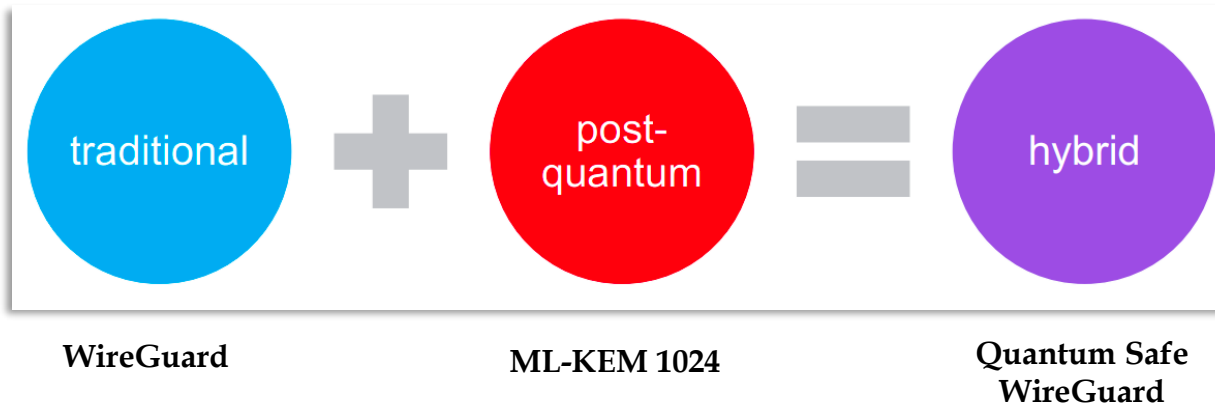
hash function BLAKE2s

prologue WireGuard v1 zx2c4 Jason@zx2c4.com (34 bytes)

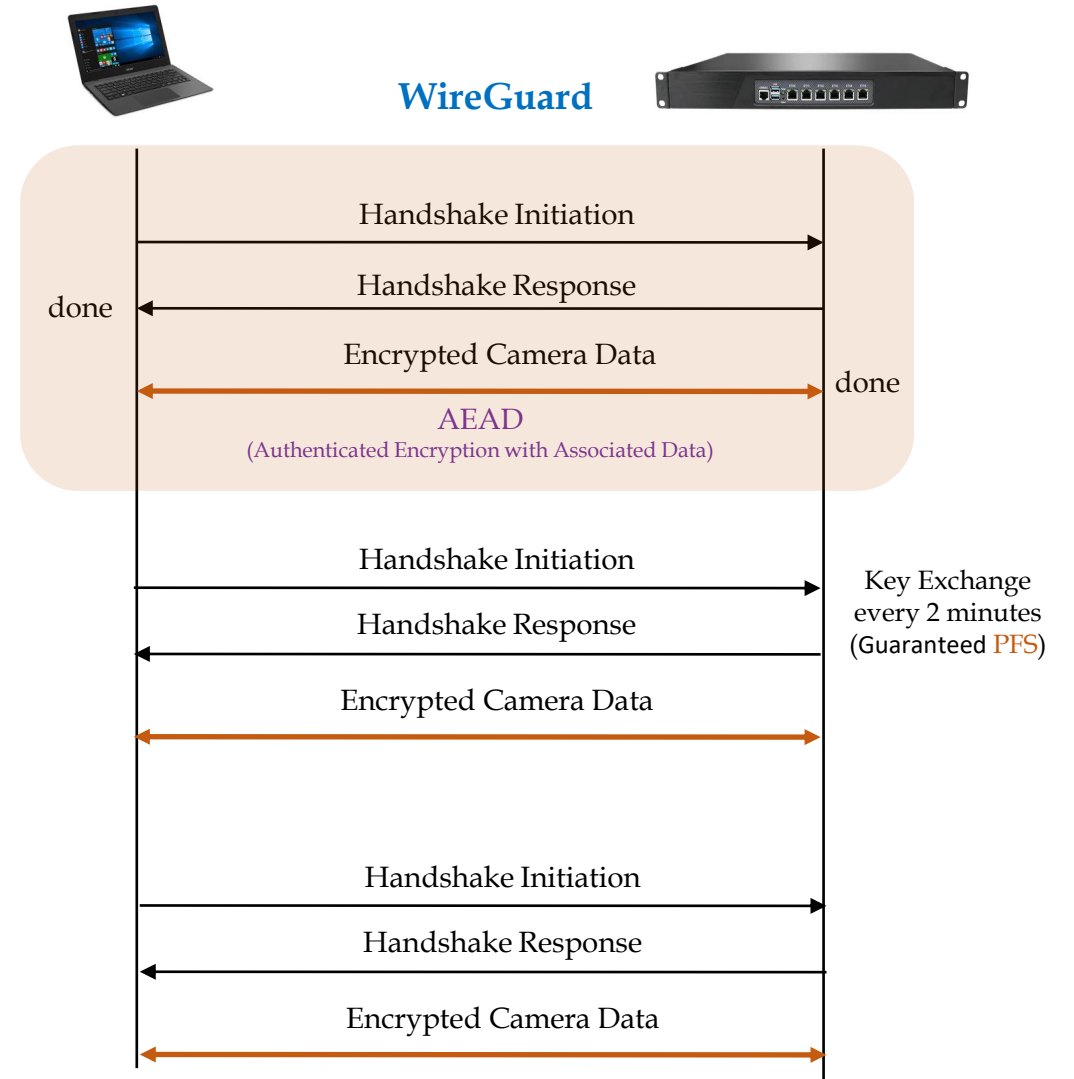
2. WireGuard(3) - Algorithms(2)

보안 알고리즘	상세 내용
NoiseIK Key 교환 방식 및 상호 인증	NoiseIK handshake 방식(Noise IKpsk2) ▪ ECDH(Diffie-Hellman) 기반 ▪ Curve25519 Public key(32 byte)를 교환 후, 이를 통해 안전하게 shared secret 생성 ▪ Static/Ephemeral private/public key(2 쌍) 이용 ▪ Key 교환 시 아래 기능 보장 ▪ <i>Perfect forward secrecy</i> 보장, <i>replay attack</i> 방지 기능 ▪ <i>Identity</i> 감춤 기능 제공, DoS 공격 완화 기능(Cookie), ▪ PSK 사용 시, <i>post-quantum security</i> 보장 Hash 알고리즘 ▪ BLAKE2 - fast secure hashing 알고리즘 ▪ SHA series 보다 빠름. 즉 MD5 수준임. SHA-3 보안 강도. PRF & MAC 및 Key Derivation 함수 ▪ SipHash4 (Hash table lookup 용으로 사용) ▪ HKDF (HMAC-based Key Derivation Function) 사용
암호 알고리즘 Authenticator 알고리즘 (ChaCha20-Poly1305 AEAD)	ChaCha20 - 256 bits stream cipher(20 round cipher Salsa20 기반) ▪ Stream cipher는 일반적인 block cipher(예: AES-256-CBC)에 비해 속도가 빠름 ▪ key(32 bytes)는 대칭키를 사용(즉, 암호화 용 키와 복호화 용 키 동일) ▪ Video/Audio 등 stream 암호화에 적합 Poly1305 - message authentication code 알고리즘(16 bytes output 생성)

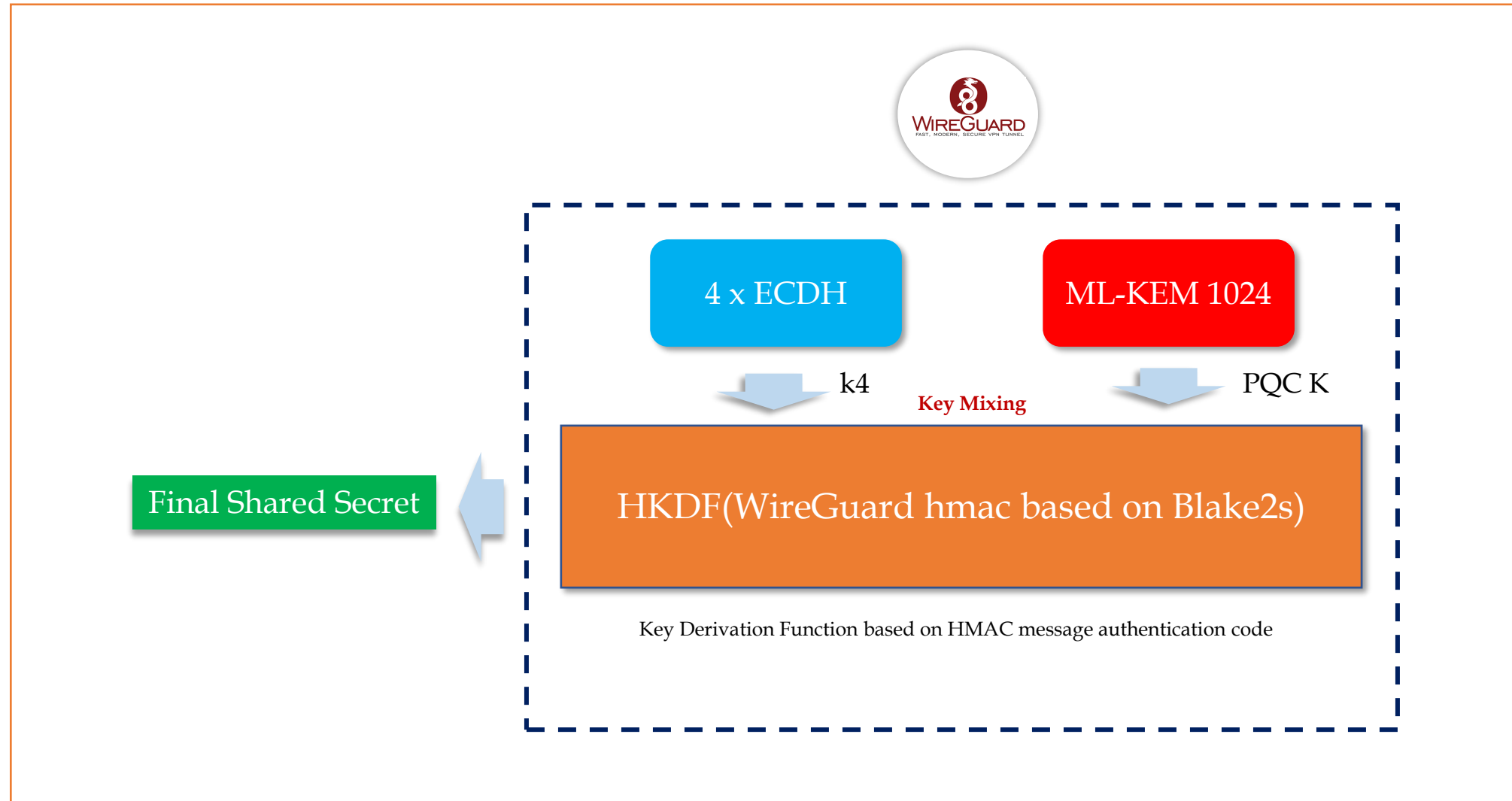
2. WireGuard(4) – PQC(1)



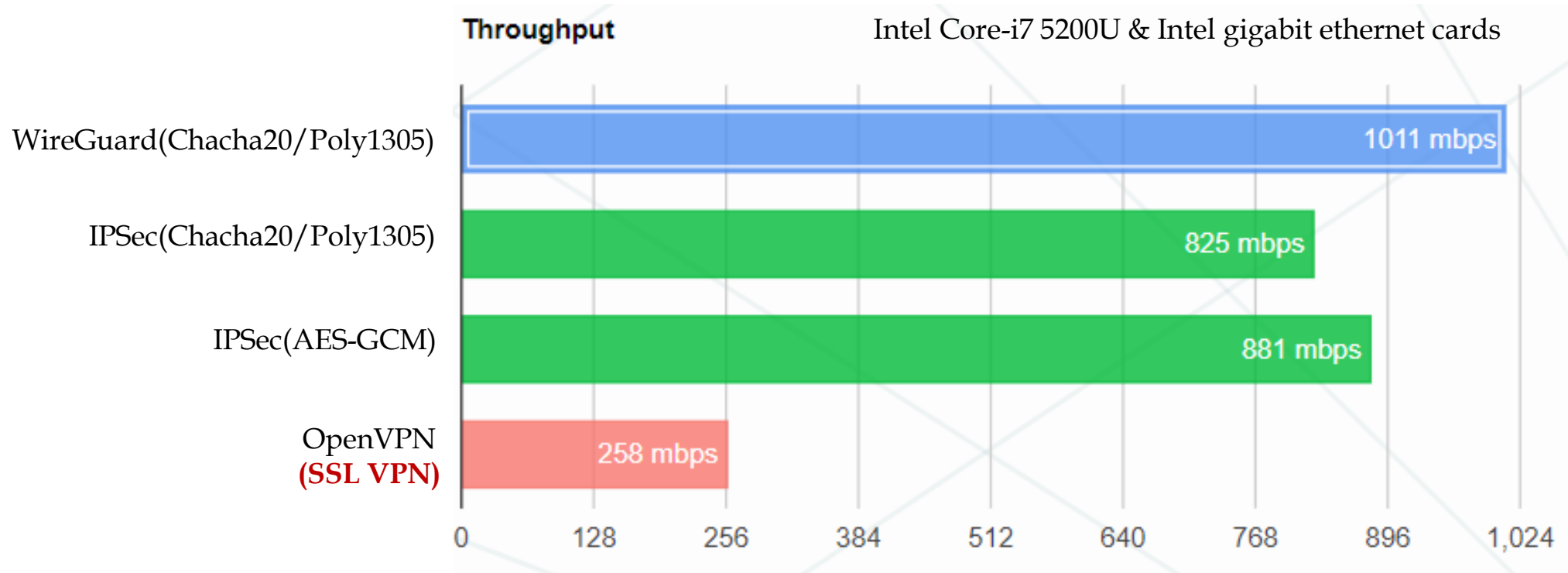
PQC 알고리즘이 통합된 WireGuard Protocol



2. WireGuard(4) – PQC(2)



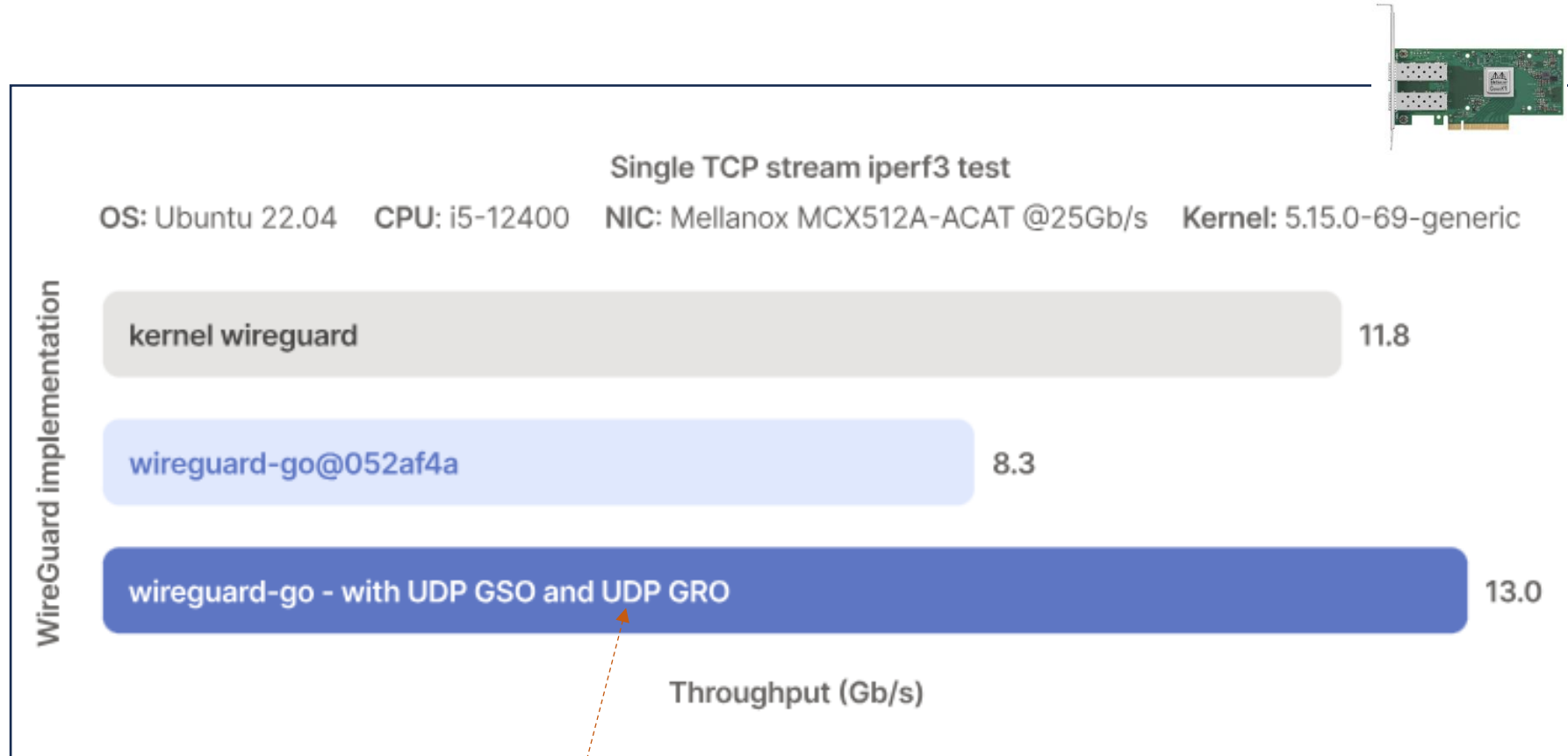
2. WireGuard(5) – Performance(1)



[출처] <https://www.wireguard.com/performance/>

[주의] OpenVPN DCO(Data Channel Offload)는 위의 그림보다는 훨씬 빠르다.

2. WireGuard(5) - Performance(2)



Packet Vector Processing
UDP GRO/GSO
Checksum Offload

그림 출처: <https://tailscale.com/blog/more-throughput>

2. WireGuard(5) - Performance(3)

■ WireGuard VPN의 우수한 성능 (Multi-Core 환경)

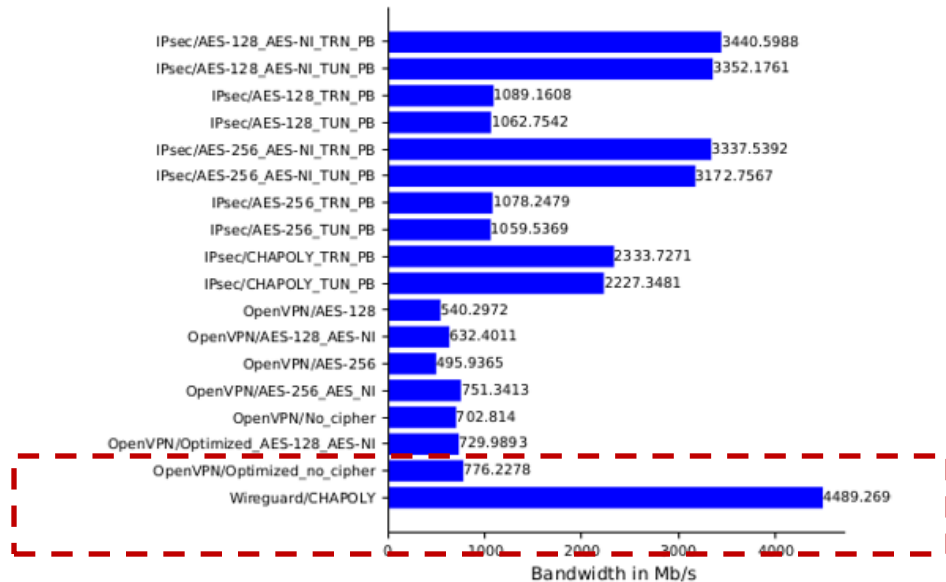
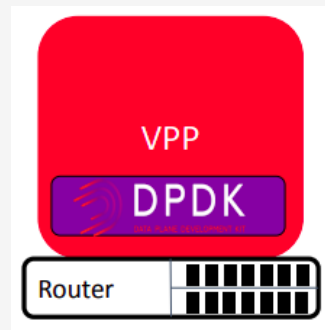


그림 출처: Performance Comparison of VPN Solutions, Lukas Osswald

- Signature 없이 인증 & 키 교환하는 AKE 기법 도입
- 최신 암호 알고리즘 및 암호 가속 기능으로 무장
 - Curve25519, Blake2s, ChaCha20, Poly1305, SipHash4
 - X86_64 AVX2, AVX512
- Perfect Forward Secrecy 보장(2분 간격 Handshaking)
 - 1-RTT만에 Handshaking 완료
- Replay Attack 방지 기능(Counter 필드 유지/관리)
- Identity Hiding 기능
- DoS 공격 완화 기능(Cookie 활용)

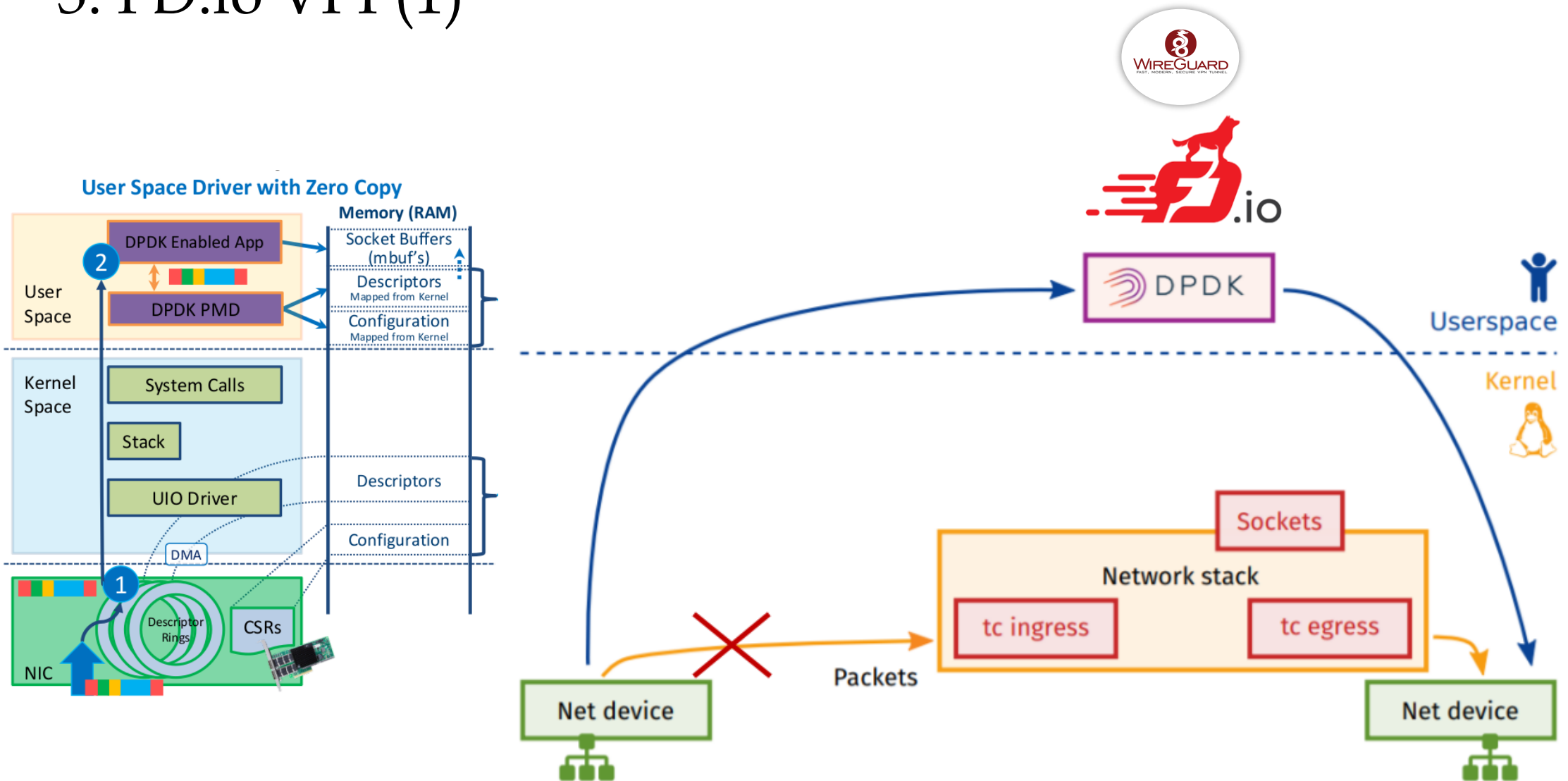
3. FD.io VPP



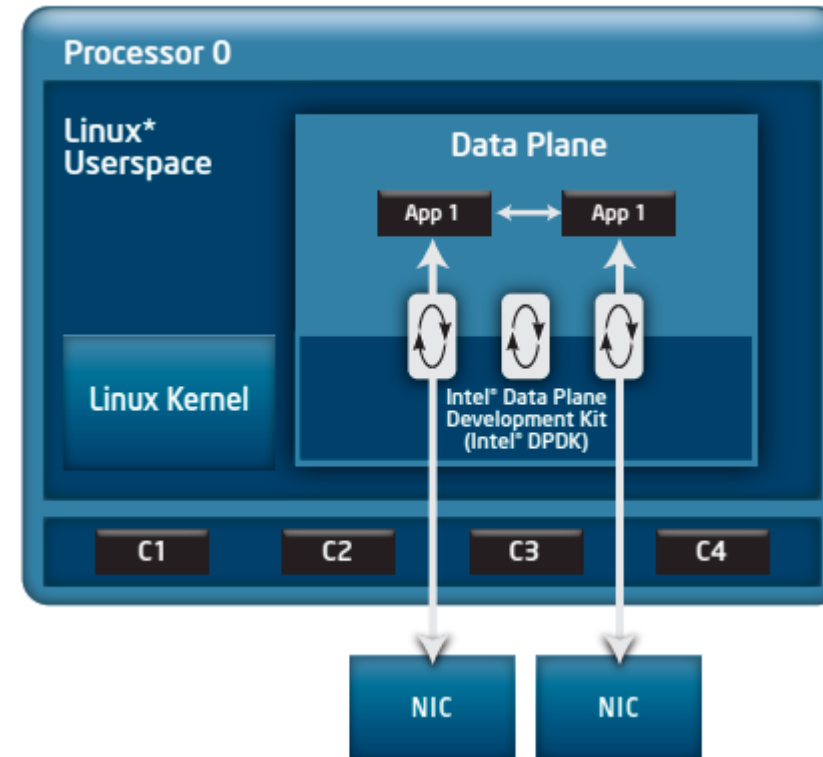
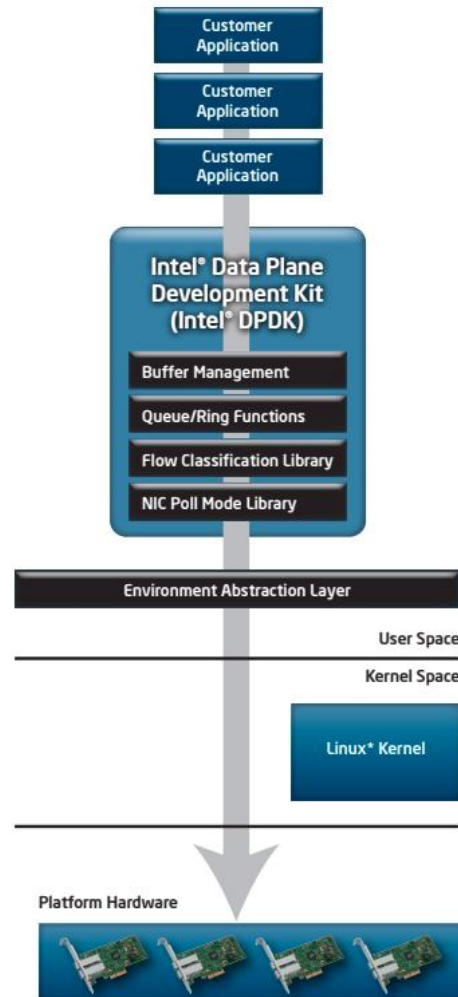
Software ASIC

FD.io VPP and FD.io logo are registered trademarks of The Fast Data Project, a series of LF Projects, LLC

3. FD.io VPP(1)



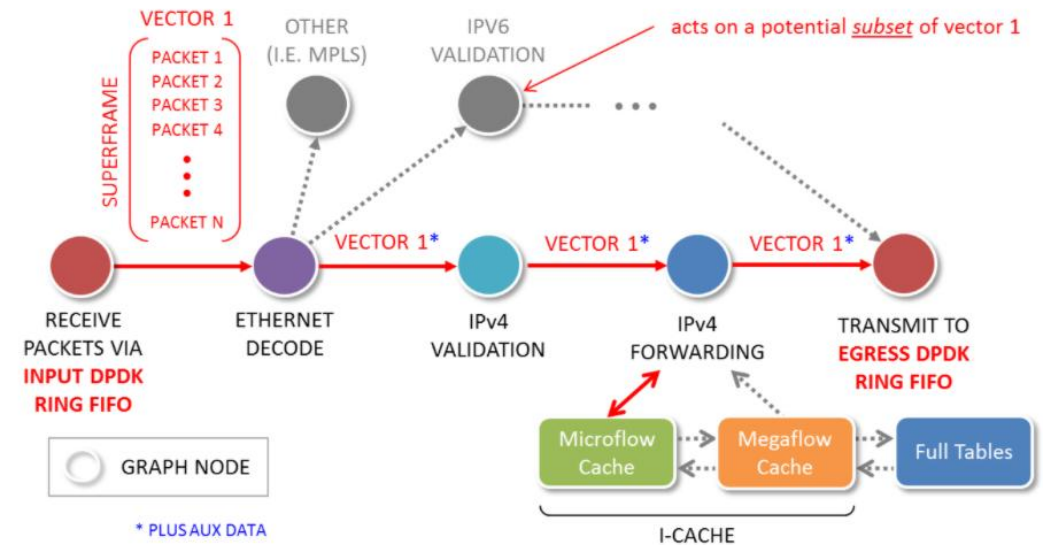
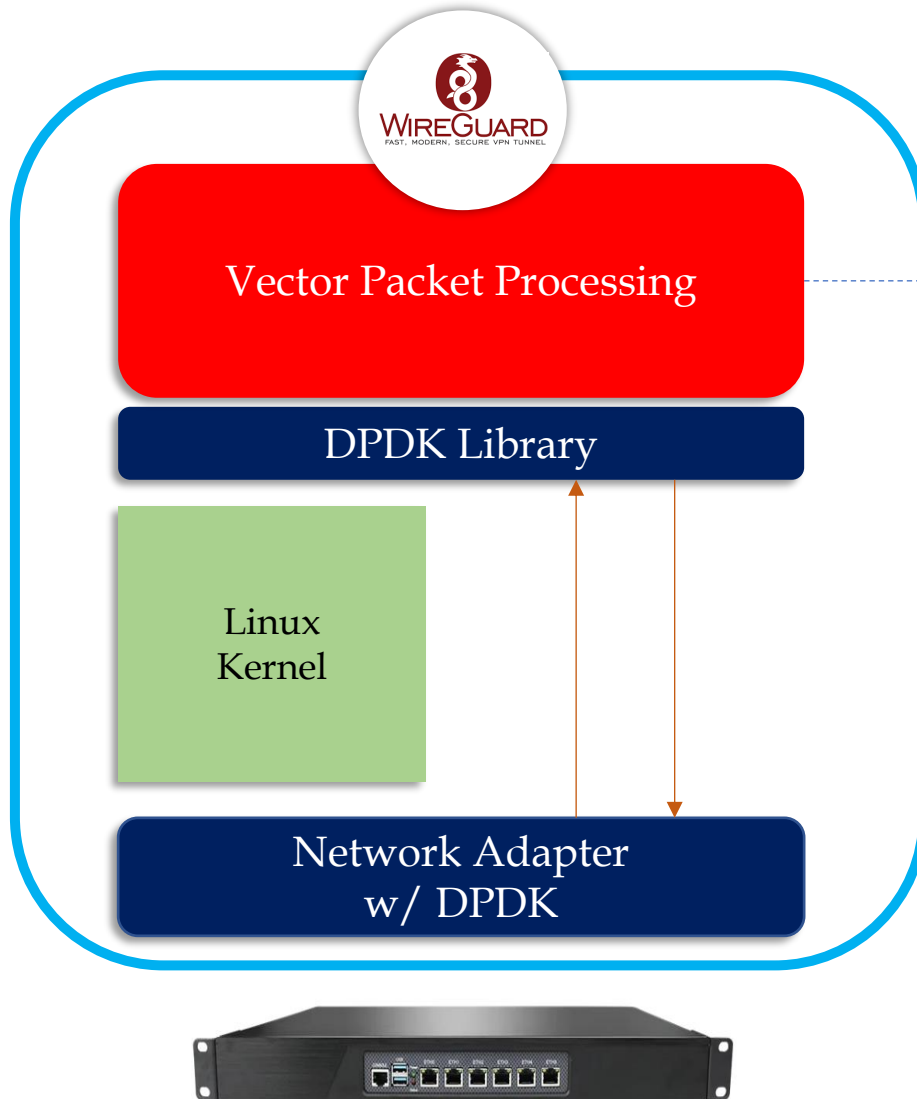
3. FD.io VPP(2)



Ref) Impressive Packet Processing Performance Enables Greater Workload Consolidation

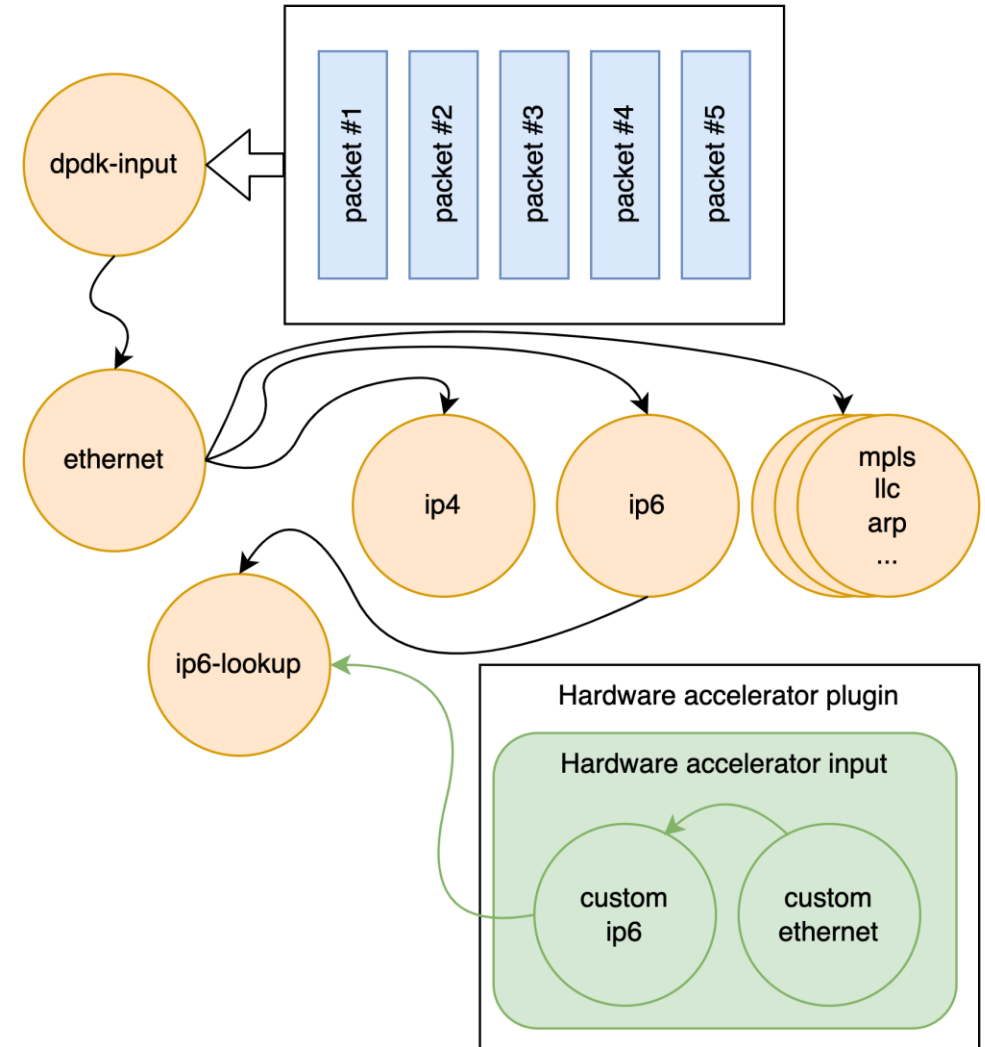
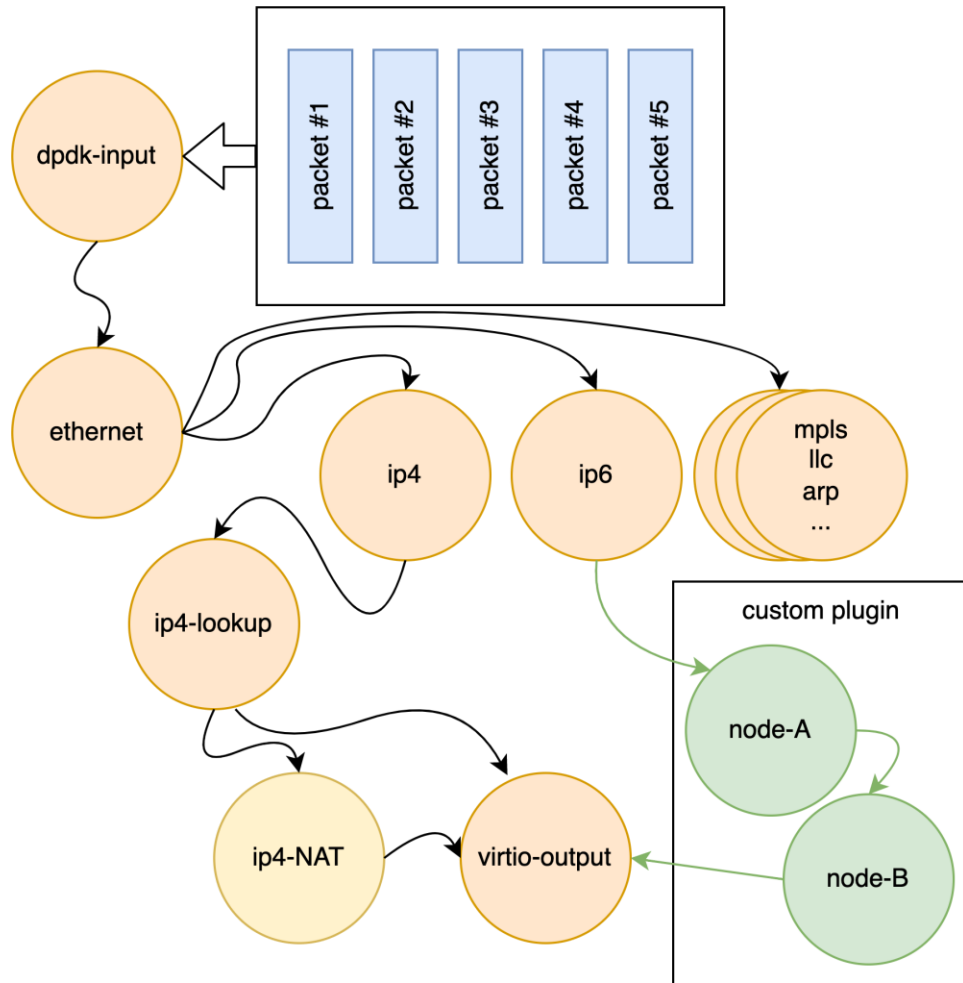
3. FD.io VPP(3)

Ref <https://fd.io/technology/>

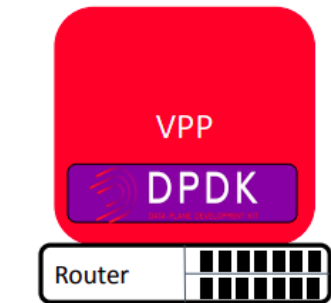


3. FD.io VPP(4)

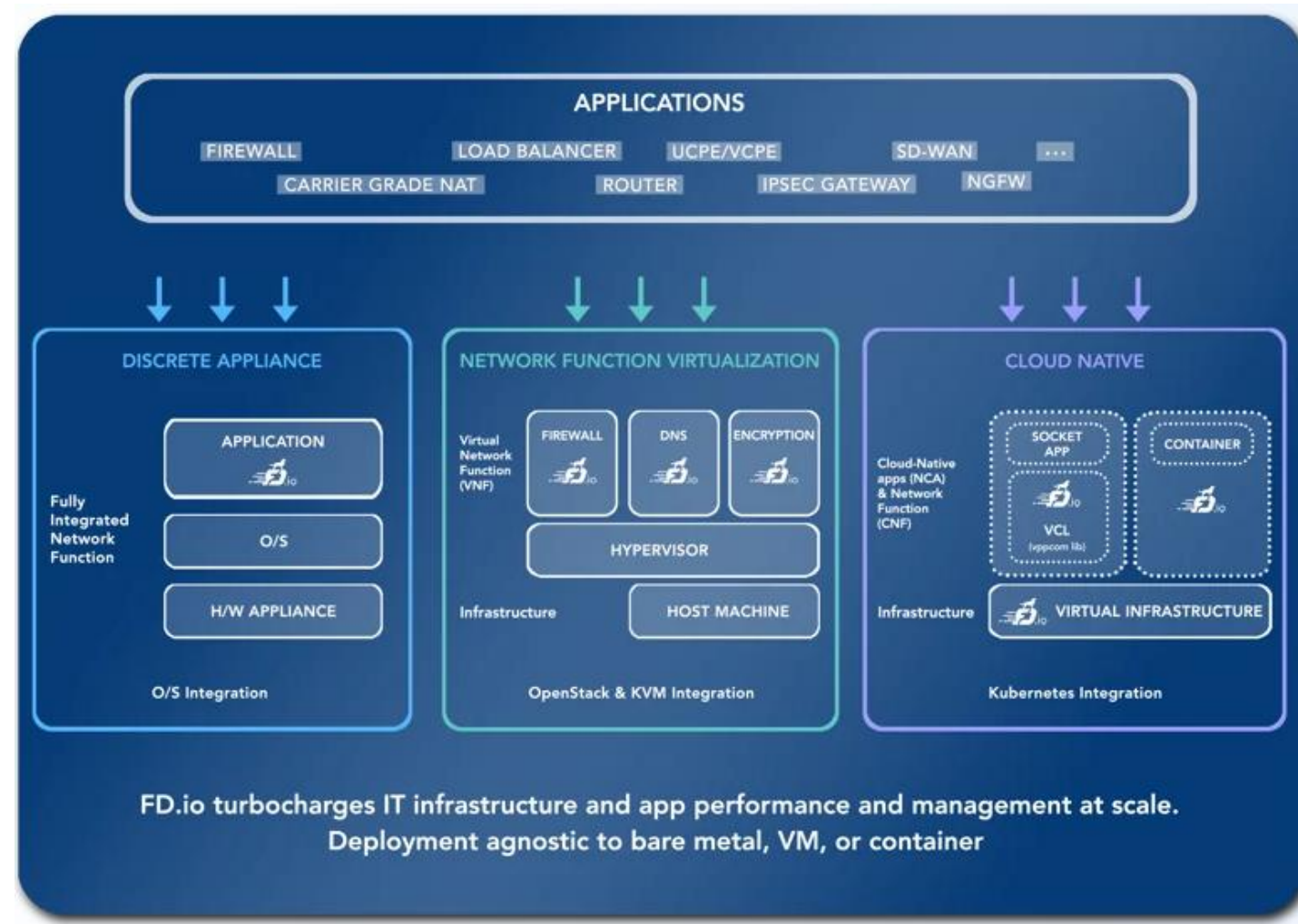
Ref <https://fd.io/technology/>



3. FD.io VPP(5)

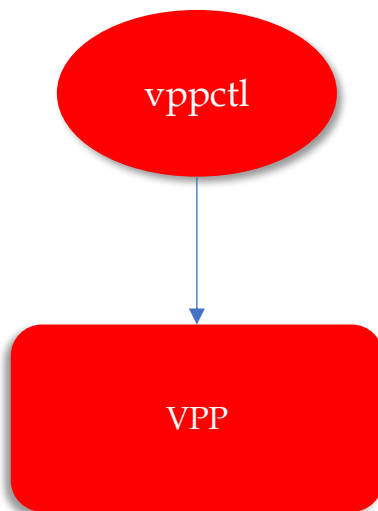


- L2 Switch
- L3 Router
- CGNAT
- L4 Switch
- WireGuard VPN
- IPSec VPN
- IPS
- VxLAN Gateway
- ...



Ref) <https://www.netgate.com/resources/articles-vector-packet-processing>

3. FD.io VPP(6)



```
/usr/bin/vpp -c /etc/vpp/startup.conf
```

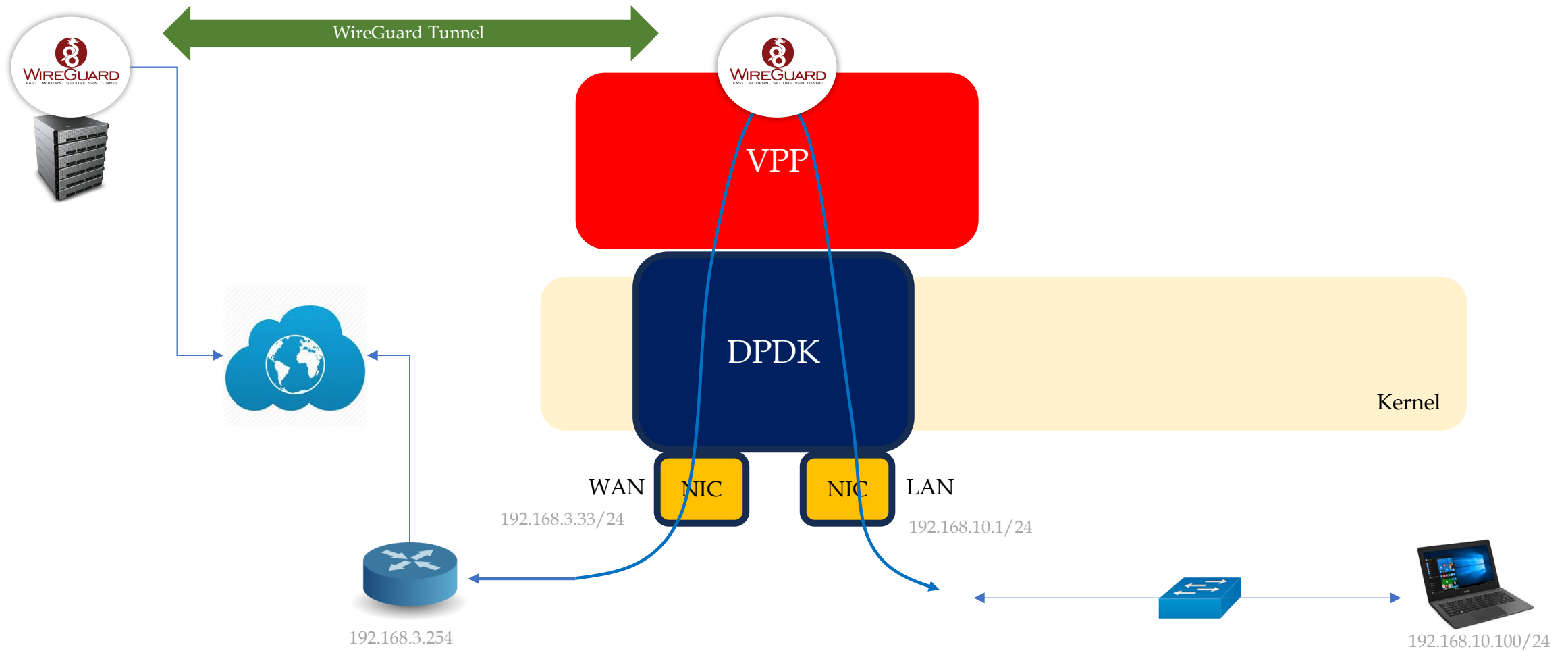
```
chy@mightykn:~/workspace/mightykn$ sudo vppctl
```

```
      _--_      _-      -   _--_      _-      _-  
    _/_/_/_\  (\)_--      | |/_/_/_\  \_/_/  
    _/_/_/_/_/_/_/_/_\_  | |/_/_/_/_/_/  
    _/_/_/_(\)_/_/_/_  |/_/_/_/_/_/_/_/_
```

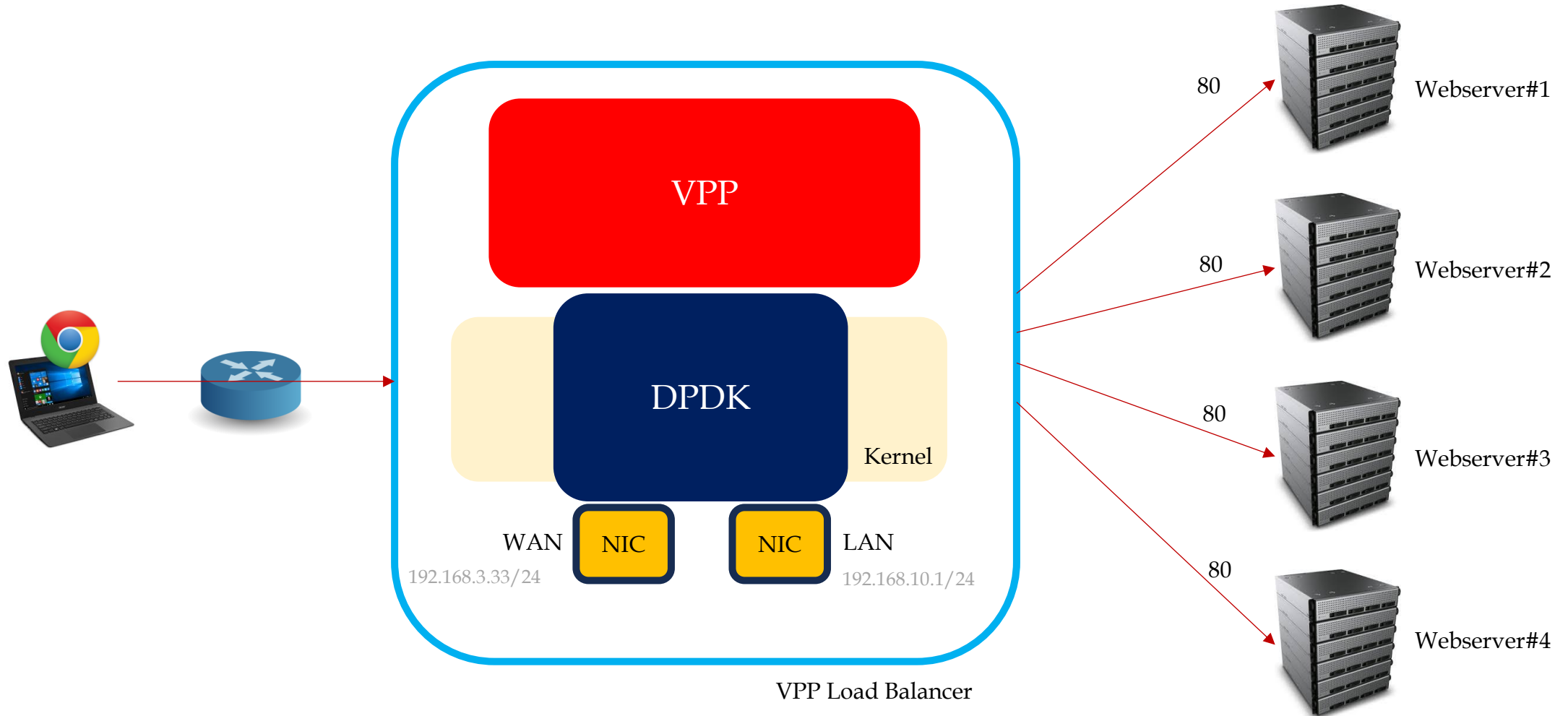
```
vpp#  
vpp# show interface
```

Name	Idx	State	MTU (L3/IP4/IP6/MPLS)	Counter	Count
enp2s0	1	up	9000/0/0/0	rx packets	1532
				rx bytes	204225
				tx packets	1398
				tx bytes	162174
				drops	165
				punt	219
				ip4	1296
				ip6	90
				tx-error	1
local0	0	down	0/0/0/0	drops	2
tap4096	2	up	9000/0/0/0	rx packets	157
				rx bytes	11050
				tx packets	198
				tx bytes	24133
				drops	14
				ip4	91
				ip6	15
tun4097	4	up	1420/0/0/0	rx packets	882
				rx bytes	50308
				tx packets	869
				tx bytes	67456
				ip4	871
wg0	3	up	1420/0/0/0	ip6	11
				rx packets	993

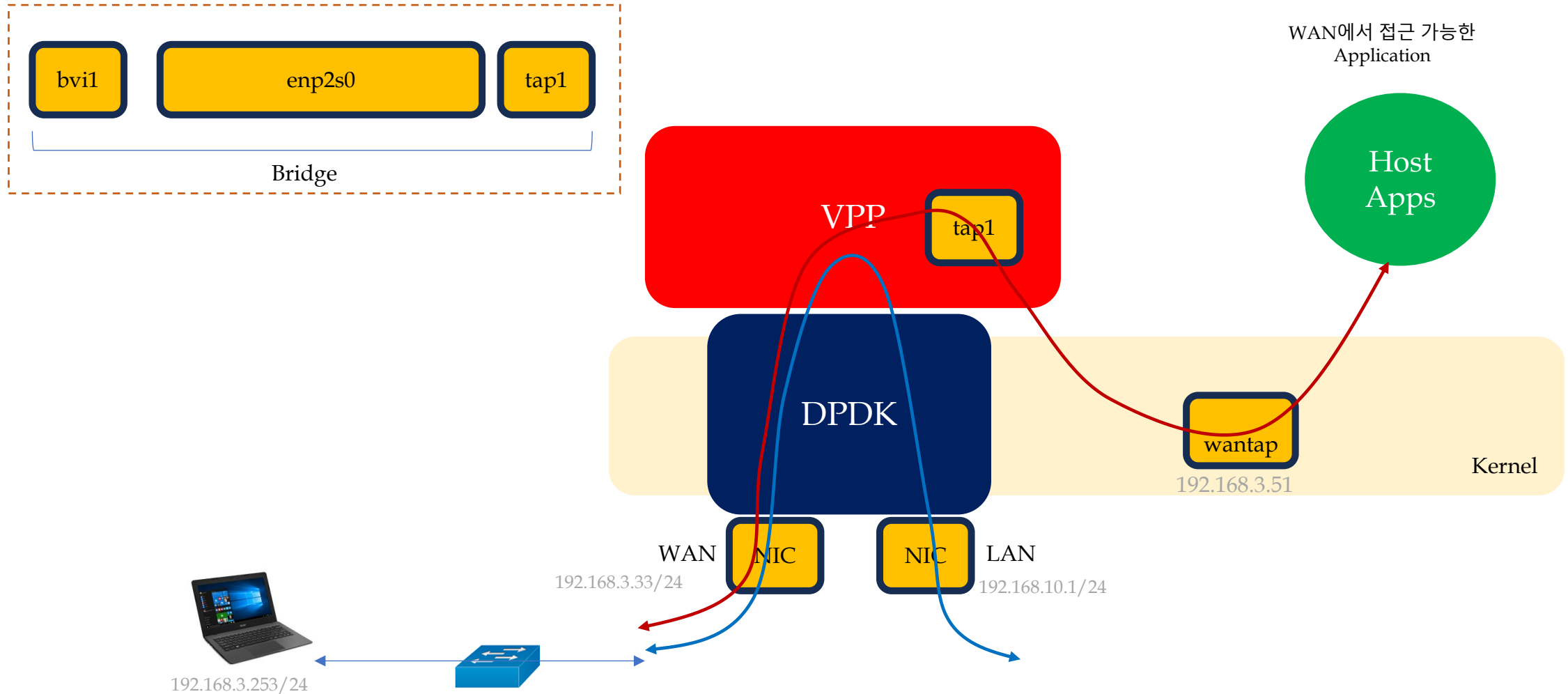
3. FD.io VPP(7) - WireGuard



3. FD.io VPP(8) – NAT & Load Balancing



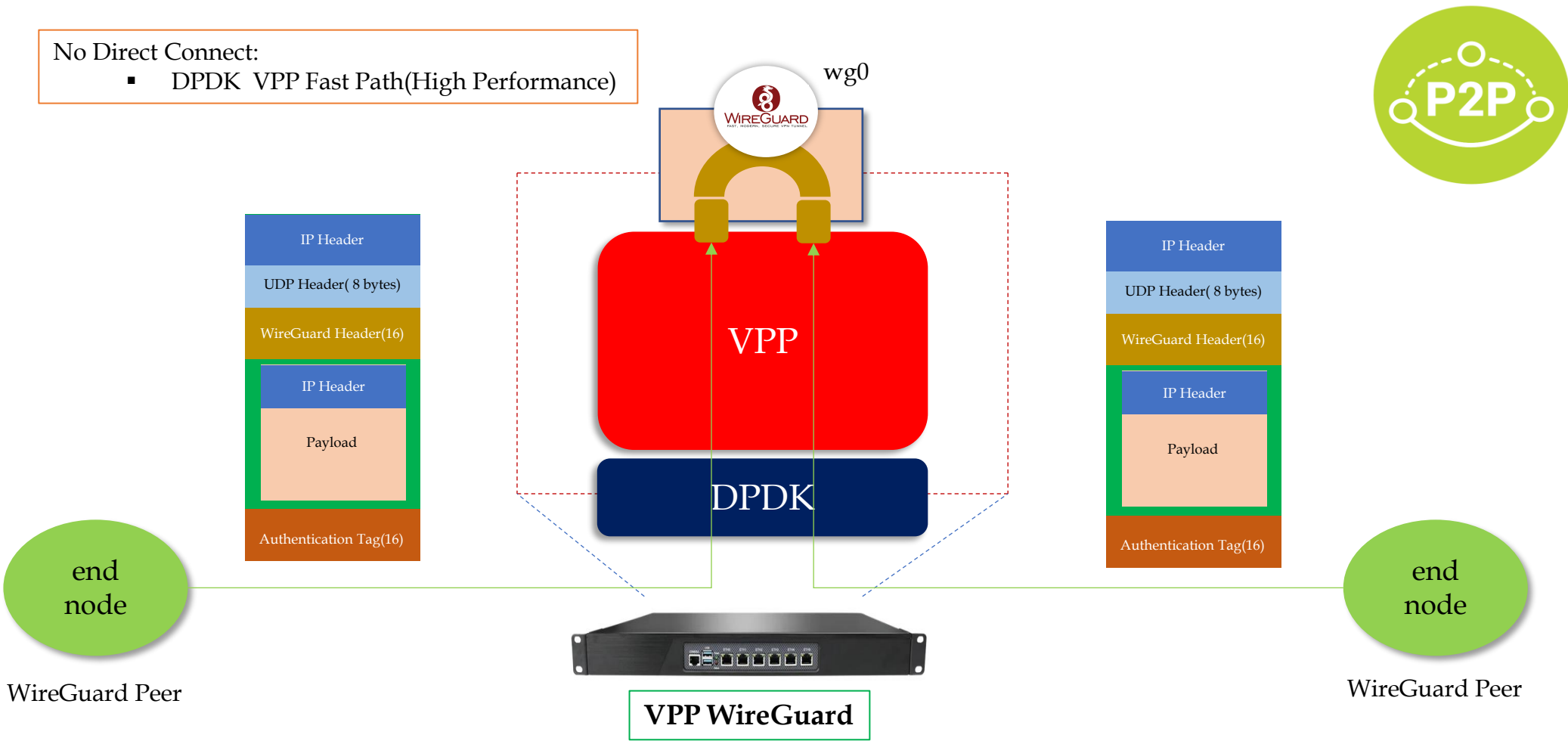
3. FD.io VPP(9) – Legacy App Access



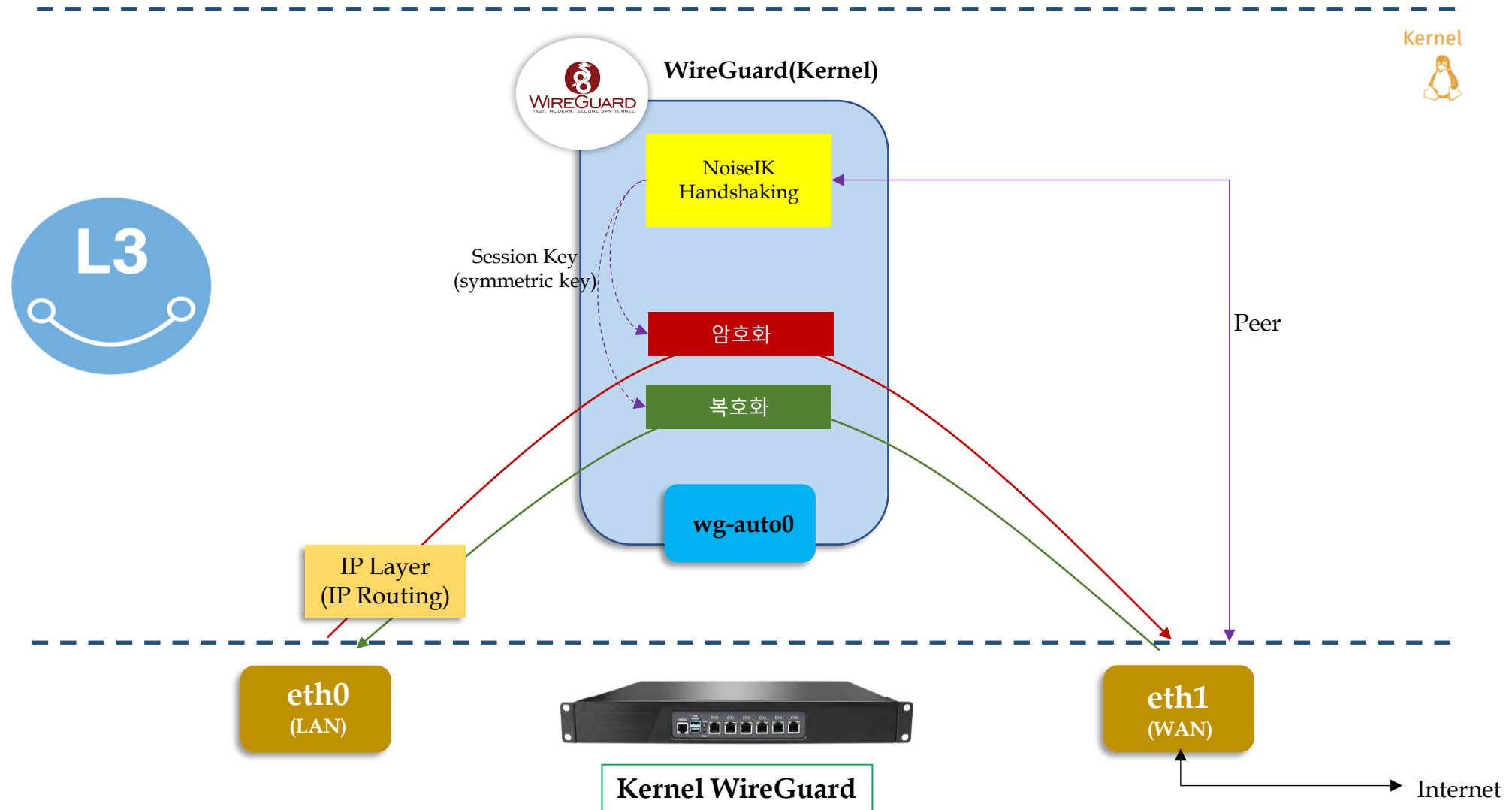
4. MightyConnect Security Box



4. MightyConnect Security Box(1) – WireGuard(1)



4. MightyConnect Security Box(1) – WireGuard(2-1)

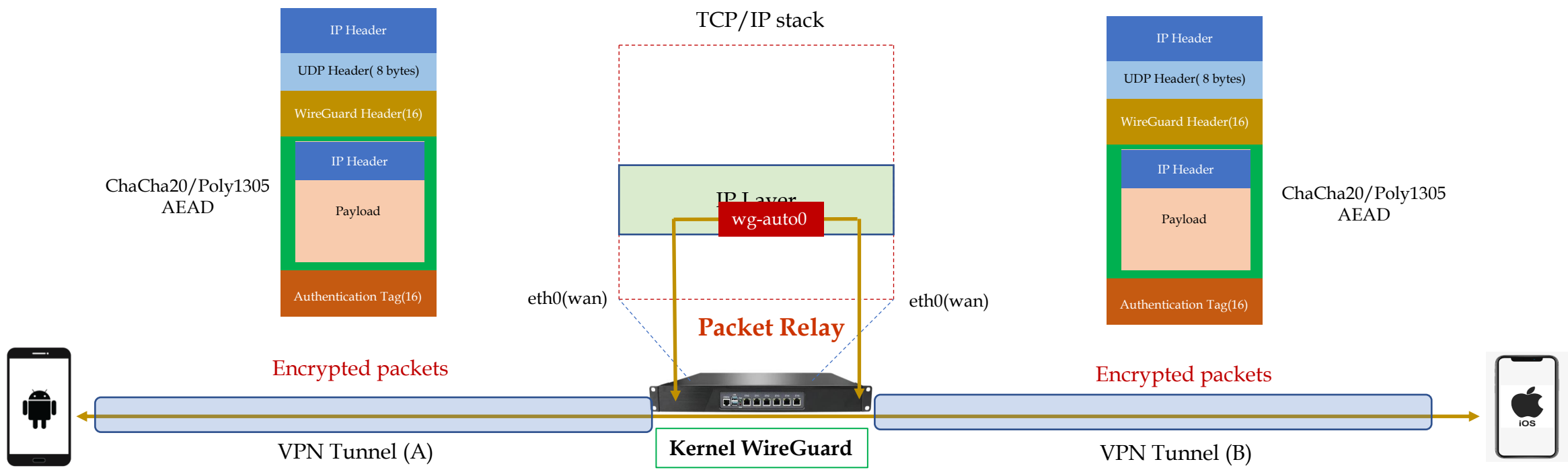


4. MightyConnect Security Box(1) – WireGuard(2-2)

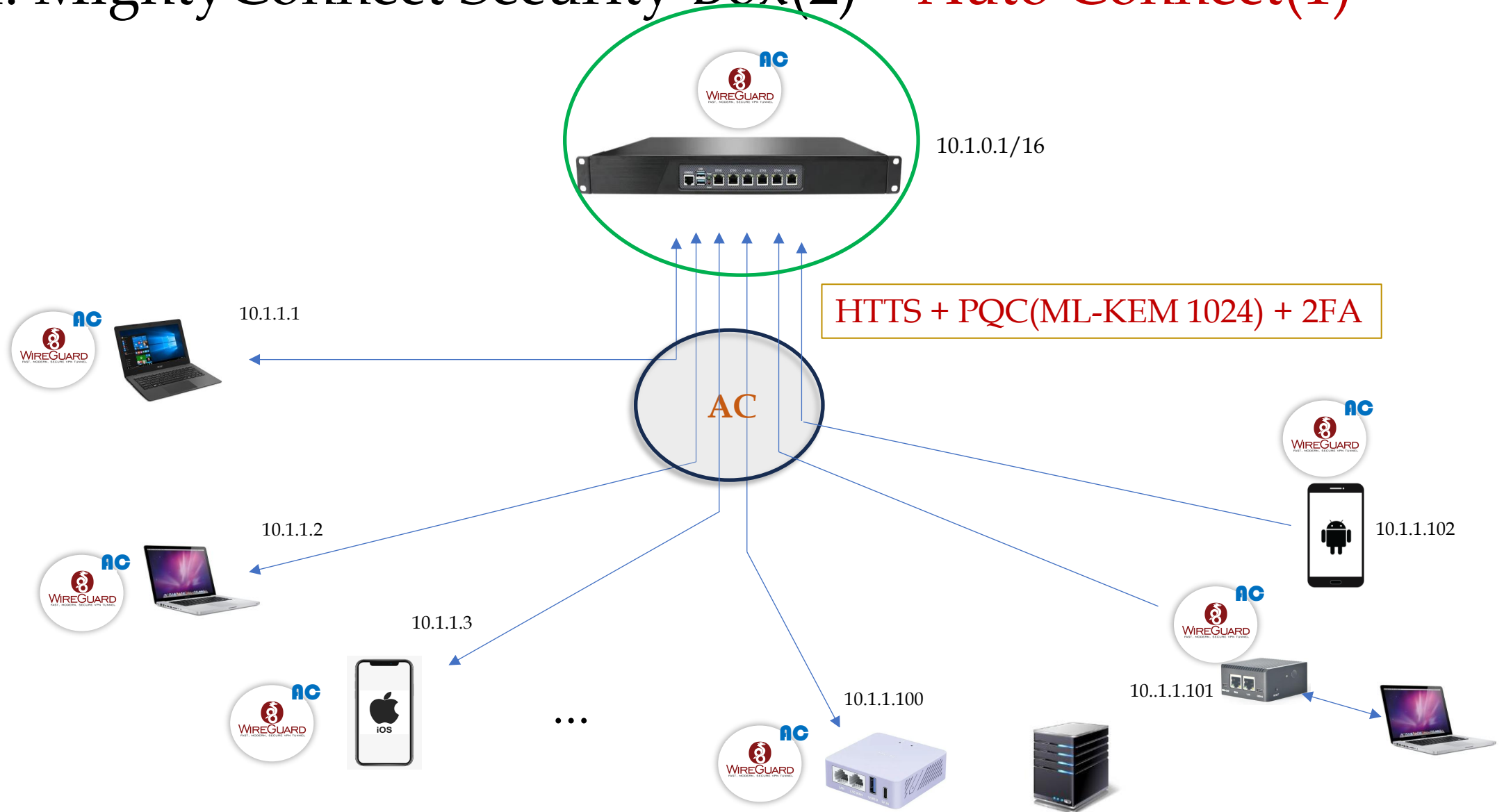


CryptoKey Routing

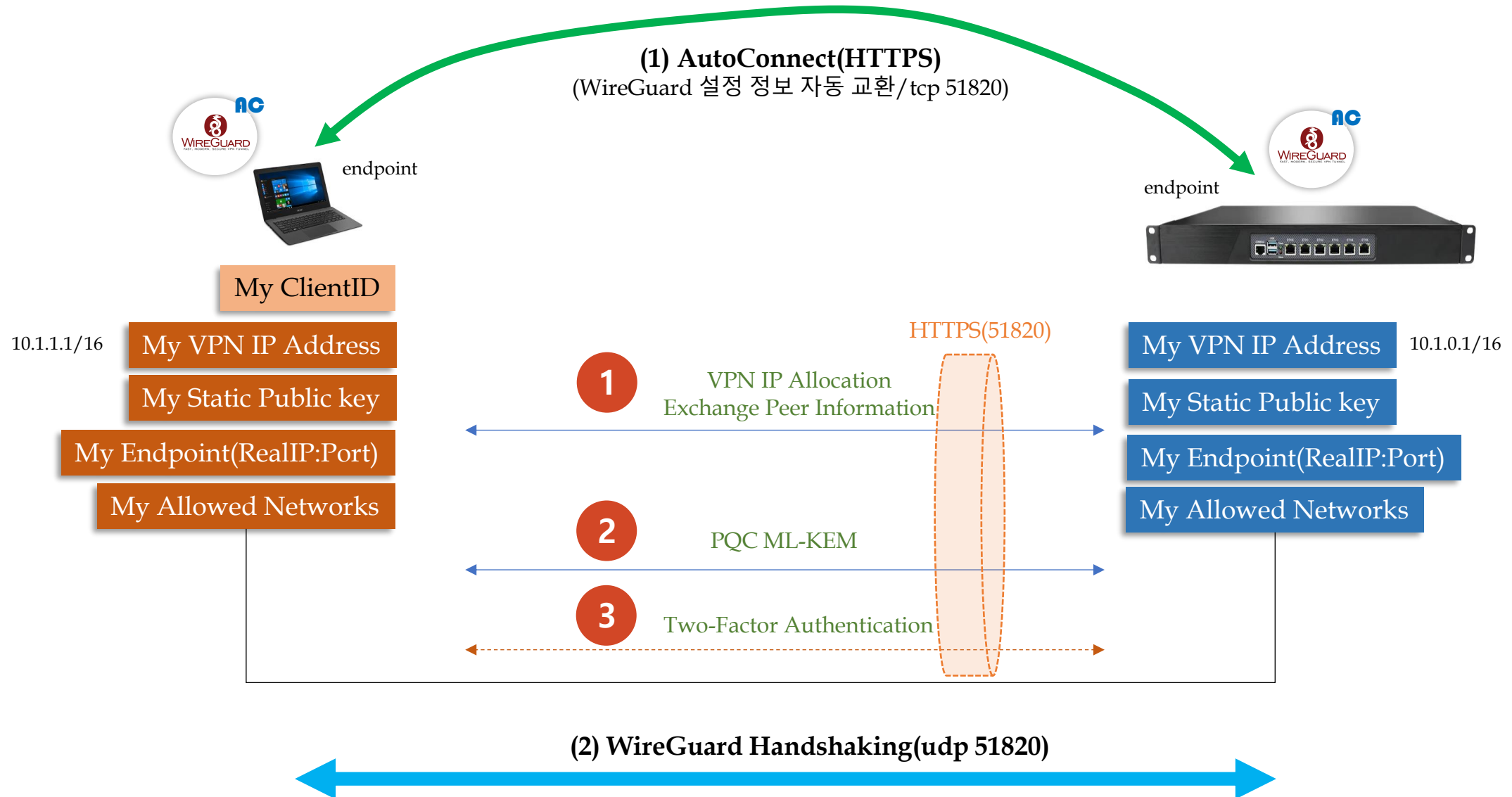
ID: Public Key::VPN IP



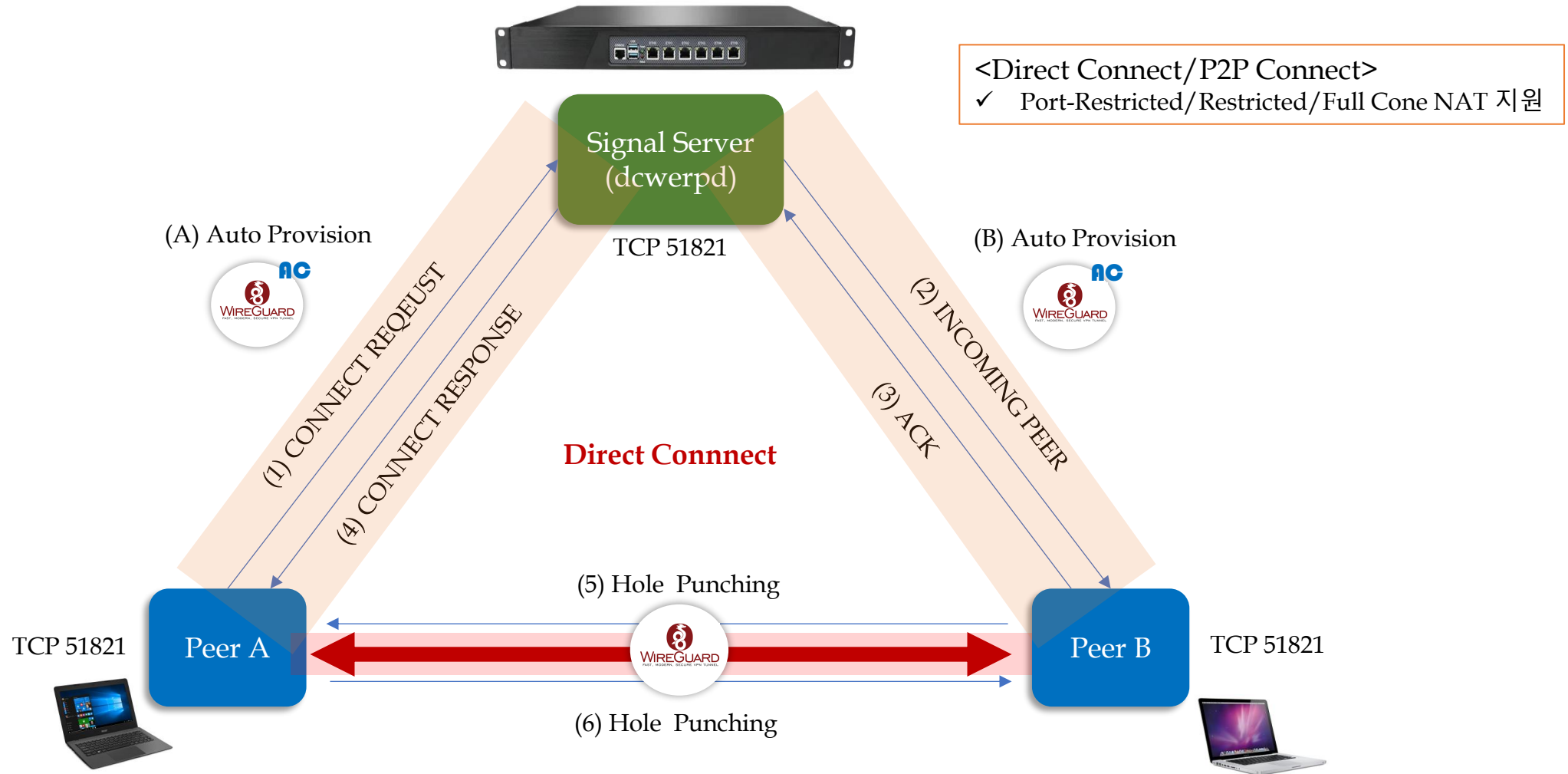
4. MightyConnect Security Box(2) – Auto Connect(1)



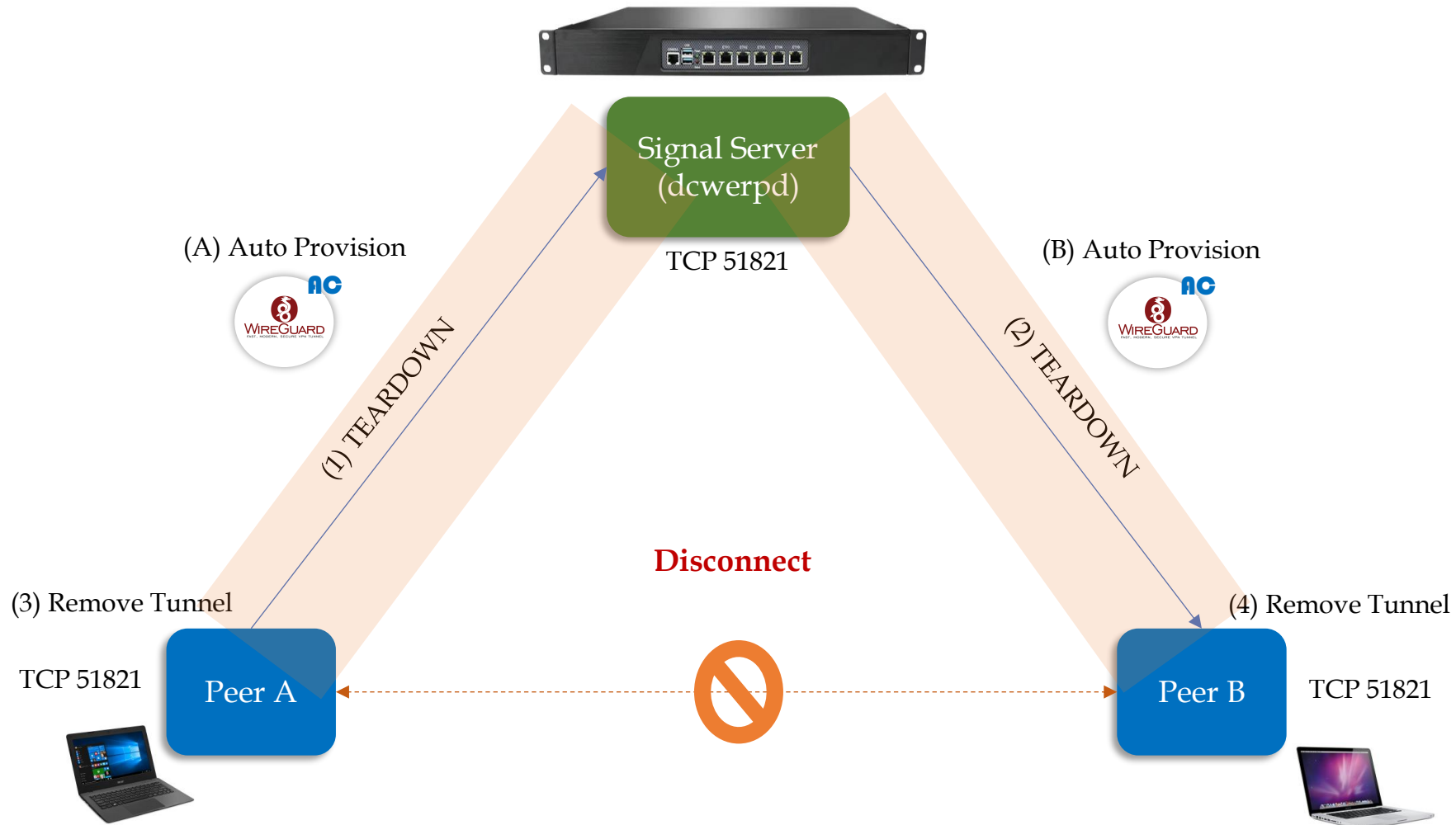
4. MightyConnect Security Box(2) – Auto Connect(2)



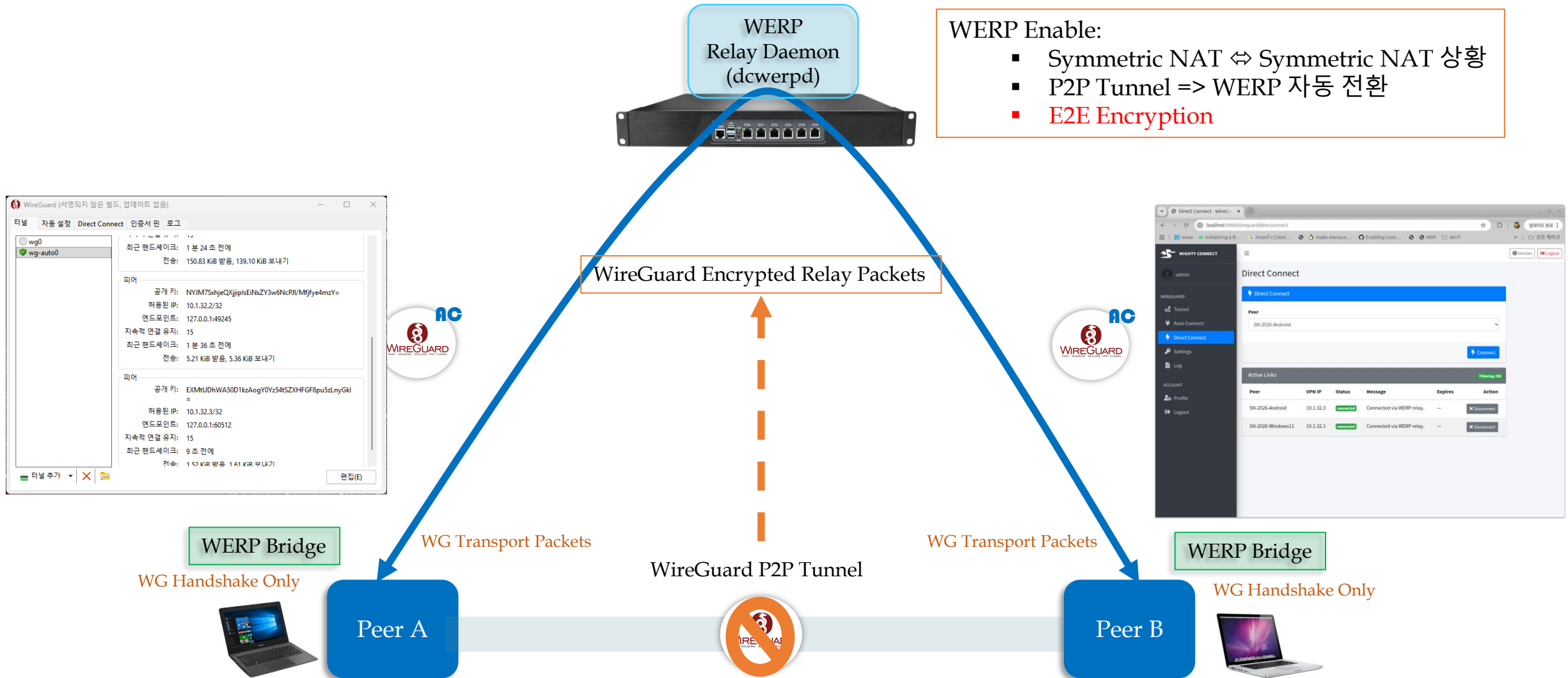
4. MightyConnect Security Box(3) – Direct Connect(1)



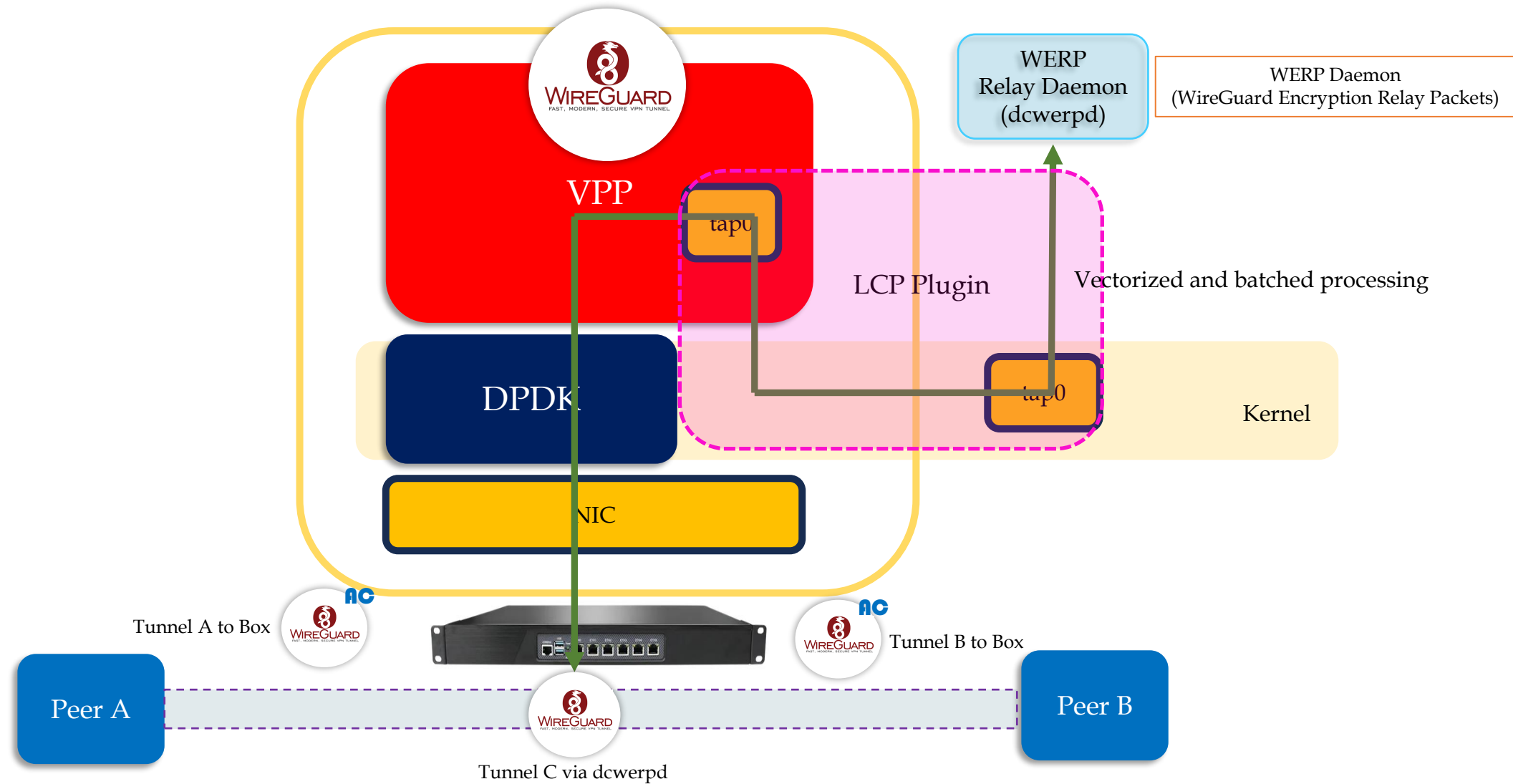
4. MightyConnect Security Box(3) – Direct Connect(2)



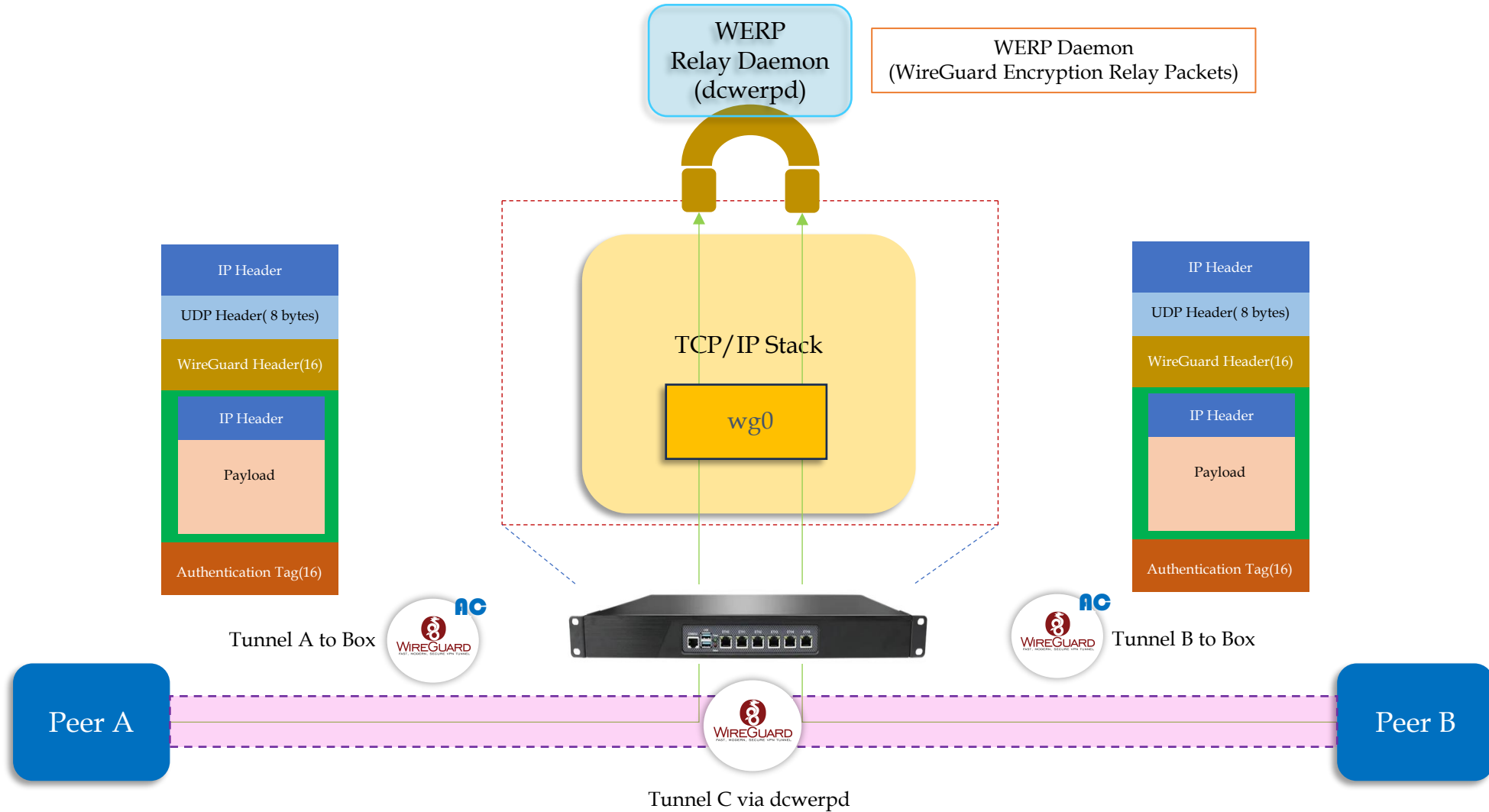
4. MightyConnect Security Box(3) – Direct Connect(3-1)



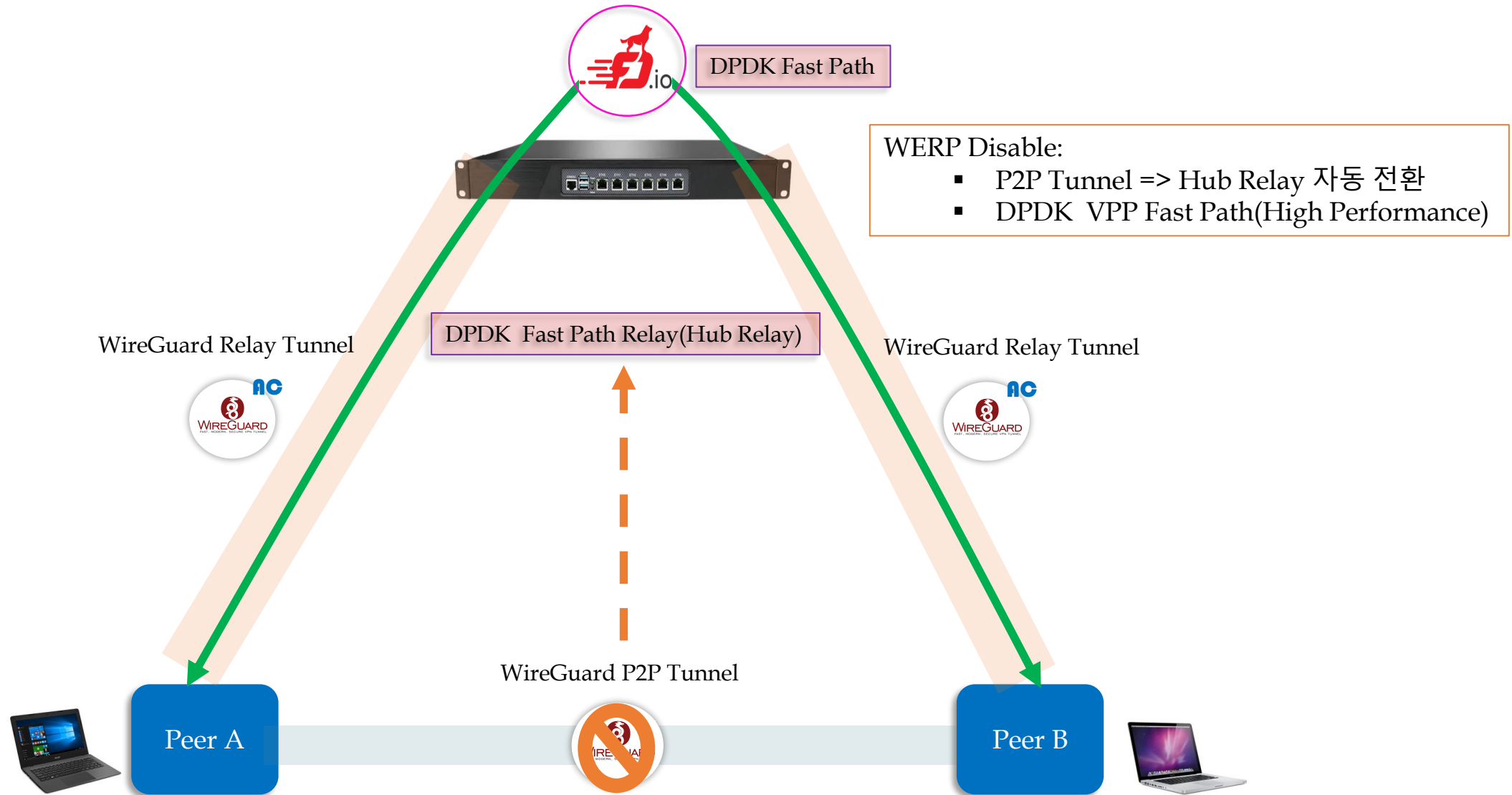
4. MightyConnect Security Box(3) – Direct Connect(3-2)



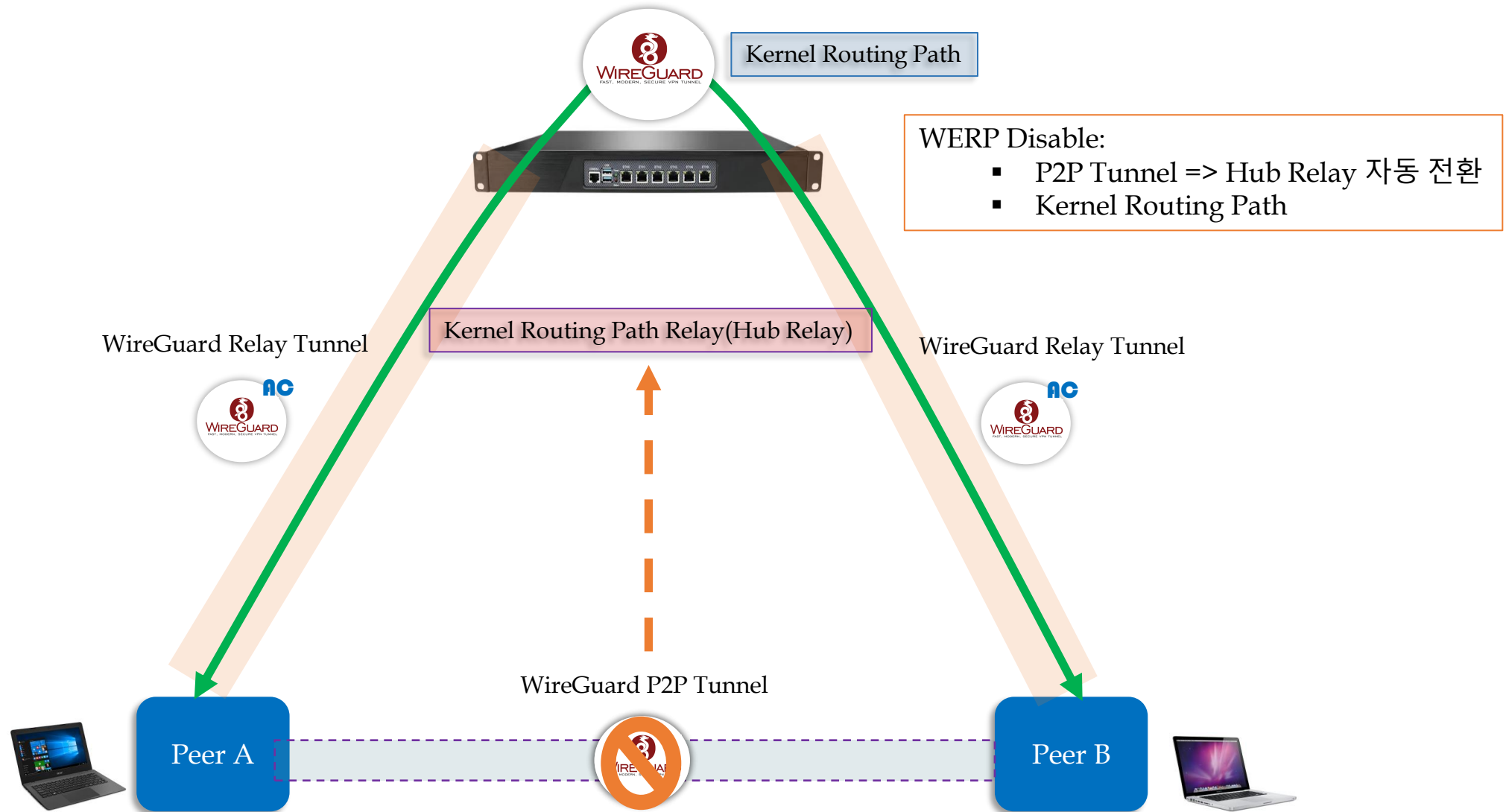
4. MightyConnect Security Box(3) – Direct Connect(3-3)



4. MightyConnect Security Box(4) – Fast Hub Relay(1)



4. MightyConnect Security Box(4) – Fast Hub Relay(2)



4. MightyConnect Security Box(5) – VPP ZTNA(1)

Mighty ZTNA Filter (Micro Segmentation)

Allow Resources per not IP Address but Client ID

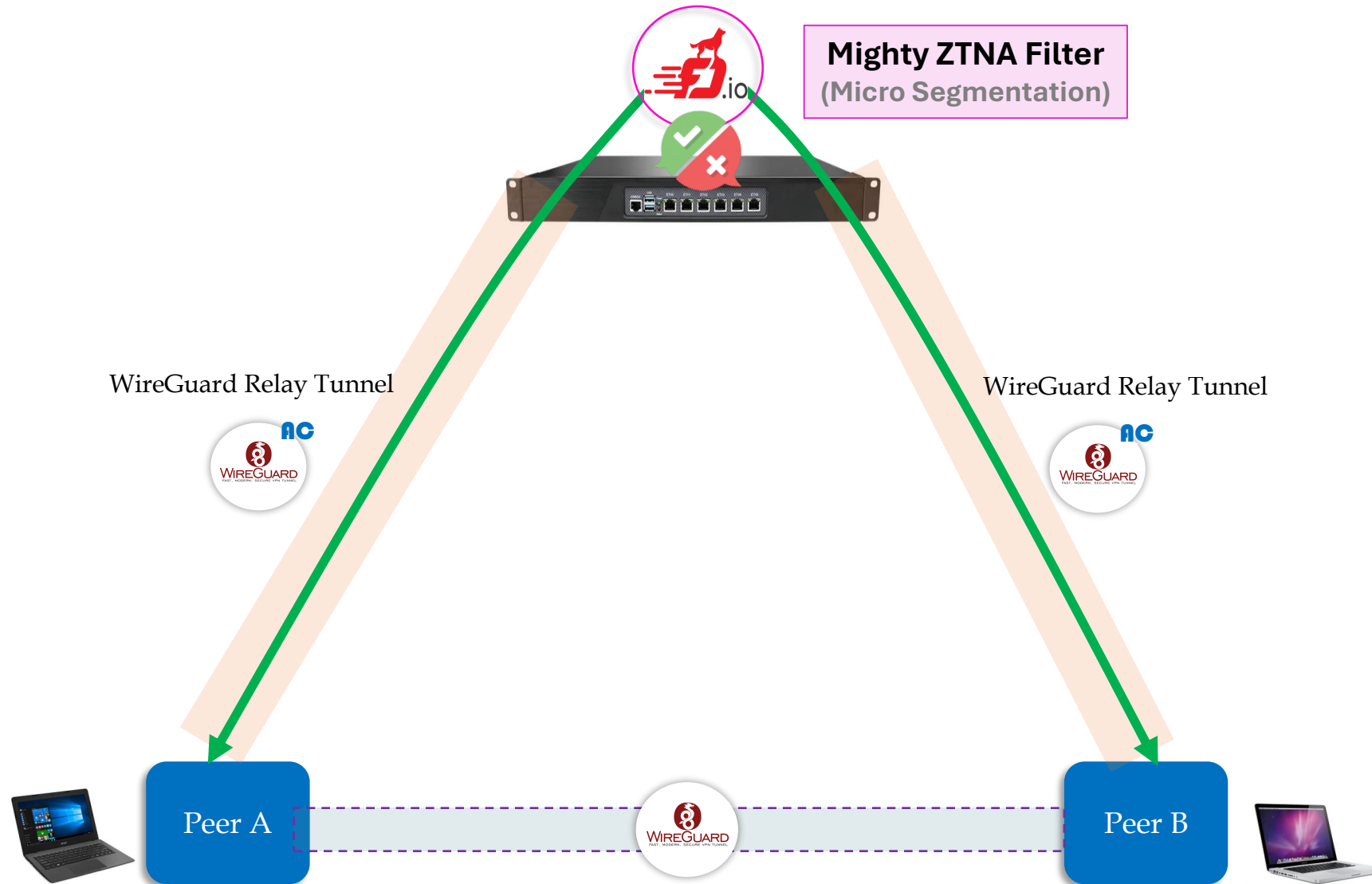


VPP ACL

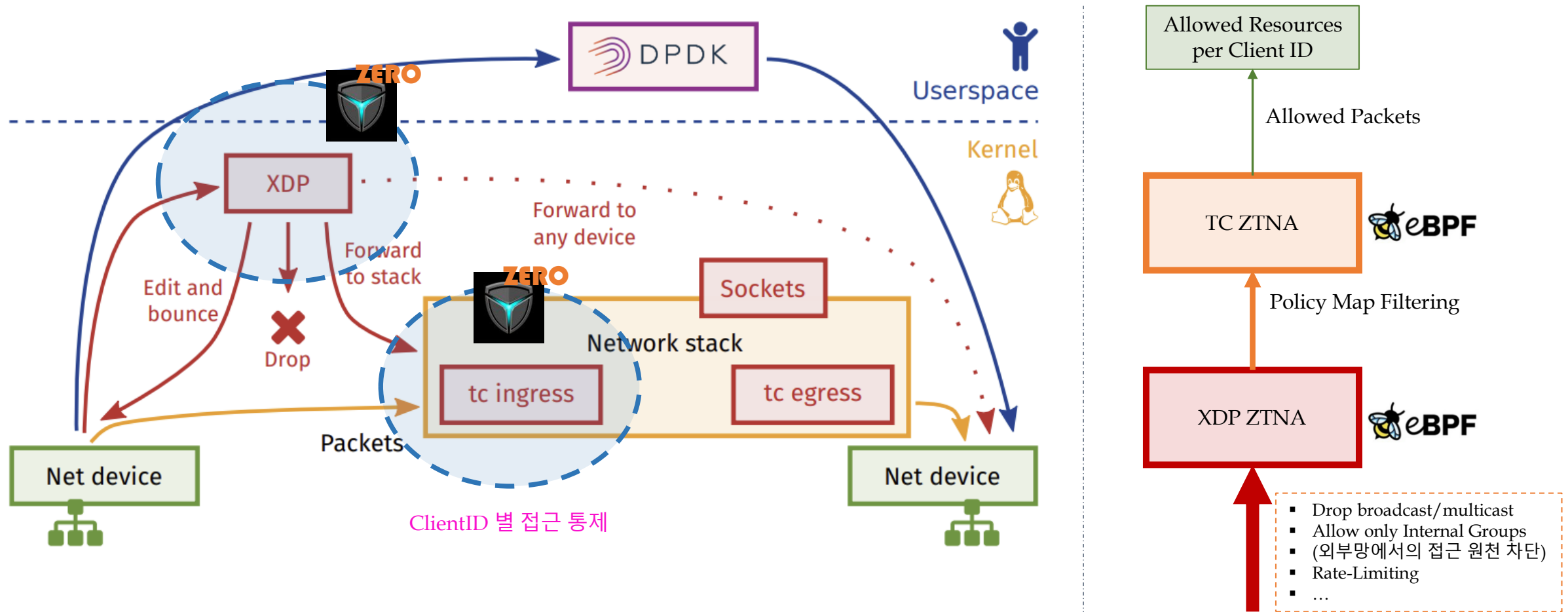
DPDK

```
VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index: 1, 2
  applied outbound on sw_if_index:
  used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index:
  applied outbound on sw_if_index: 1, 2
  used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
13: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
14: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5000
15: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
16: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
17: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
18: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
19: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
20: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
21: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
22: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
23: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
24: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
25: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5432
26: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5000
27: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 8080
  applied inbound on sw_if_index: 5
  applied outbound on sw_if_index:
  used in lookup context index: 5
acl-index 3 count 1 tag {mightyconnect-acl-egress-wg0}
0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied outbound on sw_if_index: 5
```

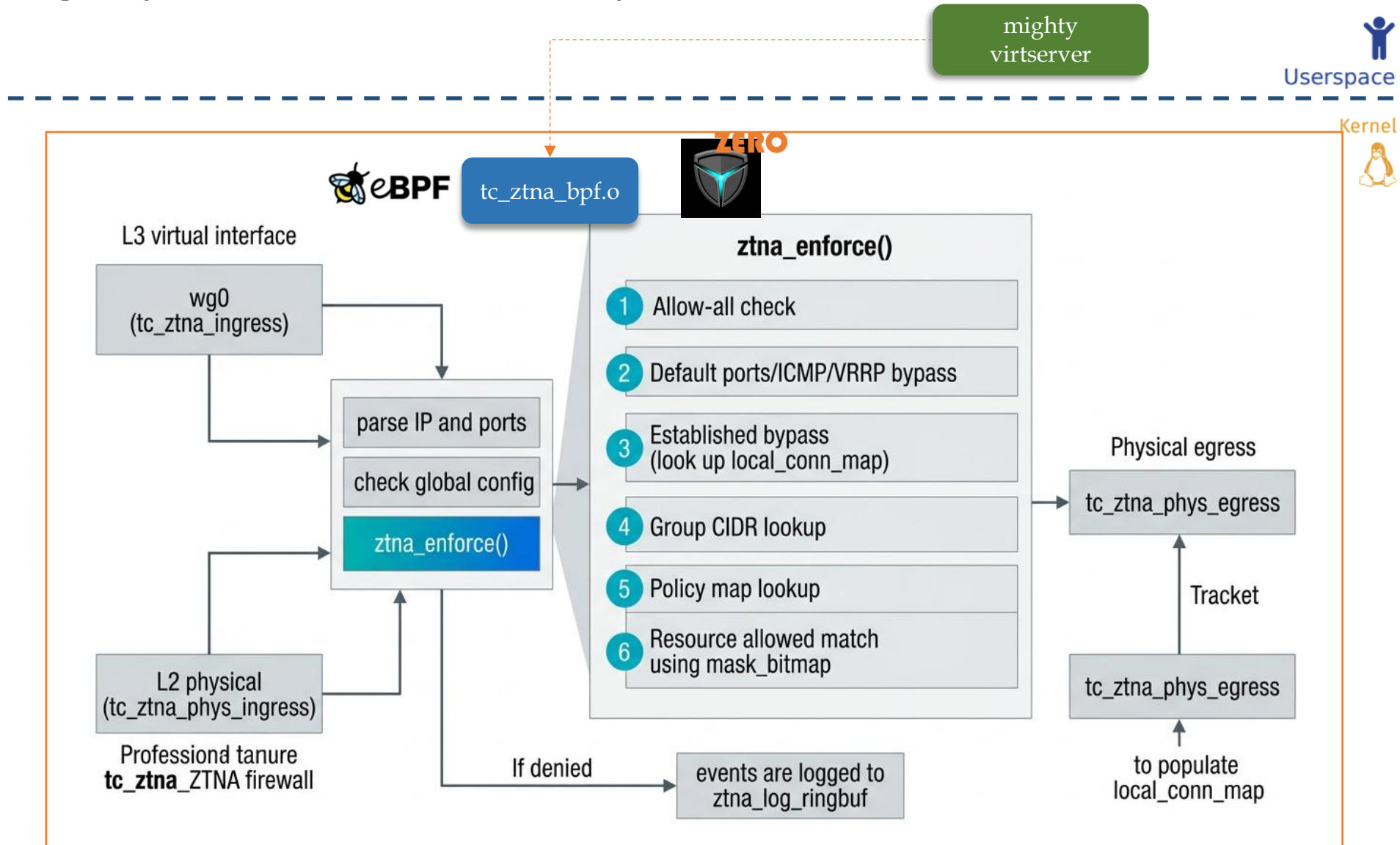
4. MightyConnect Security Box(5) – VPP ZTNA(2)



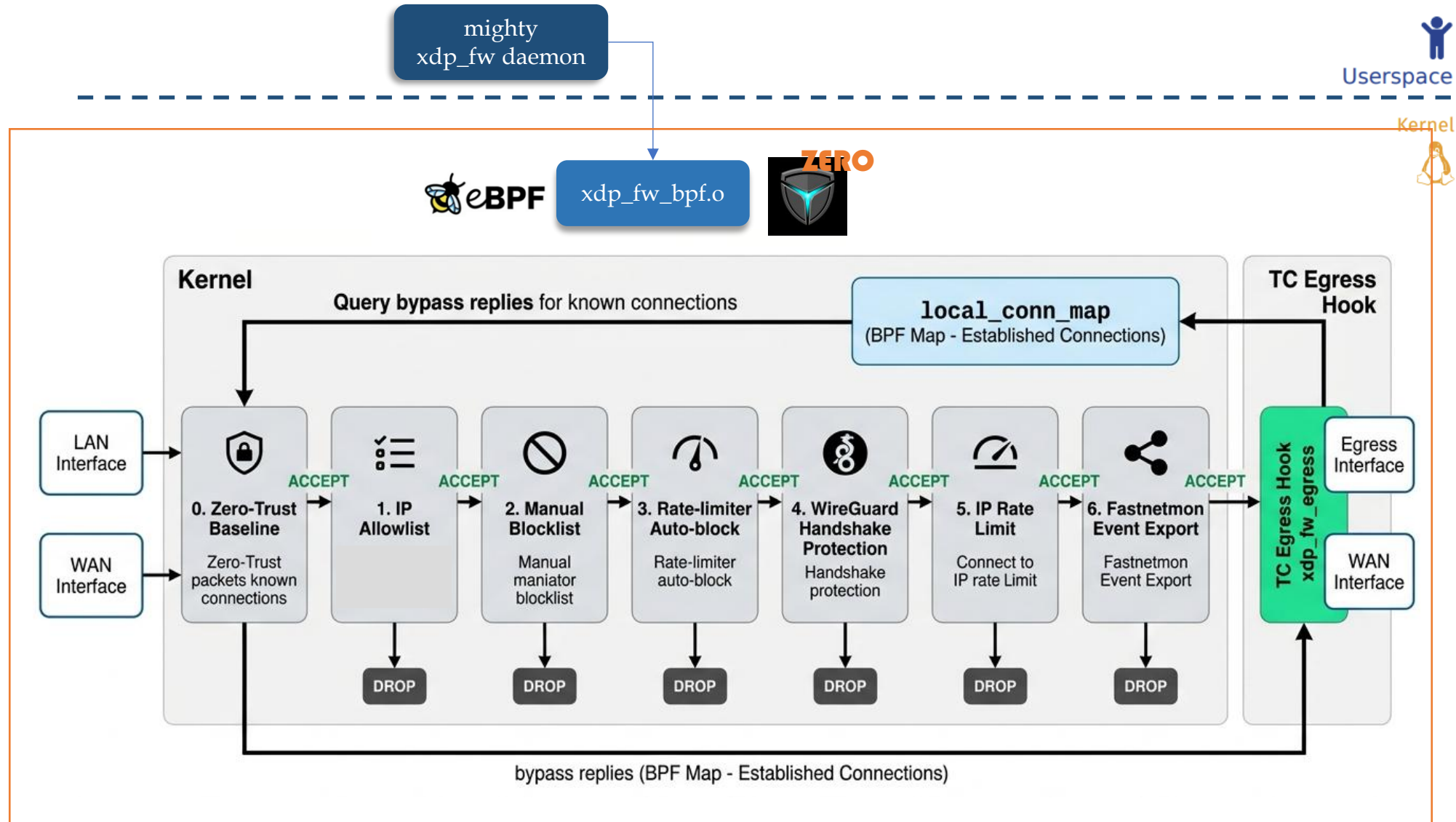
4. MightyConnect Security Box(6) – Kernel ZTNA(1)



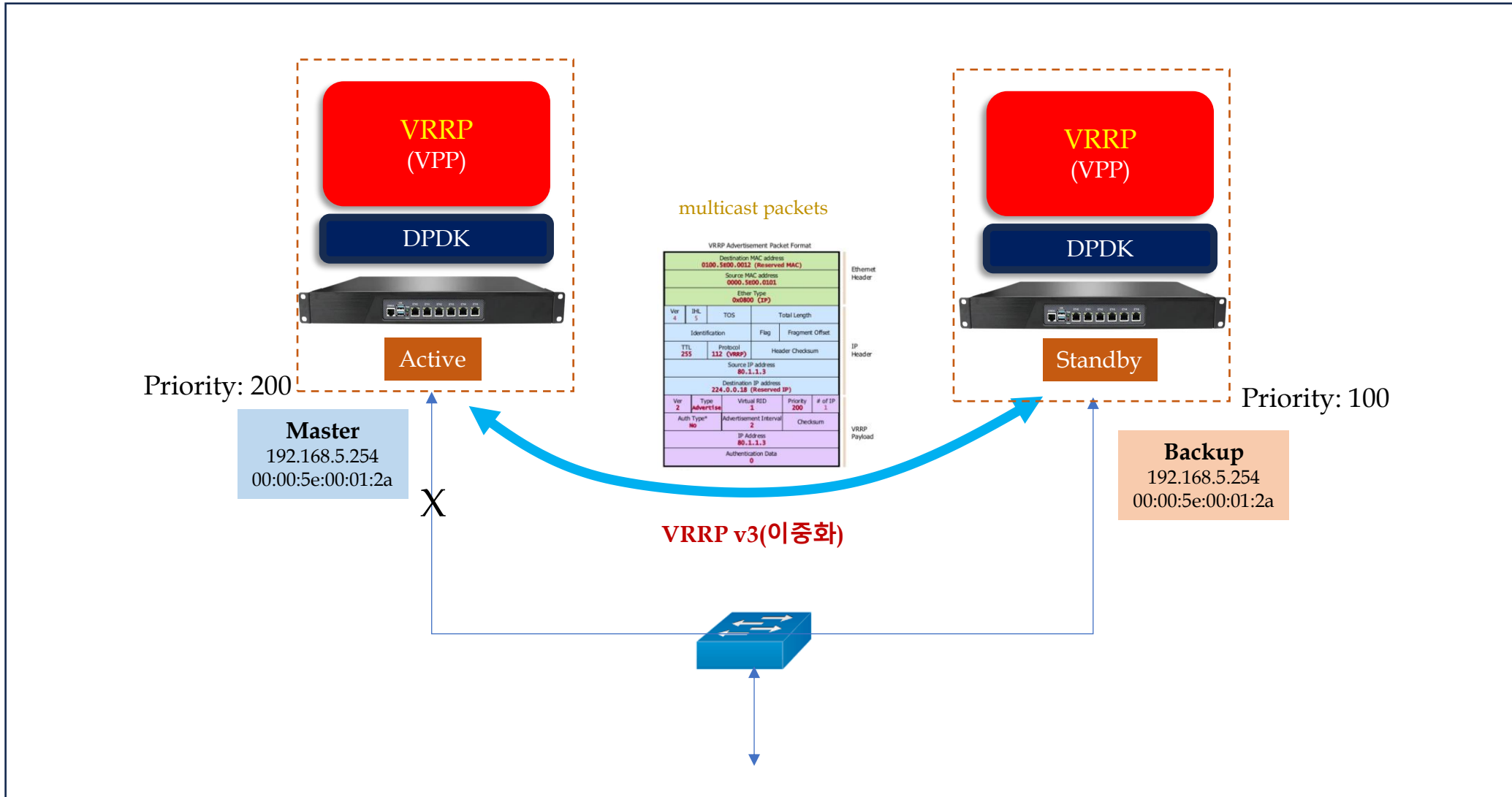
4. MightyConnect Security Box(6) – Kernel ZTNA(2)



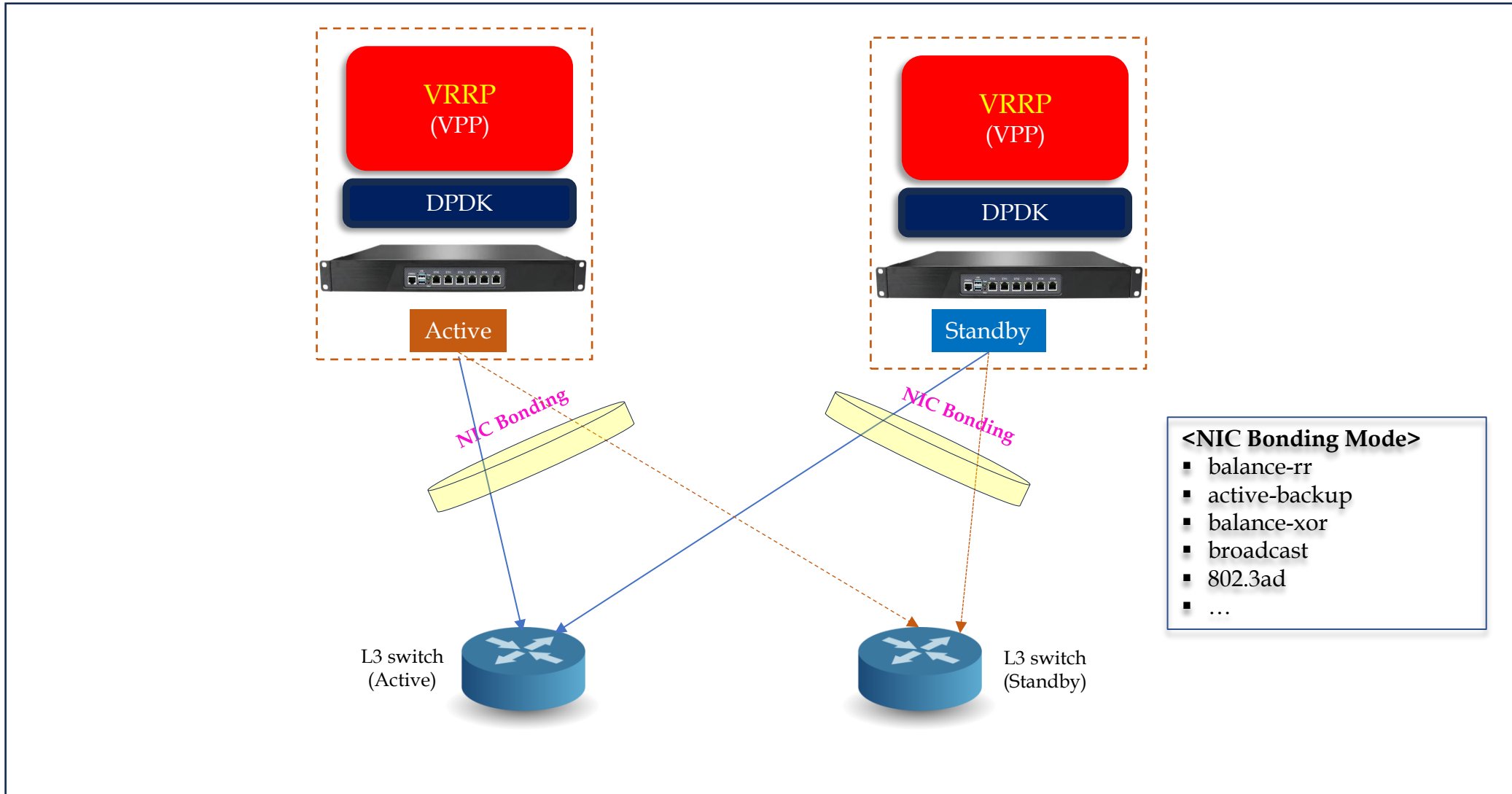
4. MightyConnect Security Box(6) – Kernel ZTNA(3)



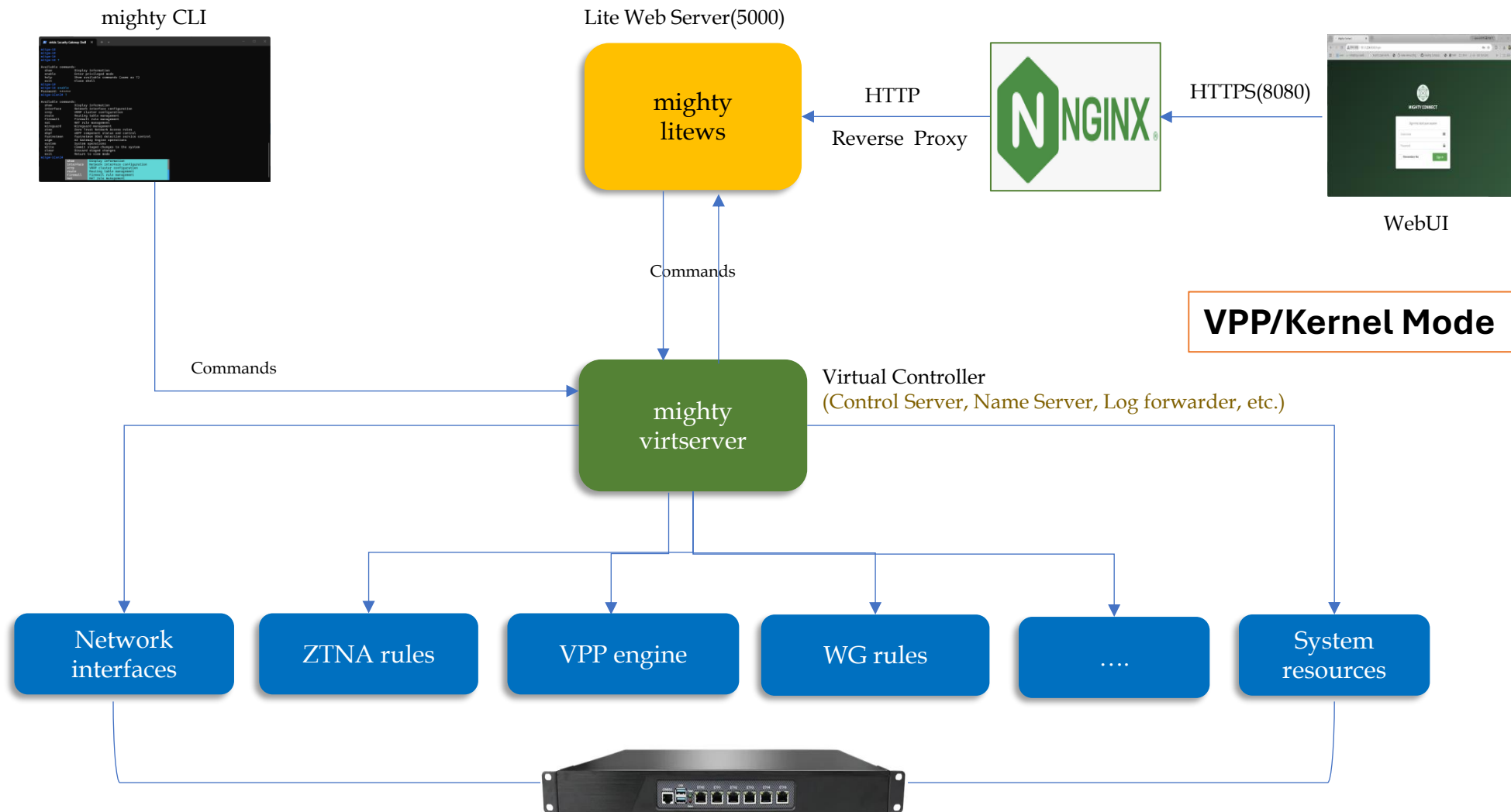
4. MightyConnect Security Box(7) – Redundancy(1)



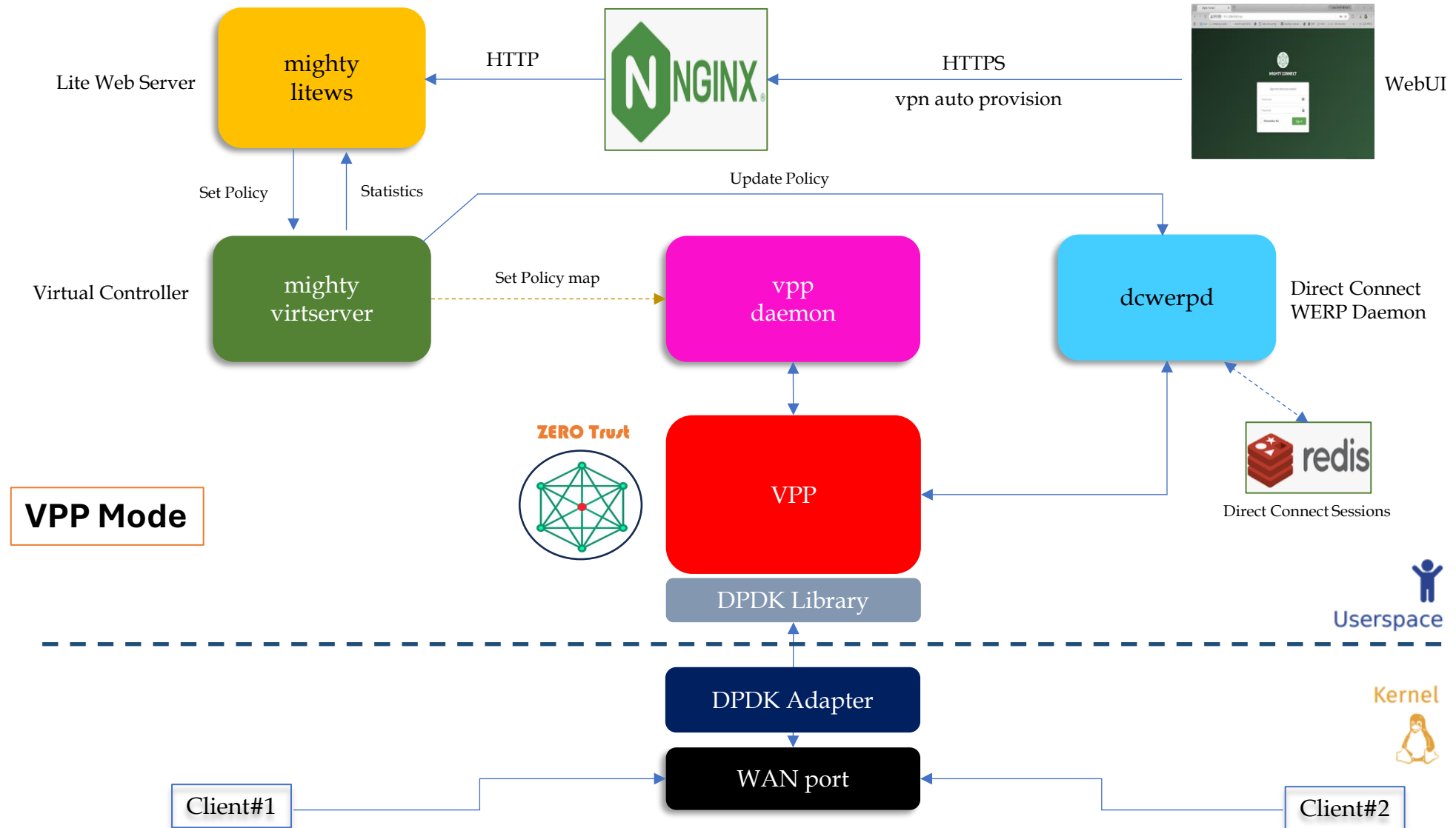
4. MightyConnect Security Box(7) – Redundancy(2)



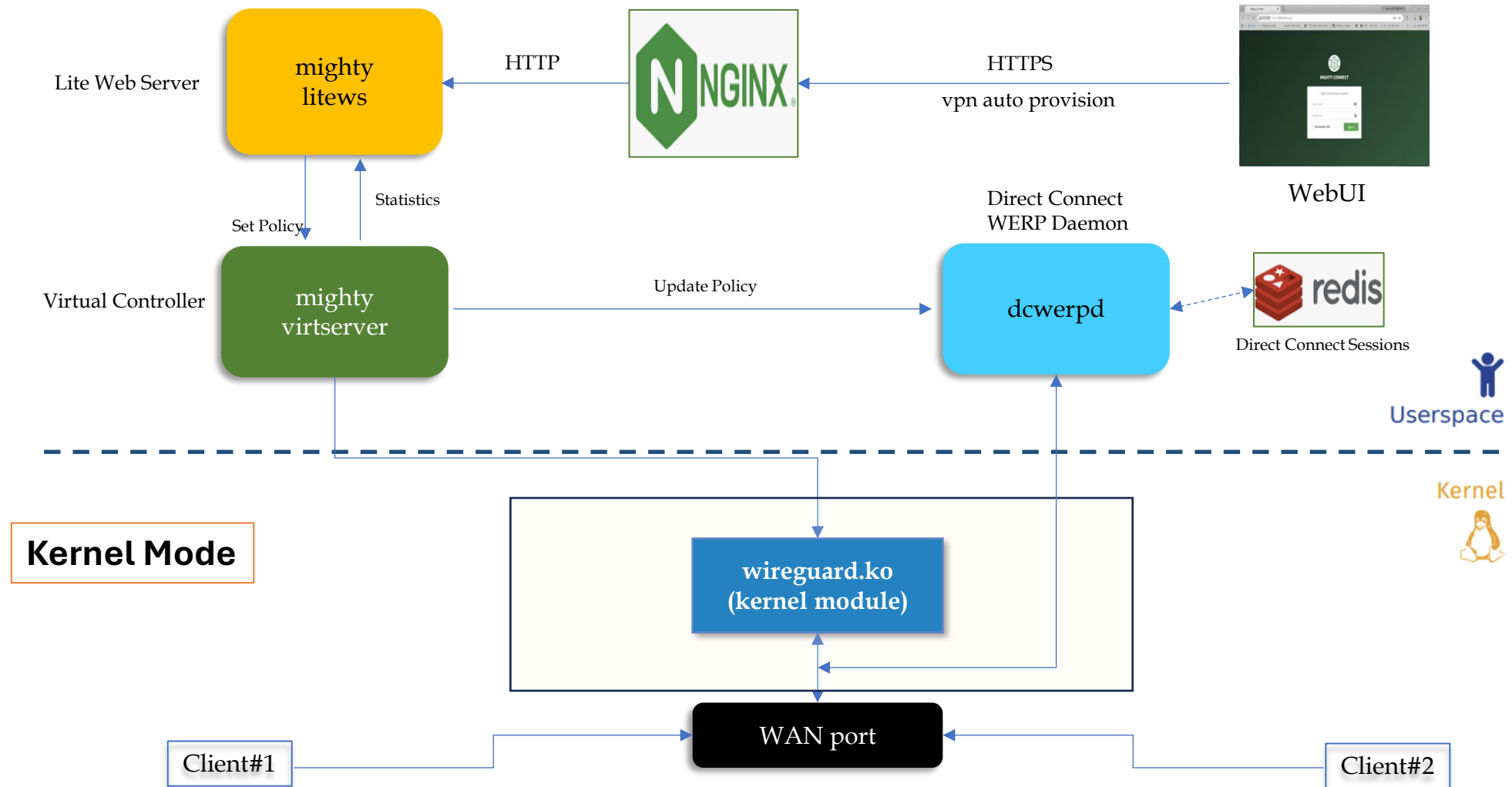
4. MightyConnect Security Box(8) – Mighty Engine(1)



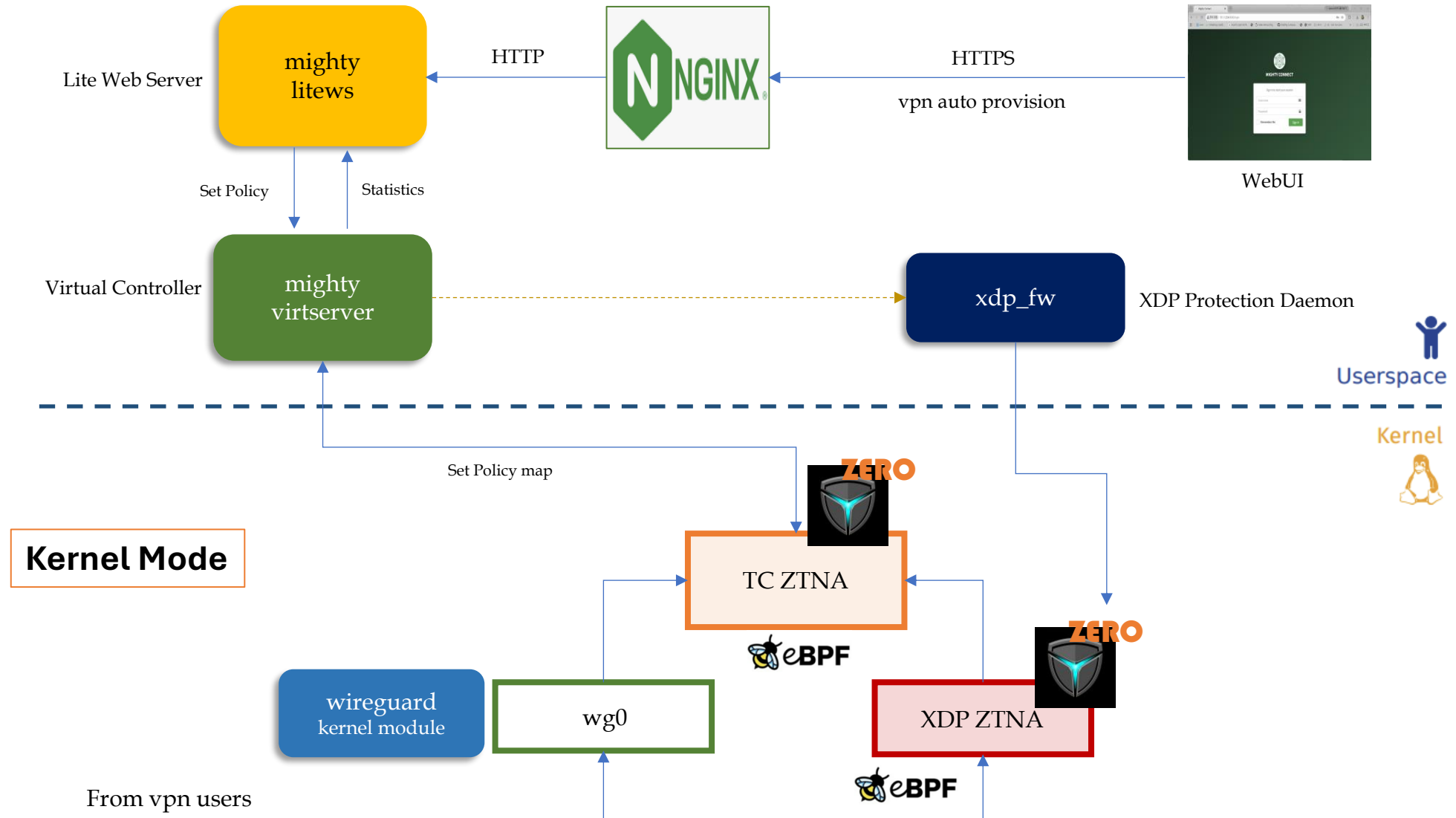
4. MightyConnect Security Box(8) – Mighty Engine(2)



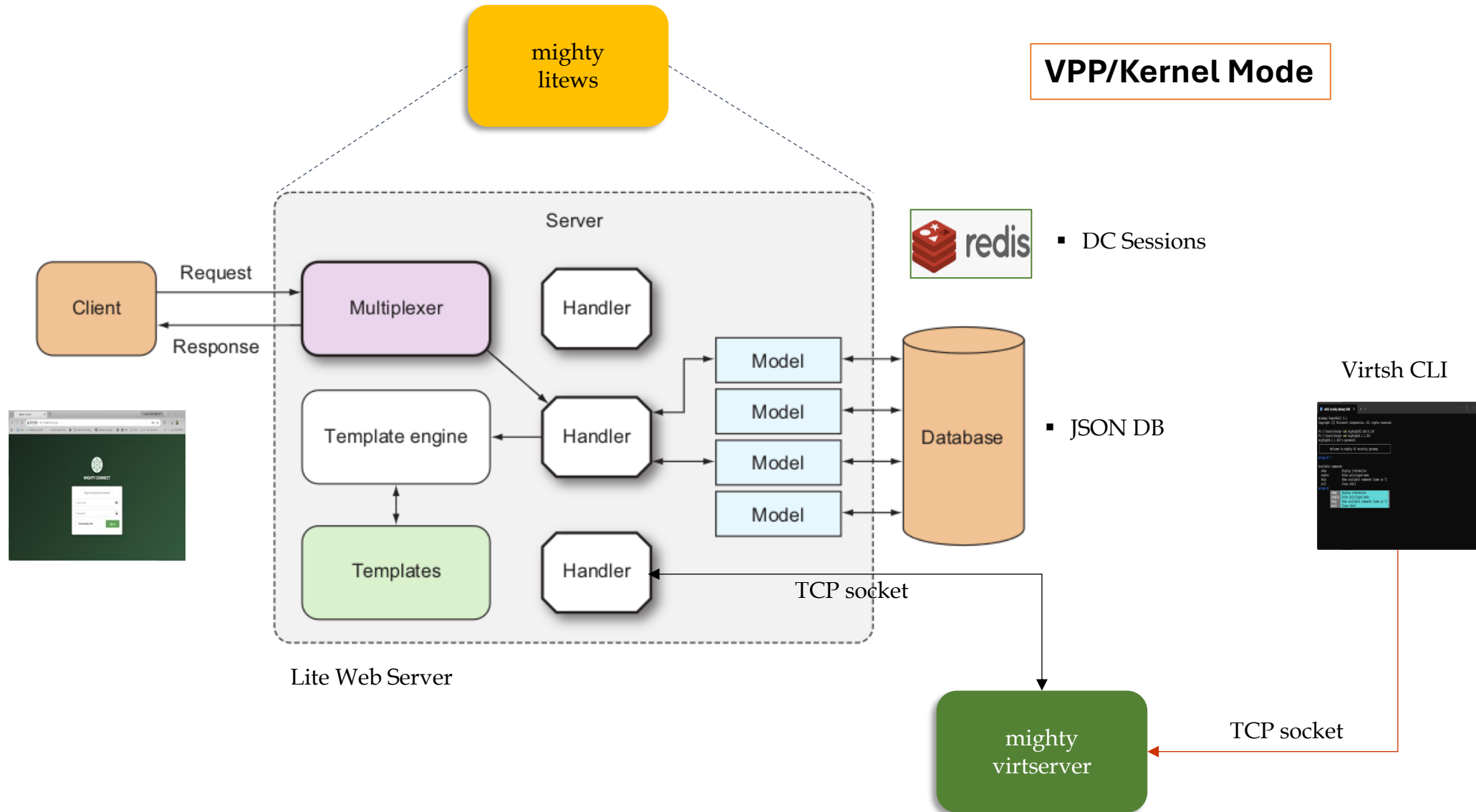
4. MightyConnect Security Box(8) – Mighty Engine(3-1)



4. MightyConnect Security Box(8) – Mighty Engine(3-2)

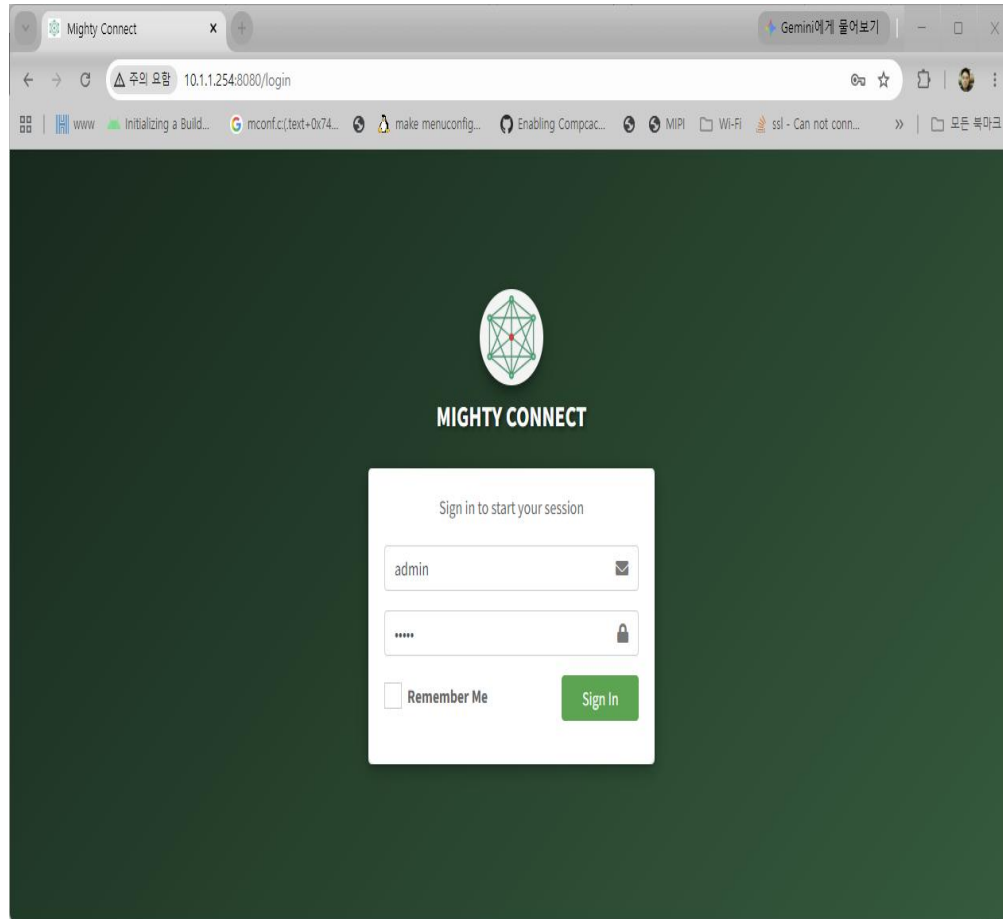


4. MightyConnect Security Box(8) – Mighty Engine(4)

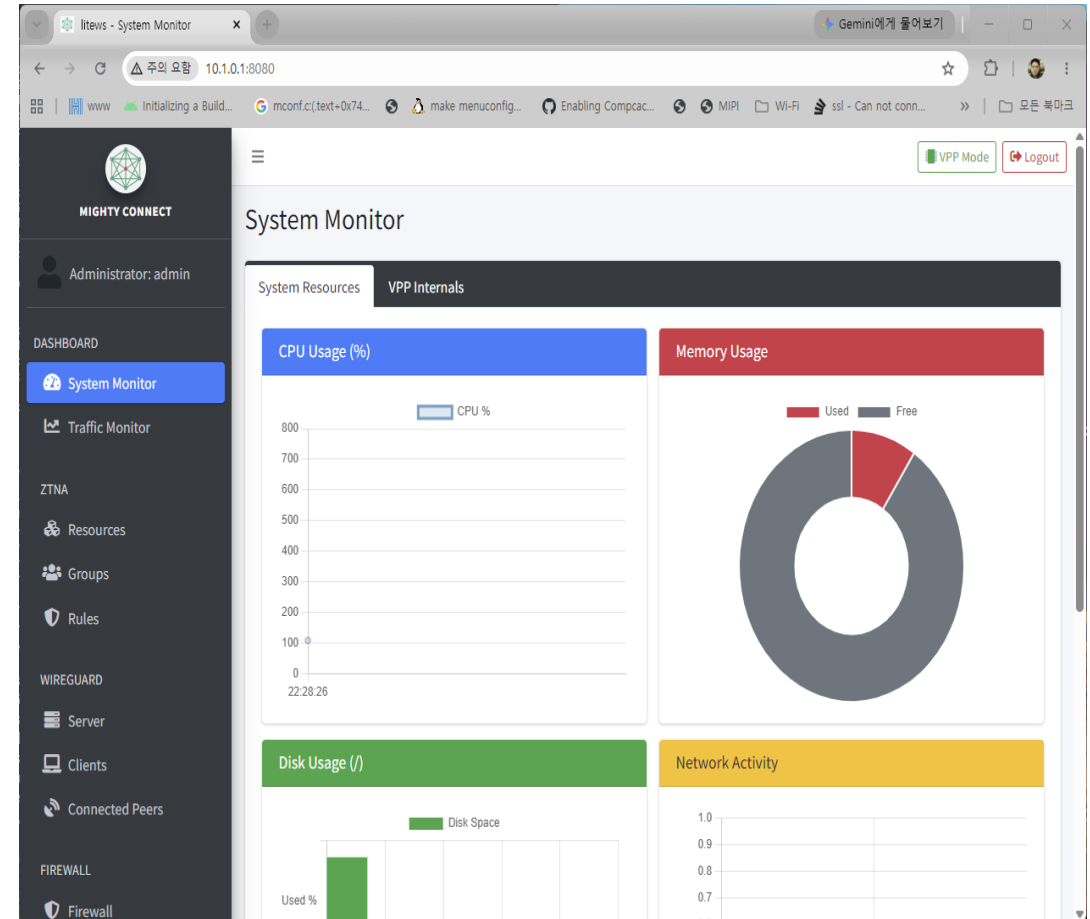


4. MightyConnect Security Box(9) – WebUI(1)

▪ WebUI Login Page



▪ Dashboard Page



4. MightyConnect Security Box(9) – WebUI(2)

▪ ZTNA VPN Groups Page

The screenshot shows the 'VPN Groups' page in the MightyConnect WebUI. The left sidebar contains navigation links: DASHBOARD, System Monitor, Traffic Monitor, ZTNA, Resources, Groups (selected), Rules, WIREGUARD, Server, Clients, Connected Peers, FIREWALL, and Firewall. The main content area has a header with 'VPN Groups' and a 'VPP Mode' indicator. Below the header is a form to 'Add New Group' with fields for 'Group Name' (e.g., Sales Team) and 'Expected Max Members' (e.g., 50), and a checkbox for 'Start with full internal mesh'. A table below lists existing groups:

Name	Dedicated Subnet	Members	Created	Removal
groupC	10.1.33.0/25 (10.1.33.1 ~ 10.1.33.126)	0 / 100	2026-08-11 15:10:08	[Remove]
groupB	10.1.34.0/23 (10.1.34.1 ~ 10.1.35.254)	1 / 500 SN-2026-Windows11	2026-08-11 14:27:25	[Remove]
groupA	10.1.32.0/24 (10.1.32.1 ~ 10.1.32.254)	4 / 200 SN-2026-Windows11, SN-2026-EmbeddedLinux, SN-2026-Android, SN-2026-GI-Net	2026-08-11 14:25:19	[Remove]

▪ ZTNA Access Rules Page

The screenshot shows the 'Access Rules' page in the MightyConnect WebUI. The left sidebar is identical to the previous page, with 'Rules' selected. The main content area has a header with 'Access Rules' and a 'VPP Mode' indicator. Below the header is a 'ZTNA Engine Status' section showing 'Bound (ACL index 2 on wg0)'. A table displays engine statistics:

Interface	Active Policies	Resource Rules
wg0	4	7

Below this, a table shows 'Allowed' (41) and 'Denied' (471) counts. A note states: 'Allowed/Denied are system-wide VPP ACL packet counts (every ACL-bound interface), not ZTNA-only.' Below this is a section for 'Access Rules (Total: 4)' with '+ Add New Access Rule' and '+ Grant Resource Access' buttons. A table lists the rules:

Source	Destination	Protocol	Port	Enabled	Expires	Actions
Group: groupC	Group: groupB	tcp	443,5000	[On]	—	[Edit] [Remove]
Group: groupC	Group: groupC	tcp	5432	[On]	—	[Edit] [Remove]
Group: groupB	Group: groupB	all	any	[On]	—	[Edit] [Remove]
Group: groupA	Group: groupA	udp, tcp, tcp	6000,7999, 22,80,5000, 8080	[On]	—	[Edit] [Remove]

4. MightyConnect Security Box(9) – WebUI(3)

■ WireGuard Server Page

The screenshot shows the 'WireGuard Server Settings' page. The left sidebar contains a navigation menu with options like Dashboard, System Monitor, Traffic Monitor, ZTNA, Resources, Groups, Rules, WireGuard (selected), Server (highlighted), Clients, Connected Peers, Firewall, and Firewall. The main content area has tabs for 'Interface/Keypair' and 'Settings'. The 'Interface' tab is active, showing fields for 'Server Interface Addresses' (10.1.0.1/16), 'Listen Port' (51820), and script fields for 'Post Up Script', 'Pre Down Script', and 'Post Down Script'. The 'Key Pair' tab shows 'Private Key' and 'Public Key' fields, with a 'Generate' button.

■ WireGuard Client Auto Provision Page

The screenshot shows the 'WireGuard Client Auto Provision' page. The left sidebar is identical to the previous page. The main content area has a tab for 'Clients' and a section for 'Enable ZTNA Access Control'. Below this is a table titled 'Authorized Devices' with columns: Device ID, Comment, Status, 2FA, Created At, and Action. The table lists several devices, including 'SN-2026-NanoP15', 'SN-2026-Android', 'SN-2026-G1-iNet', 'SN-2026-EmbeddedLinux', 'SN-2026-Windows11', 'SN-2026-X1022', and 'SN-2026-X1021'. Each device has a 'Registered' status and a 'Disabled' 2FA status. The 'Action' column contains a trash icon for each device.

Device ID	Comment	Status	2FA	Created At	Action
SN-2026-NanoP15	wireguard linux nanopi	Registered	Disabled	2026-08-06 11:28	
SN-2026-Android	wireguard android	Registered	Disabled	2026-08-03 09:18	
SN-2026-G1-iNet	wireguard linux gl-inet	Registered	Disabled	2026-08-03 09:17	
SN-2026-EmbeddedLinux	wireguard linux	Registered	Disabled	2026-08-03 09:16	
SN-2026-Windows11	wireguard windows	Registered	Disabled	2026-08-03 09:16	
SN-2026-X1022	Bulk Imported	Pending	Disabled	2026-08-03 09:00	
SN-2026-X1021	Bulk Imported	Pending	Disabled	2026-08-03 09:00	

4. MightyConnect Security Box(9) – WebUI(4)

■ WireGuard Connected Peers Page

Connected Peers Status

Auto-refreshing every 10s

Peer Status Direct Connect Status Map View

Device: wg0

#	Name	Allocated IPs	OS	Endpoint	Location	Received	Transmitted	Status
0	SN-2026-GI-iNet	10.1.1.4/32	OpenWrt 21.02-SNAPSHOT	192.168.8.239:51820	Internal/Private	794.93 KB	662.31 KB	Connected
1	SN-2026-Windows11	10.1.1.2/32	Windows 11 (Build 26200)	192.168.8.103:59719	Internal/Private	1.72 MB	20.18 MB	Connected

■ Network Interfaces Page

Network Interfaces

Administrator: admin

DASHBOARD

- System Monitor
- Traffic Monitor

ZTNA

- Resources
- Rules

WIREGUARD

- Server
- Clients
- Connected Peers

FIREWALL

- Firewall

Interfaces VRRP

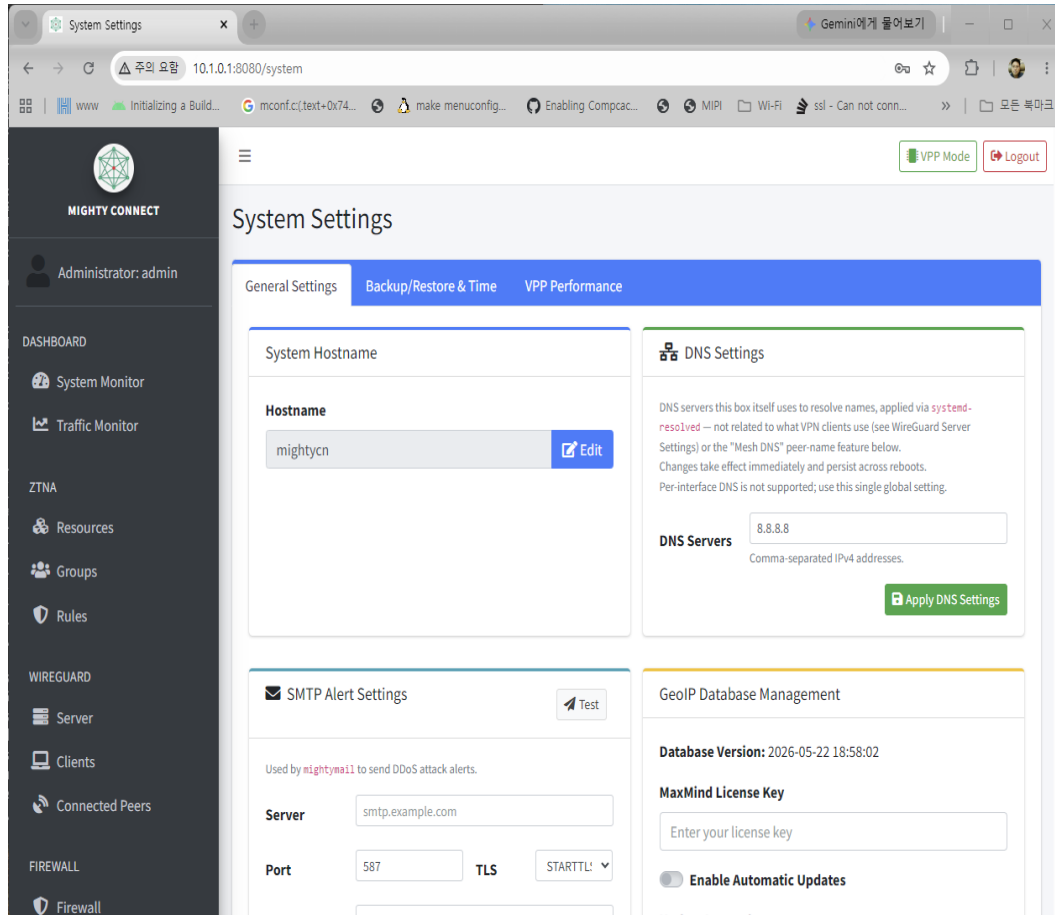
Ethernet Interfaces

Restart Interfaces Create Bonding

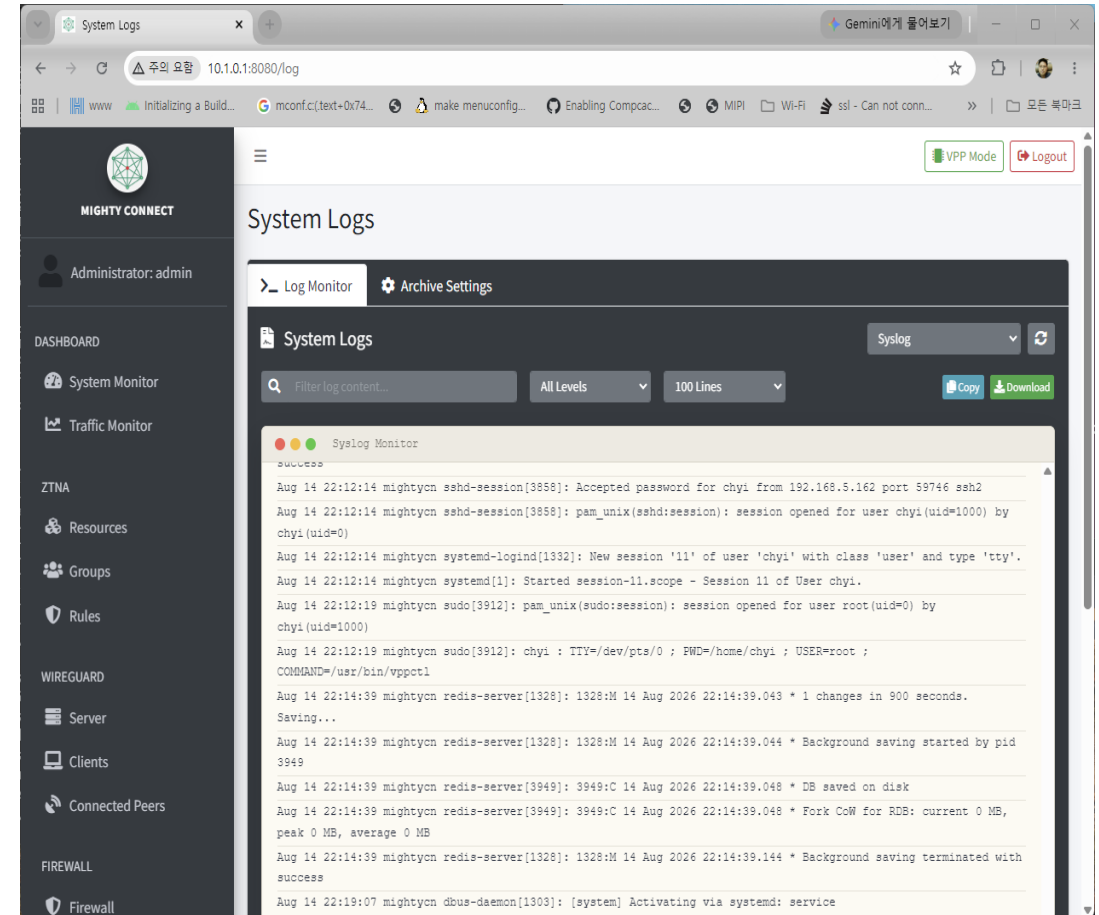
Interface	Status	IP Address	Netmask	Role	DPDK	Link Info	Action
enp2s0	UP	192.168.8.114	255.255.255.0	LAN	DPDK	Link 1000Mb/s Full	Edit
enp6s0	UP	192.168.10.1	255.255.255.0	LAN2	DPDK	—	Edit
enp1s0	UP	192.168.5.1	255.255.255.0	MGMT	—	Link 2500Mb/s Full Port: TP Max: 2500baseT/Full Auto-Neg: On	Edit
enp3s0	DOWN	—	—	—	—	No Link Port: TP Max: 2500baseT/Full Auto-Neg: On	Edit
enp4s0	DOWN	—	—	—	—	No Link Port: TP Max: 2500baseT/Full Auto-Neg: On	Edit

4. MightyConnect Security Box(9) – WebUI(5)

■ System Settings Page



■ Logs Page



4. MightyConnect Security Box(10) – CLI

```
PS C:\Users\chung> ssh mighty@10.1.1.254
mighty@10.1.1.254's password:

Welcome to mighty AI security gateway

mightycn# ?

Available commands:
show          Display information
enable       Enter privileged mode
help         Show available commands (same as ?)
exit        Close shell

mightycn# show interface
Interface Status IP Address DDPK
-----
enp2s0 UP 192.168.8.114/255.255.255.0 yes
enp6s0 UP 192.168.10.1/255.255.255.0 yes
tap4096 UP - yes
tap4097 UP - yes
tun4098 UP - yes
wg0 UP 10.1.1.254/255.255.255.0 yes
enp1s0 UP 192.168.5.1/255.255.255.0 -
enp3s0 DOWN -
enp4s0 DOWN -
enp5s0 DOWN 192.168.20.1/255.255.255.0 -

VPP interfaces (show interface):
Name Idx State MTU (L3/IP4/IP6/MPLS) Counter Count
enp2s0 1 up 9000/0/0/0 rx packets 83438
rx bytes 31067596
tx packets 96924
tx bytes 69994133
drops 1197
punt 2101
ip4 82166
ip6 286
tx-error 2
tx-error 1
drops 4
rx packets 547
rx bytes 163088
tx packets 862
tx bytes 191013
drops 16
ip4 459
ip6 17
tap4097 4 up 9000/0/0/0 rx packets 41921
rx bytes 40105173
tx packets 27211
tx bytes 2196120
ip4 41903
ip6 18
wg0 5 up 1420/0/0/0 rx packets 79471
```

```
mightycn# show ztna
Network ZTNA Rules (L4 Access Control):
Client Name (Label) Client IP Authorized Resources Status
-----
SN-2026-Android 10.1.1.3/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443) Enabled
SN-2026-EmbeddedLinux 10.1.1.1/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80) Enabled
SN-2026-Gl-iNet 10.1.1.4/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80), litews (tcp/10.1.1.0/24/5000) Enabled
SN-2026-NanoPi5 10.1.1.5/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80) Enabled
SN-2026-Windows11 10.1.1.2/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80), postgresql (tcp/10.1.1.0/24/5432), litews (tcp/10.1.1.0/24/5000), boxlitews (tcp/10.1.1.254/32/8080) Enabled

Always Allowed (all clients, bypasses policy_map entirely): DNS tcp/53, DNS udp/53, ICMP

VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
applied inbound on sw_if_index: 1, 2
applied outbound on sw_if_index:
used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
applied inbound on sw_if_index:
applied outbound on sw_if_index: 1, 2
used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443

--More--
```

4. MightyConnect Security Box(11) – H/W Spec(1)

고급(상급) 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-5000 / 2U Rackmount
CPU	Intel® Xeon® Scalable Processor Gold 5218 2CPU(32 cores, 64 threads), 2.3GHz
RAM	DDR4 64GB
Storage(SSD/HDD)	480GB/4TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 10 GbE RJ45 2 ports
Power Input	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium

H/W 스펙은 수시로 변경될 수 있음. 특히, CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

4. MightyConnect Security Box(11) – H/W Spec(2)

중급 모델

항목	사양
모델명 / 베이슬롯	mighty-connect-2500 / 2U Rackmount
CPU	Intel® Xeon® D-2779 1 CPU(16 core, 32 threads)
RAM	DDR4 64GB
Storage(SSD/HDD)	480GB/1TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 10 GbE RJ45 2 ports
Power Input	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium

H/W 스펙은 수시로 변경될 수 있음. 특히, CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

4. MightyConnect Security Box(11) – H/W Spec(3)

하급 모델

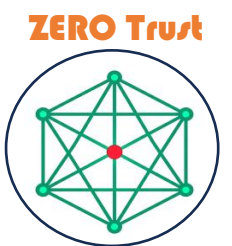
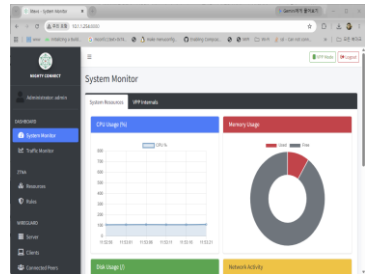
항목	사양
모델명 / 베이슬롯	mighty-connect-1000 / 1U Rackmount
CPU	Intel® Xeon® D-2146NT 1 CPU(8 core, 16 threads)
RAM	DDR4 32GB
Storage(SSD/HDD)	480GB/1TB (OS/Log 저장용)
Ethernet Ports	1 GbE RJ45 2 ports + 1 GbE RJ45 2 ports
Power Input	200W Low-noise AC-DC power supply

H/W 스펙은 수시로 변경될 수 있음. 특히, CPU Core 종류는 단종 등의 이유로 변경될 수 있음.

5. MightyConnect Devices

MightyConnect Devices

5. MightyConnect Devices



MightyConnect™ Box



MightyConnect™ Mini



MightyConnect™ Windows



MightyConnect™ Ubuntu



MightyConnect™ macOS/iOS



MightyConnect™ Android

5.1 MightyConnect Mini

MightyConnect Mini Security Gateway



GL-iNet Series
<https://www.gl-inet.com/>

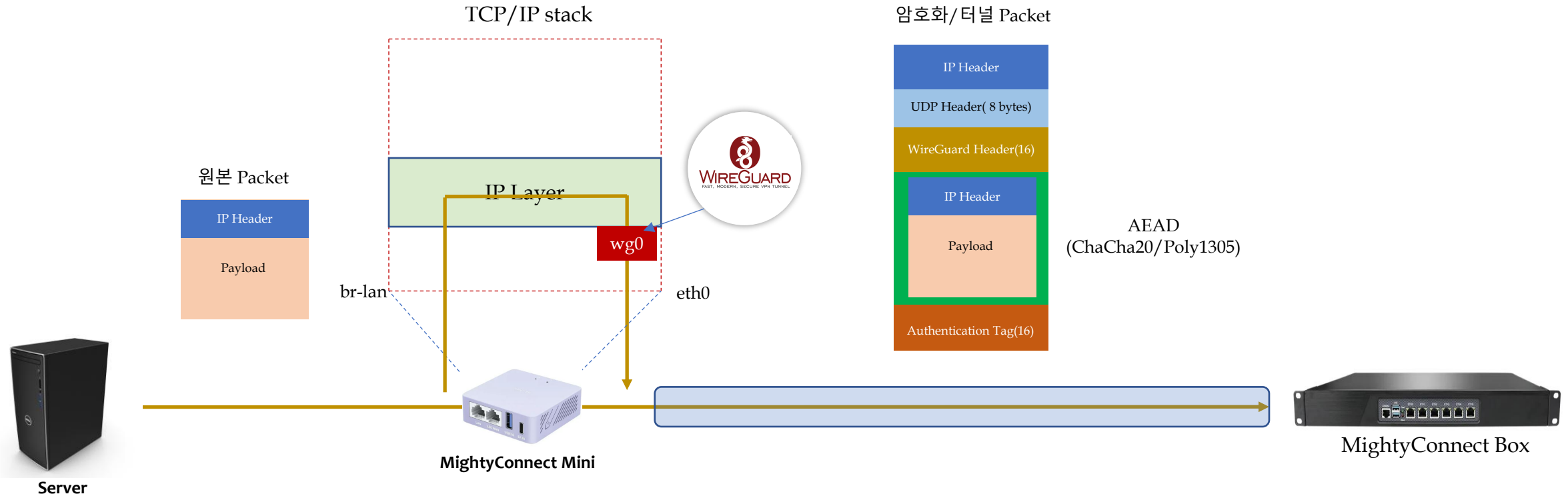


NanoPi Series
<https://www.friendlyelec.com/>

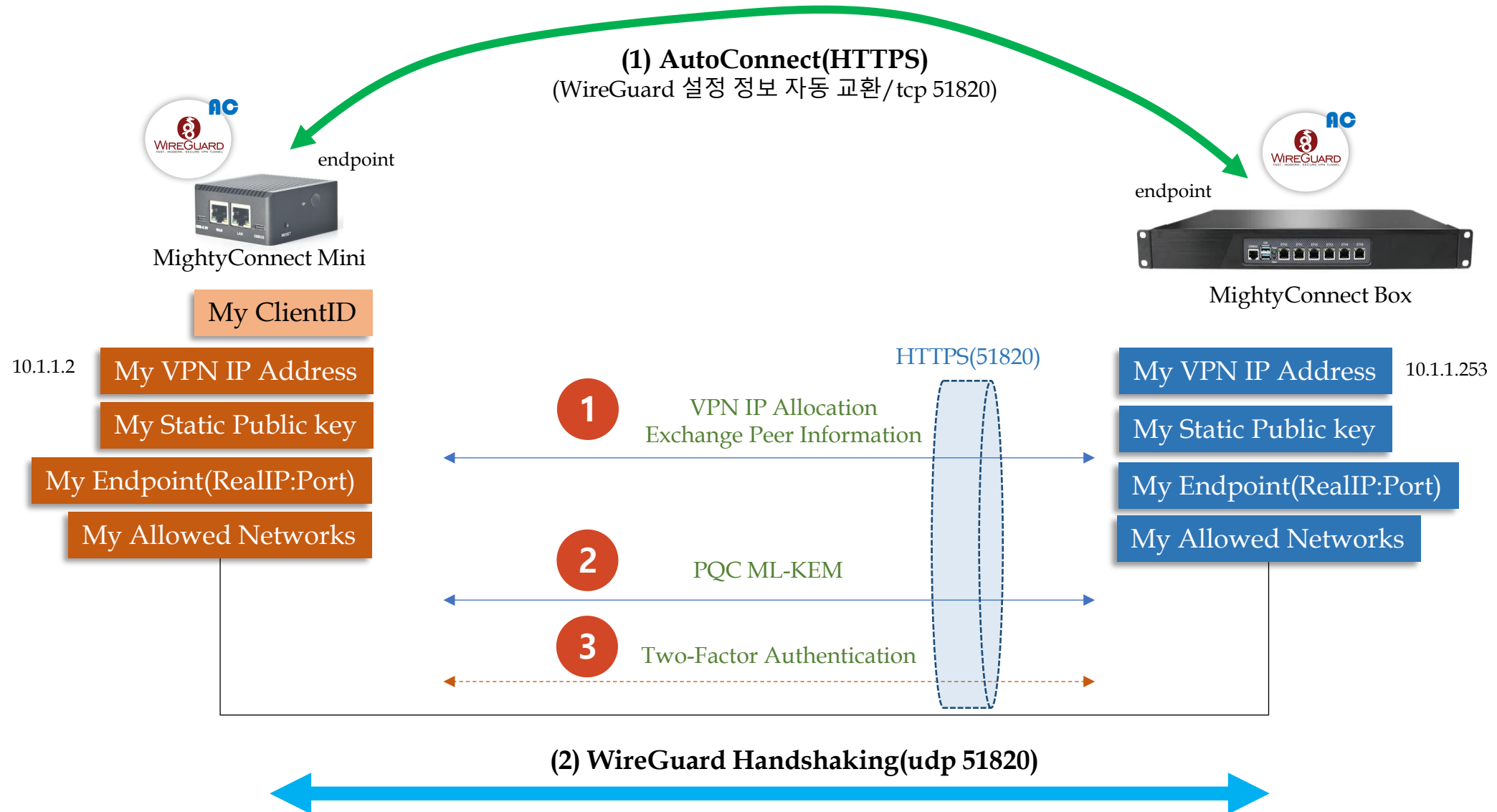
5.1 MightyConnect Mini(1) – Overview

CryptoKey Routing

ID: Public Key::VPN IP

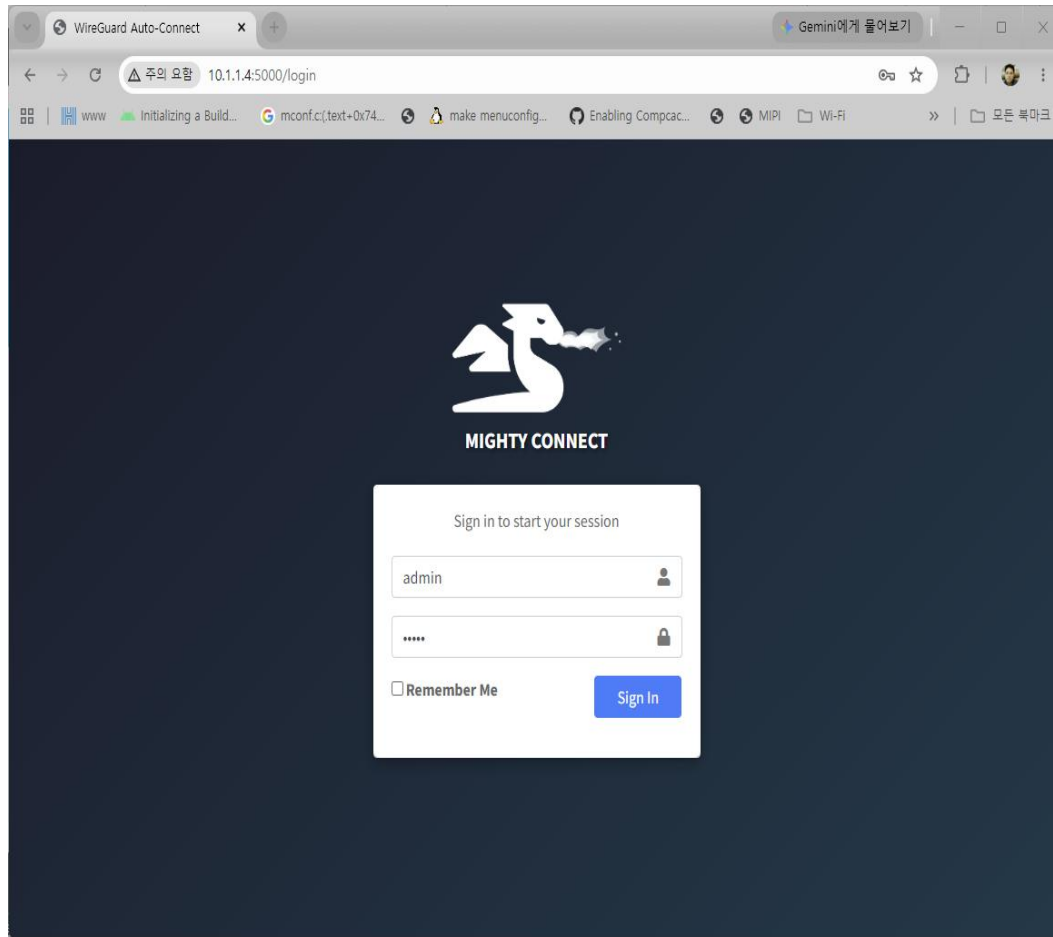


5.1 MightyConnect Mini(2) – Auto Connection

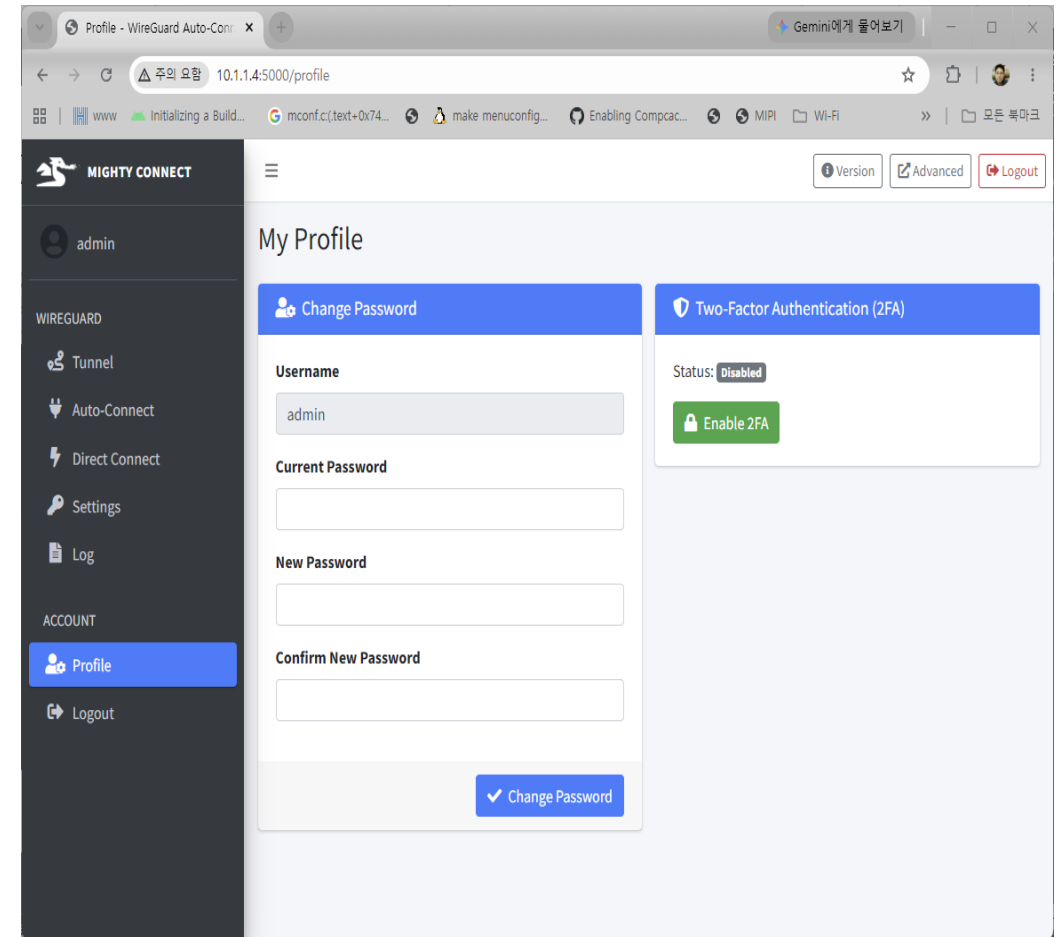


5.1 MightyConnect Mini(3) – WebUI(1)

■ Login Page



■ Admin User Page



5.1 MightyConnect Mini(3) – WebUI(2)

■ Tunnel Status Page

The screenshot shows the 'Tunnel Status' page of the WG Auto-Connect web interface. The left sidebar contains a menu with 'admin', 'WIREGUARD', 'Tunnel' (selected), 'Auto-Connect', 'Settings', 'Log', 'ACCOUNT', 'Profile', and 'Logout'. The main content area displays the 'Tunnel Status' for the 'wg-auto0' interface. A 'Restart Daemon' button is visible in the top right of the status section. The status is 'running'. The public key is 'wac8svbrogFLtgZ1M3IXZ0qXLxA07XwyfKI+KGqJlc=' and the private key is '(hidden)'. The listening port is 51820, and the assigned IP is 10.1.1.12/24. The peer is '9JpQ7SGTfy7o+8P1SXaf40J4wJ8Kq19o5CPRc8jJbg4=' and the preshared key is '(hidden)'. The endpoint is 192.168.8.129:51820, allowed IPs are 10.1.1.0/24, the latest handshake was 46 seconds ago, and the transfer shows 2.1 MiB received and 2.11 MiB sent. The persistent keepalive is set to every 15 seconds.

Parameter	Value
daemon status	running
public key	wac8svbrogFLtgZ1M3IXZ0qXLxA07XwyfKI+KGqJlc=
private key	(hidden)
listening port	51820
assigned ip	10.1.1.12/24
peer	9JpQ7SGTfy7o+8P1SXaf40J4wJ8Kq19o5CPRc8jJbg4=
preshared key	(hidden)
endpoint	192.168.8.129:51820
allowed ips	10.1.1.0/24
latest handshake	46 seconds ago
transfer	2.1 MiB received, 2.11 MiB sent
persistent keepalive	every 15 seconds

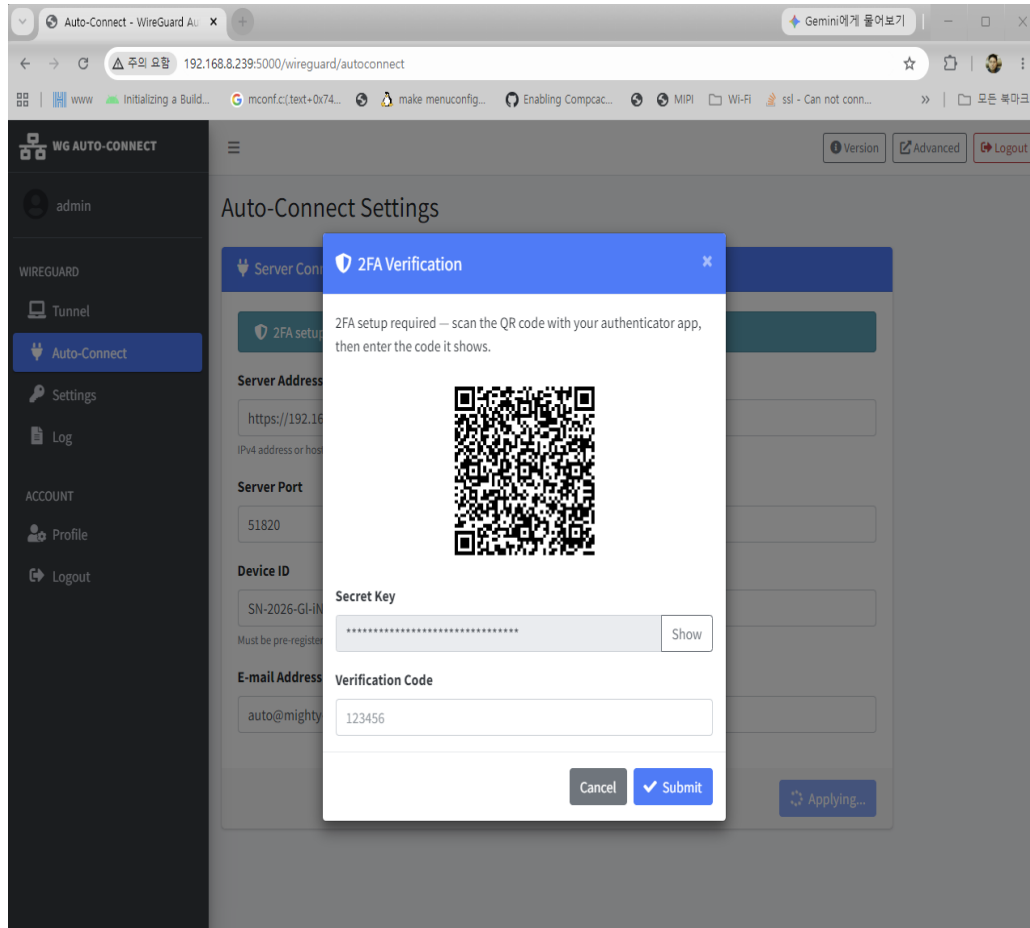
■ Auto Connection Page

The screenshot shows the 'Auto-Connect Settings' page of the WG Auto-Connect web interface. The left sidebar is identical to the previous page, with 'Auto-Connect' selected. The main content area displays the 'Auto-Connect Settings' for the 'wg-auto0' interface. A 'Connected' status bar is shown at the top. The settings include: 'Server Address' (https://192.168.8.129), 'Server Port' (51820), 'Device ID' (SN-2026-GI-iNet), and 'E-mail Address' (auto@mighty-sg.local). An 'Apply' button is at the bottom right.

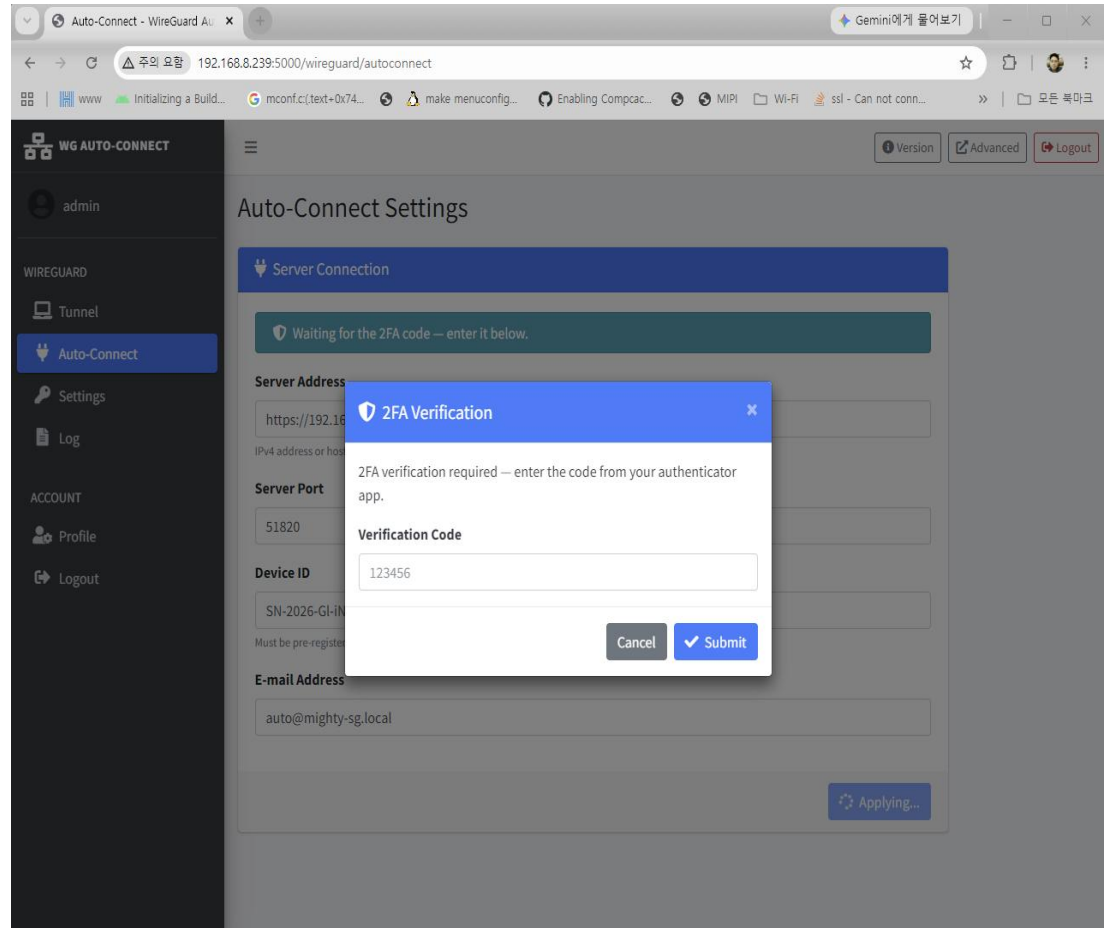
Setting	Value
Server Address	https://192.168.8.129
Server Port	51820
Device ID	SN-2026-GI-iNet
E-mail Address	auto@mighty-sg.local

5.1 MightyConnect Mini(3) – WebUI(3)

- Auto Connection 2FA Step1 Page



- Auto Connection 2FA Step2 Page



5.1 MightyConnect Mini(3) – WebUI(4)

▪ Direct Connect Page

Direct Connect

Peer: SN-2026-Android

Connect

Active Links

Peer	VPN IP	Status	Message	Expires	Action
SN-2026-Windows11	10.1.1.2	connected	Connected directly.	—	Disconnect

▪ Direct Connect Filter Rules

Direct Connect

Peer: SN-2026-Android

Connect

Active Links

Outbound traffic to the VPN is restricted to these services (plus the box itself):

ICMP TCP/22 TCP/443 TCP/80 TCP/5000

Other direct-connected peers may reach these services on this device:

TCP/5432 TCP/5000 ICMP TCP/22 TCP/443 TCP/80

Peer	VPN IP	Status	Message	Expires	Action
SN-2026-Windows11	10.1.1.2	connected	Connected directly.	—	Disconnect

5.1 MightyConnect Mini(3) – WebUI(5)

■ Settings Page

Settings - WireGuard Auto-Connect

10.1.1.4:5000/wireguard/settings

MIGHTY CONNECT

admin

WIREGUARD

- Tunnel
- Auto-Connect
- Direct Connect
- Settings
- Log

ACCOUNT

- Profile
- Logout

Auto-Connect Certificate Pin

TLS Certificate Pin

Pinned Certificate (SPKI SHA-256)

sha256//cuKwPCvgk04PNk7xHw4ao7UOD5+jCf2NXikMjdPZqY=

Must match the auto-connect server's certificate pin exactly, in curl's CURLOPT_PINNEDPUBLICKEY format (sha256//<base64>). Get the current value from the gateway's System page or scripts/generate_autoconnect_cert.sh output. Leave empty to disable pinning.

Apply

Internal PC IP (DMZ)

Traffic reaching this gateway's own VPN IP — whether relayed through the box or reached via a Direct Connect link — is redirected to the internal PC below. Leave empty to disable.

Internal PC IP

192.168.3.139

Save

■ Log Page

Log - WireGuard Auto-Connect

10.1.1.4:5000/wireguard/log

MIGHTY CONNECT

admin

WIREGUARD

- Tunnel
- Auto-Connect
- Direct Connect
- Settings
- Log

ACCOUNT

- Profile
- Logout

Daemon Log

Log Viewer | Log Size Limit

Auto-Refresh wg-auto-daemon.log 200 lines

/var/log/mightysg/wg-auto-daemon.log

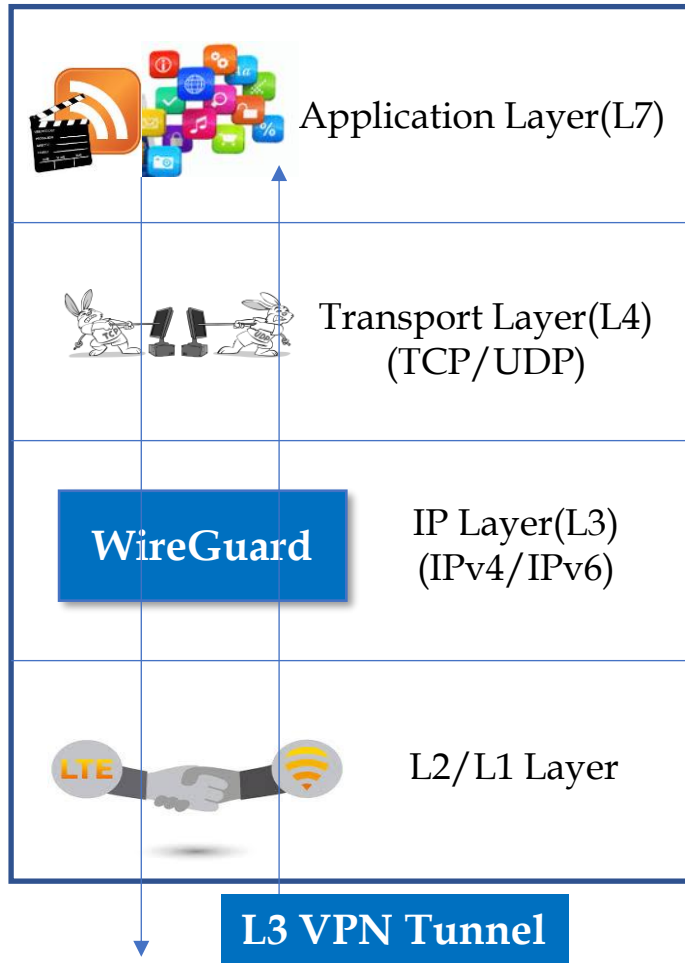
```
[2026-08-08 15:47:01] [DEBUG] Executing: wg set wg-auto0 listen-port 51820 private-key /usr/local/mightysg/etc/wg-auto-private.key peer sSBt+Cd1Vbjz0n9f3fY2VP2Ux6nM7jfj1Q18GEBKpAE= preshared-key /tmp/wg_psk_wg-auto0 endpoint 192.168.8.114:51820 allowed-ips 10.1.1.0/24 persistent-keepalive 15
[2026-08-08 15:47:01] [INFO] WireGuard tunnel wg-auto0 is UP and configured.
[2026-08-08 15:47:01] [INFO] Tunnel restored successfully.
[2026-08-08 15:47:01] [INFO] DirectConnectClient: signaling listener up on 10.1.1.4:51821
[2026-08-08 15:47:02] [INFO] DirectConnectFilter: applied outbound filter to wg-auto0 (5 allowed, 6 exposed service(s))
[2026-08-08 15:47:03] [INFO] OpenWrtForwardFilter: applied (5 allowed, 6 exposed service(s)); wgauto<->lan forwarding restricted to these, zone default REJECT backstops the rest
[2026-08-08 15:47:03] [INFO] OpenWrtDnat: applied - traffic to 10.1.1.4/32 (this gateway's VPN IP) redirected to 192.168.3.139, except ports reserved for this gateway itself (Direct Connect signaling 51821, litews UI 5000)
[2026-08-08 15:47:03] [INFO] ControlServer: listening on 127.0.0.1:51821
[2026-08-08 16:03:24] [INFO] Direct-connect: adding peer Pn17WjIG1+gc5f4KVkBDWTPghBQ5ctGLXFDorCjh908= (10.1.1.2/32 via 192.168.8.103:50855)
[2026-08-08 16:03:24] [DEBUG] Executing: wg set wg-auto0 peer Pn17WjIG1+gc5f4KVkBDWTPghBQ5ctGLXFDorCjh908= endpoint 192.168.8.103:50855 allowed-ips 10.1.1.2/32 persistent-keepalive 15
[2026-08-08 16:03:24] [DEBUG] NatPuncher: probing 10.1.1.2 via wg-auto0
[2026-08-08 16:03:26] [DEBUG] NatPuncher: probing 10.1.1.2 via wg-auto0
[2026-08-08 16:03:27] [DEBUG] NatPuncher: probing 10.1.1.2 via wg-auto0
```


5.2 MightyConnect Ubuntu



5.2 MightyConnect Ubuntu(1) – WireGuard

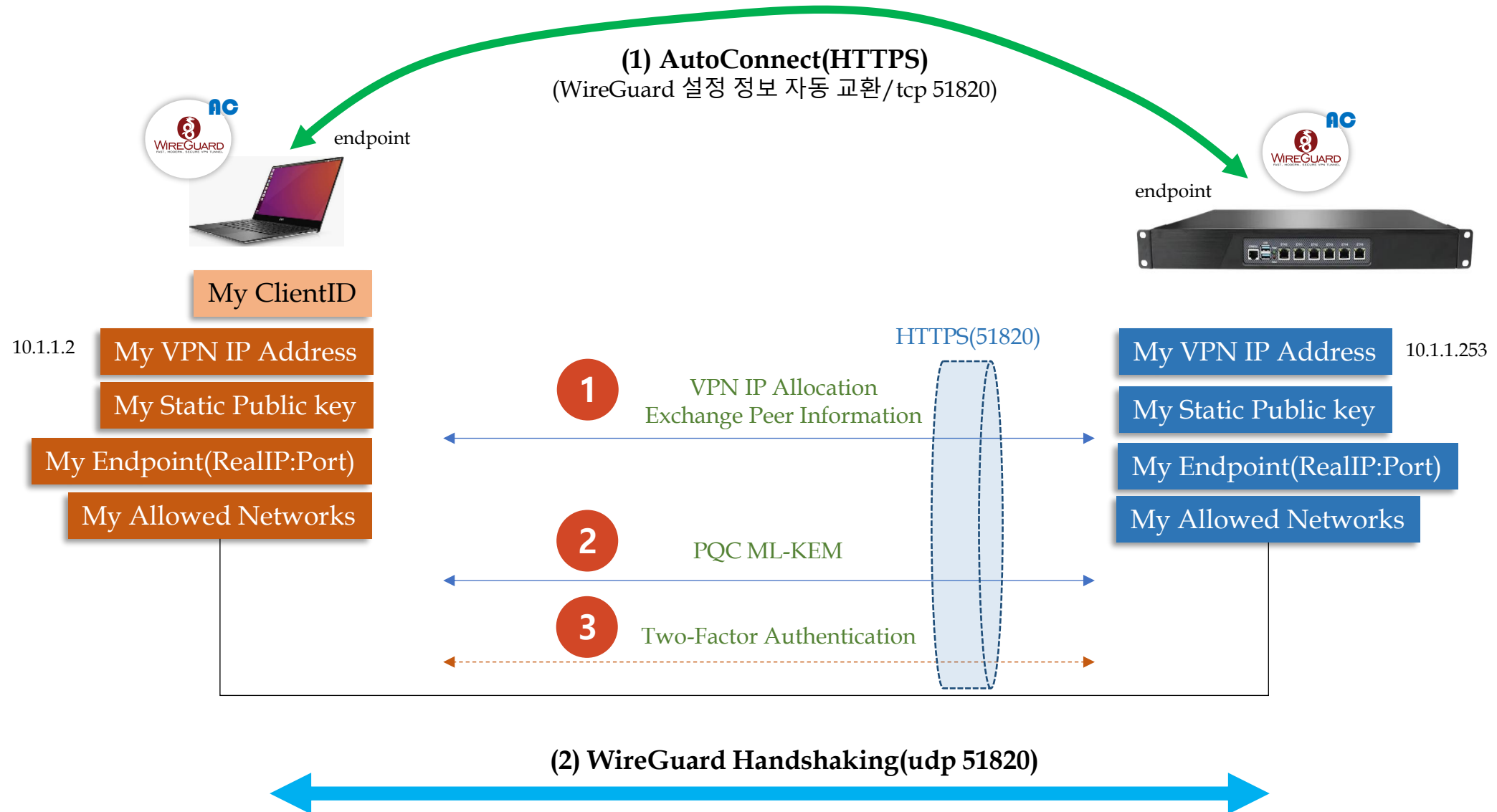
Cryptokey Routing



```
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ sudo wg show
interface: wg-auto0
  public key: NYJM7SxhjeQXjjipIsEiNxZY3w6NcPJl/Mfjfye4mzY=
  private key: (hidden)
  listening port: 51820

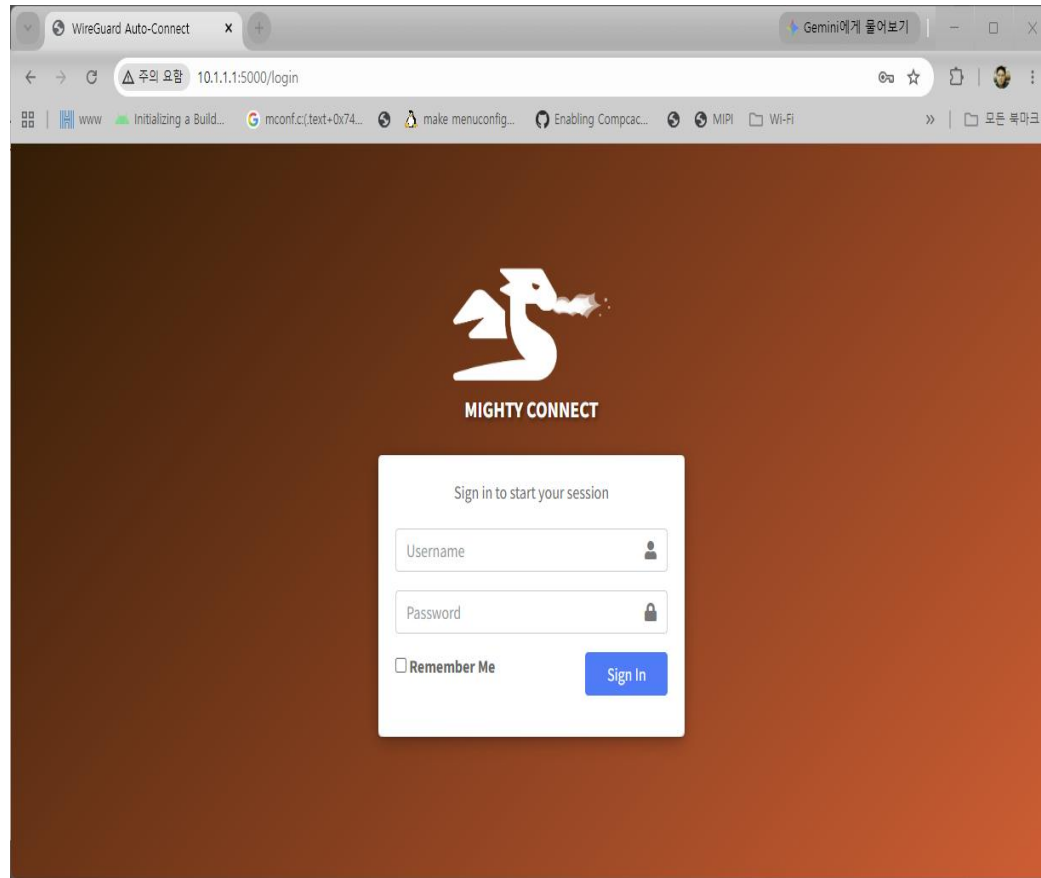
peer: 9JPq7SGTFy7o+8PLSXaf40J4wjBKql9o5CPRc8jJbg4=
  preshared key: (hidden)
  endpoint: 192.168.8.129:51820
  allowed ips: 10.1.1.0/24
  latest handshake: 1 minute, 40 seconds ago
  transfer: 316 B received, 180 B sent
  persistent keepalive: every 15 seconds
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ ping 10.1.1.253
PING 10.1.1.253 (10.1.1.253) 56(84) bytes of data:
64 bytes from 10.1.1.253: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 10.1.1.253: icmp_seq=2 ttl=64 time=2.94 ms
^C
--- 10.1.1.253 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.698/2.817/2.936/0.119 ms
```

5.2 MightyConnect Ubuntu(2) – Auto Connection

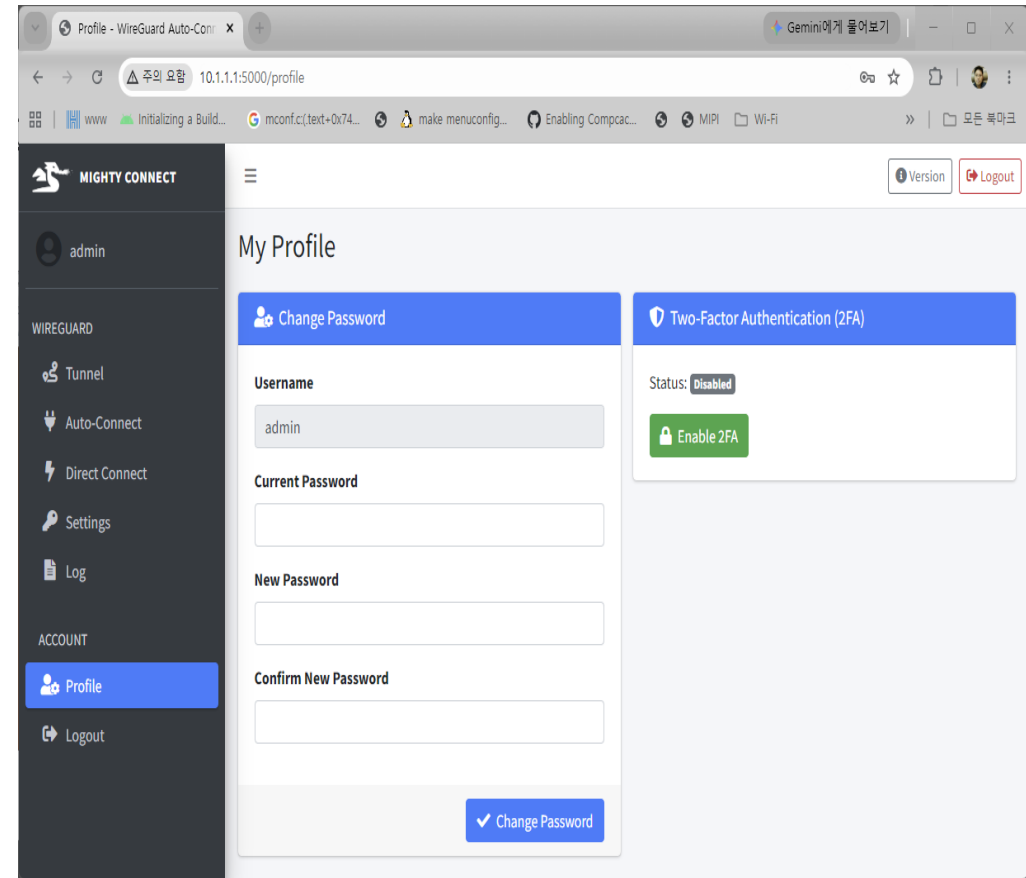


5.2 MightyConnect Ubuntu(3) – WebUI(1)

■ Login Page

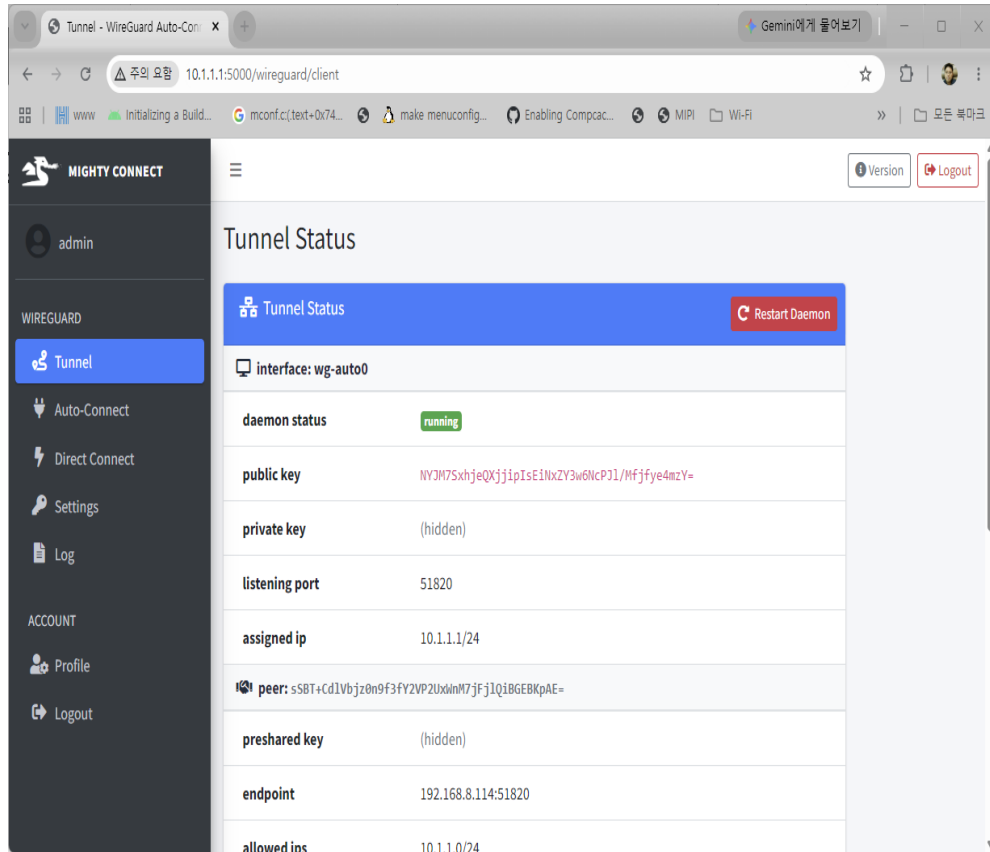


■ Admin User Page

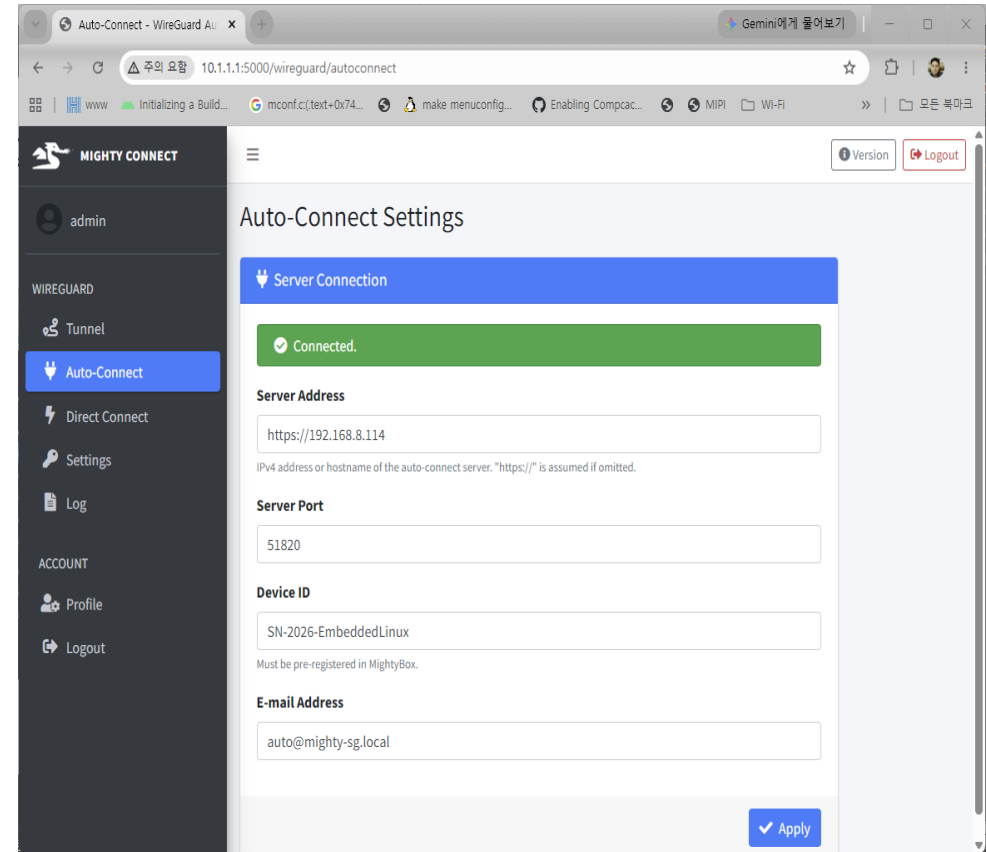


5.2 MightyConnect Ubuntu(3) – WebUI(2)

■ Tunnel Status Page

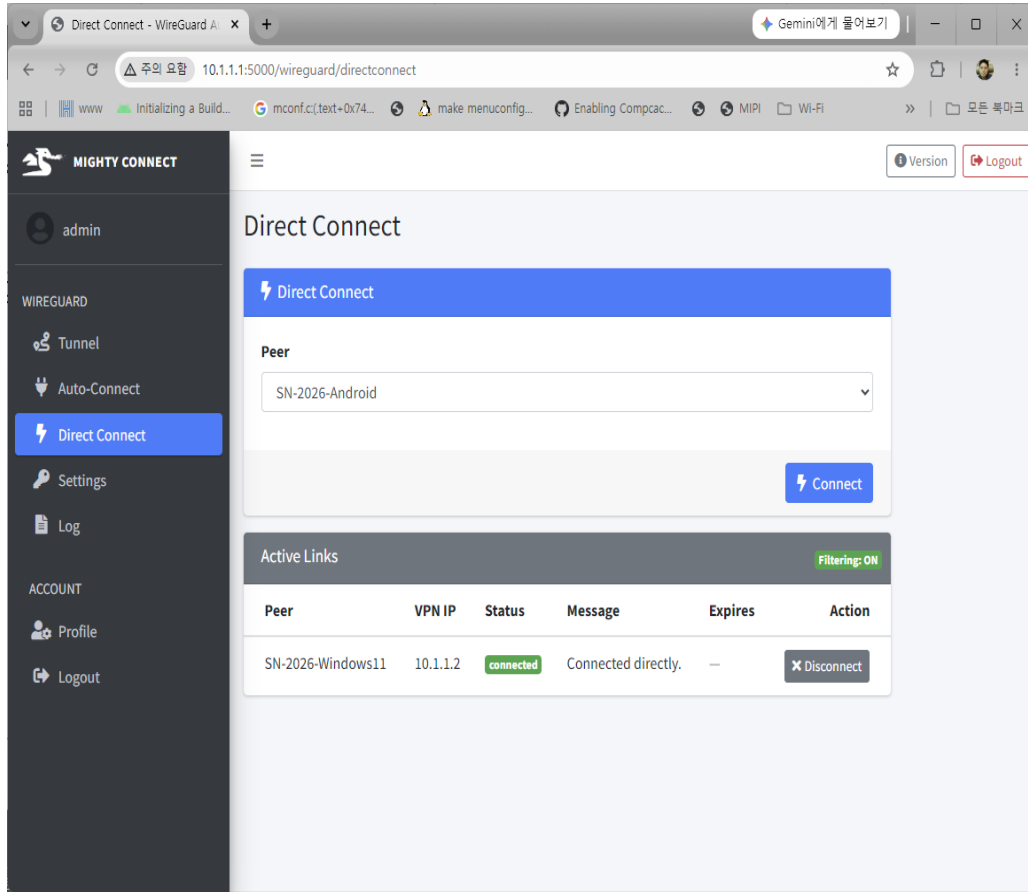


■ Auto Connection Page



5.2 MightyConnect Ubuntu(3) – WebUI(3)

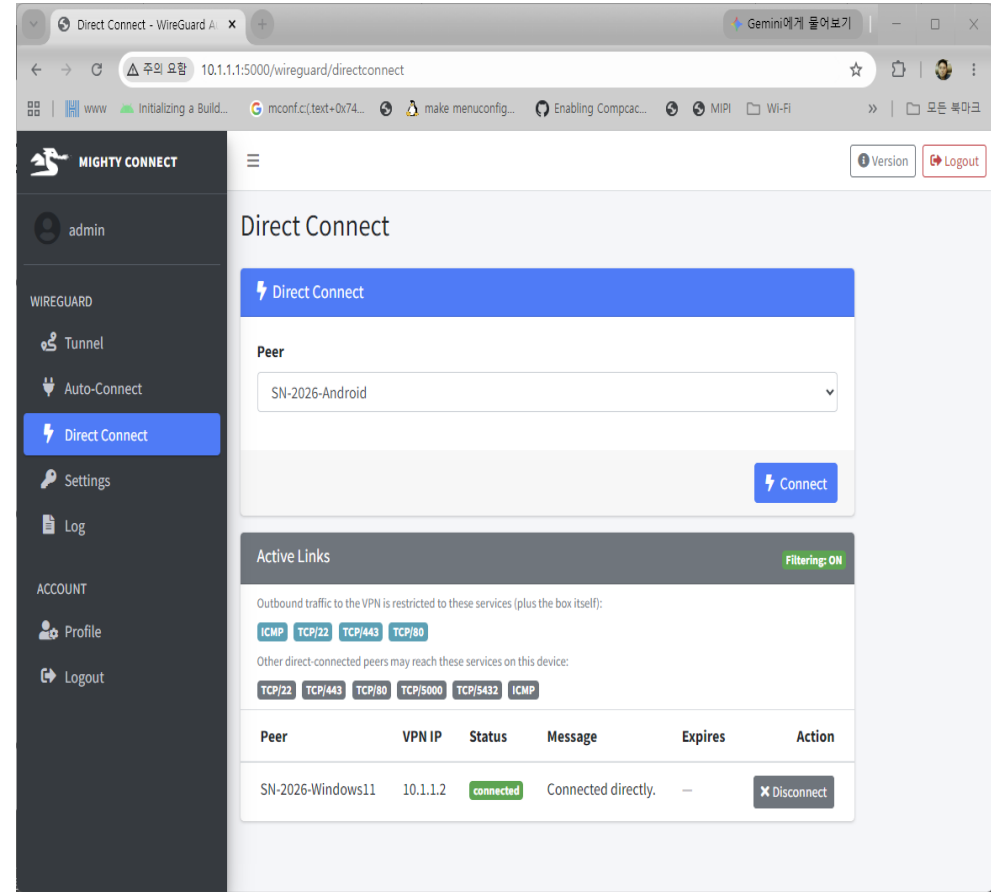
▪ Direct Connect Page



The screenshot shows the 'Direct Connect' page in the MightyConnect WebUI. The left sidebar contains navigation links for 'admin', 'WIREGUARD', 'Tunnel', 'Auto-Connect', 'Direct Connect' (highlighted), 'Settings', 'Log', 'ACCOUNT', 'Profile', and 'Logout'. The main content area has a 'Direct Connect' header and a 'Peer' dropdown menu set to 'SN-2026-Android'. A 'Connect' button is visible. Below the 'Connect' button is an 'Active Links' section with a table showing one active connection.

Peer	VPN IP	Status	Message	Expires	Action
SN-2026-Windows11	10.1.1.2	connected	Connected directly.	—	Disconnect

▪ Direct Connect Filtering



The screenshot shows the 'Direct Connect' page with filtering options. The 'Active Links' section includes a list of outbound traffic restrictions (ICMP, TCP/22, TCP/443, TCP/80) and a table of active connections.

Peer	VPN IP	Status	Message	Expires	Action
SN-2026-Windows11	10.1.1.2	connected	Connected directly.	—	Disconnect

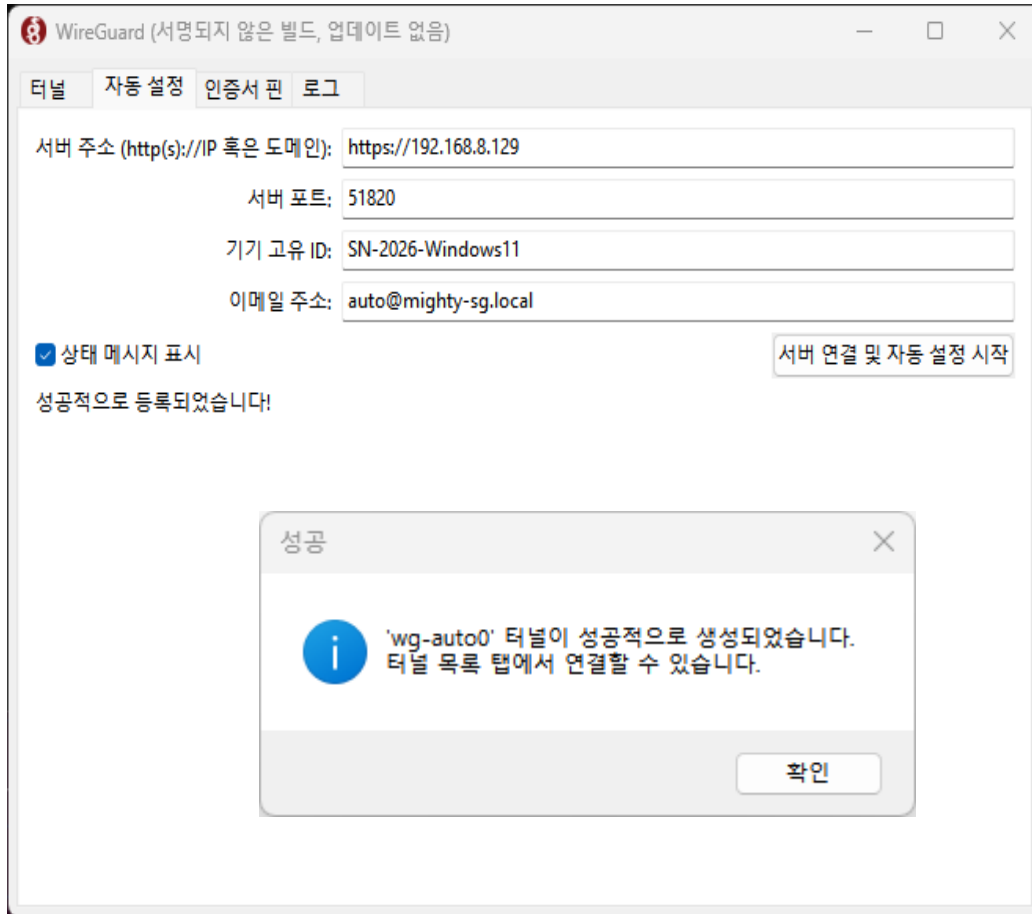
5.3 MightyConnect Windows

MightyConnect Windows Agent

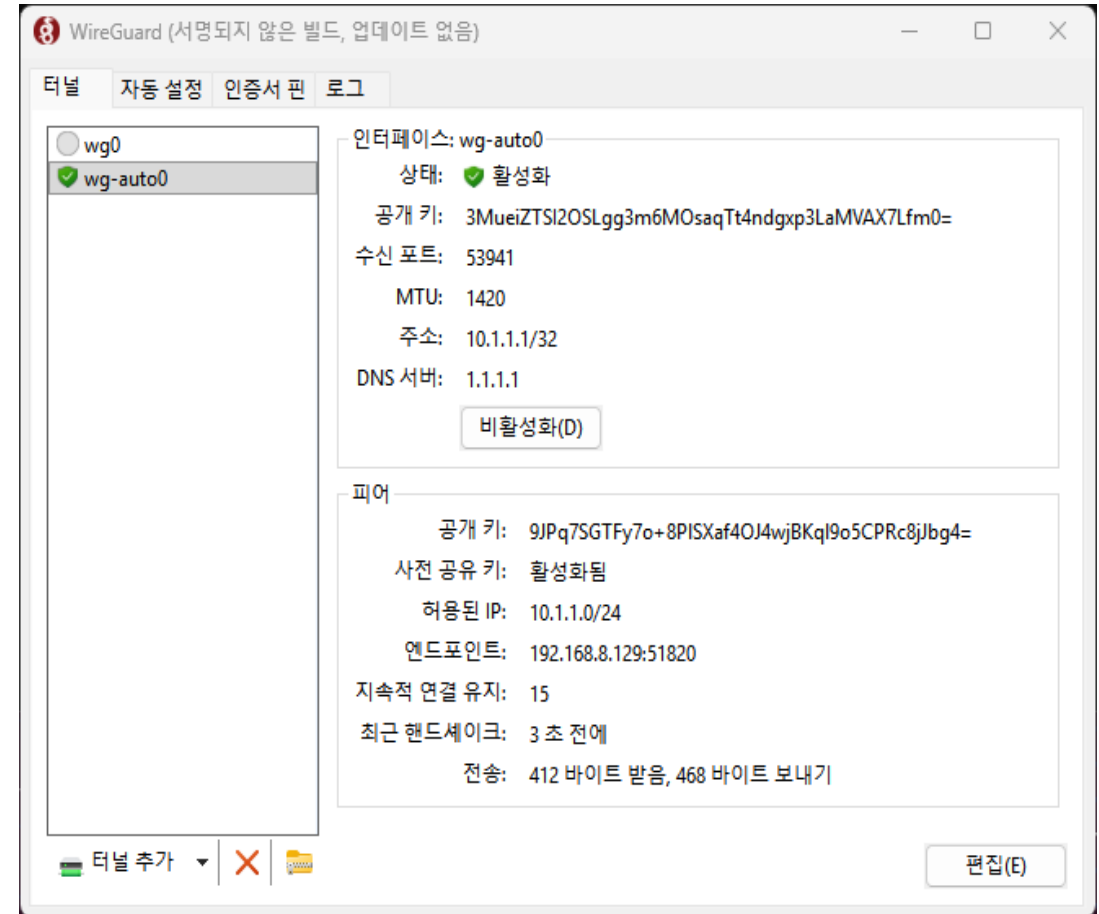


5.3 MightyConnect Windows(1) – User Interface(1)

- Auto Connection Page

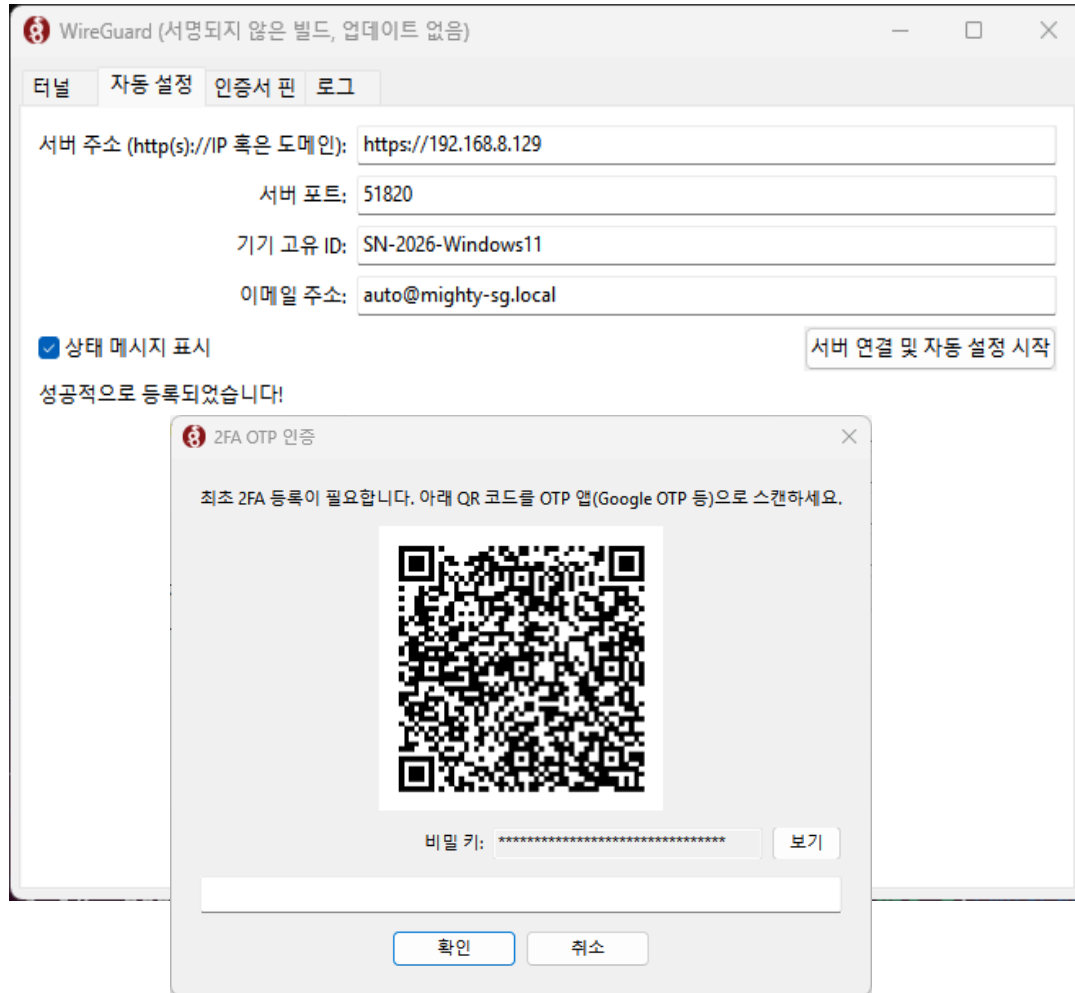


- wg-auto0 interface 생성 및 터널 활성화 Page

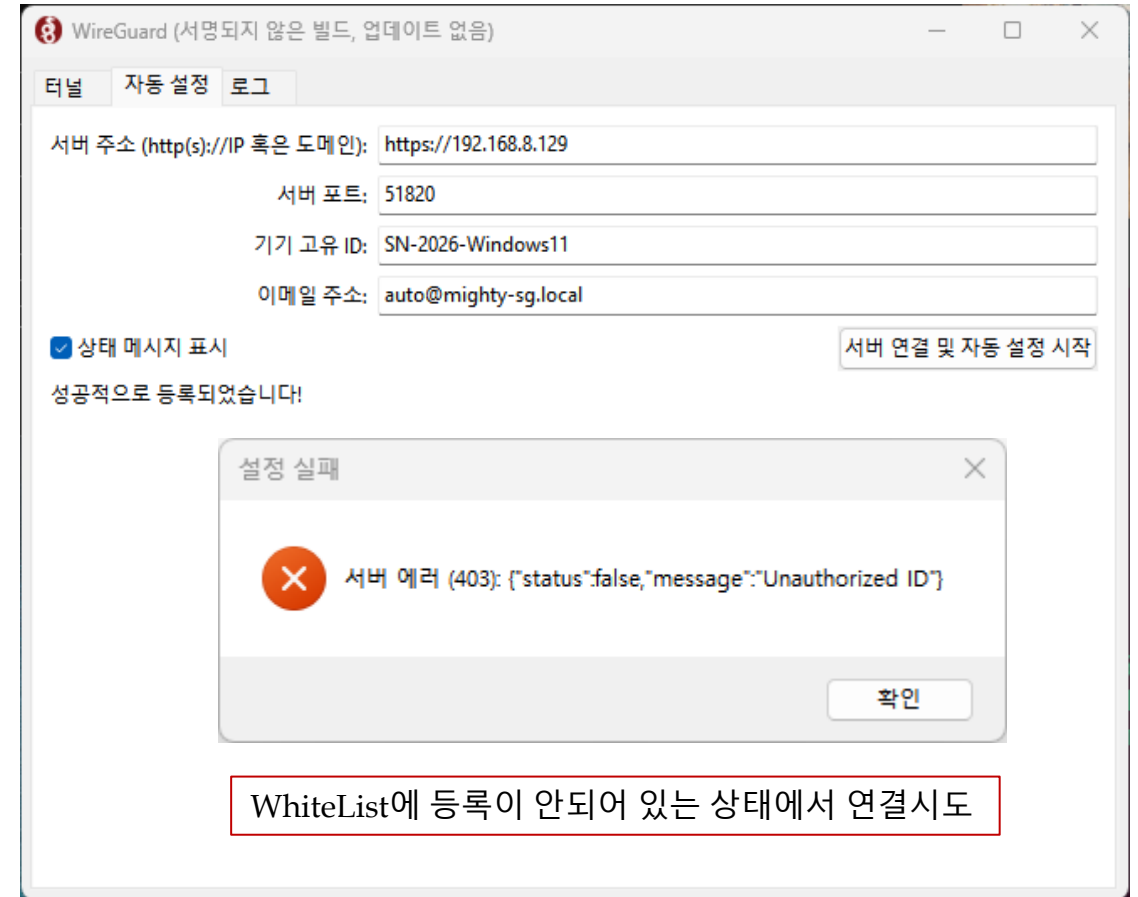


5.3 MightyConnect Windows(1) – User Interface(2)

- Auto Connection – 2FA 인증 Page

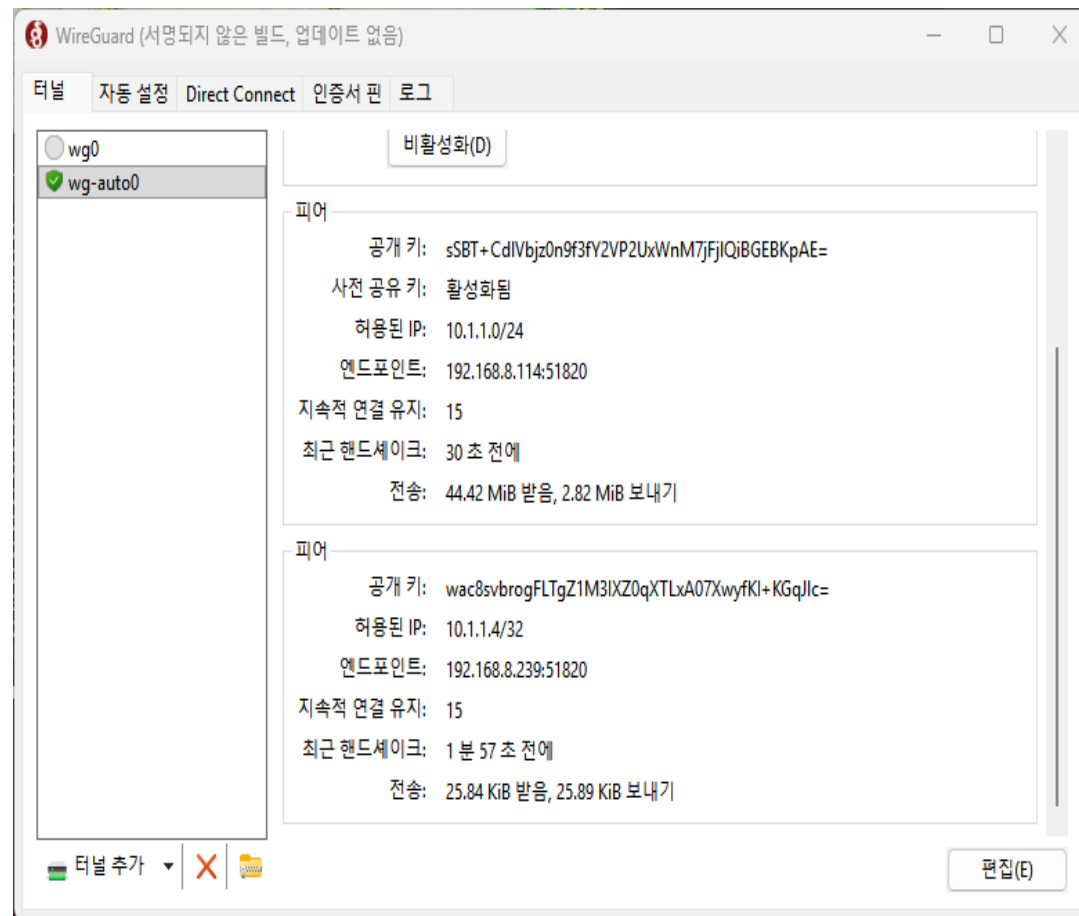


- Auto Connection – 인증 실패 예시



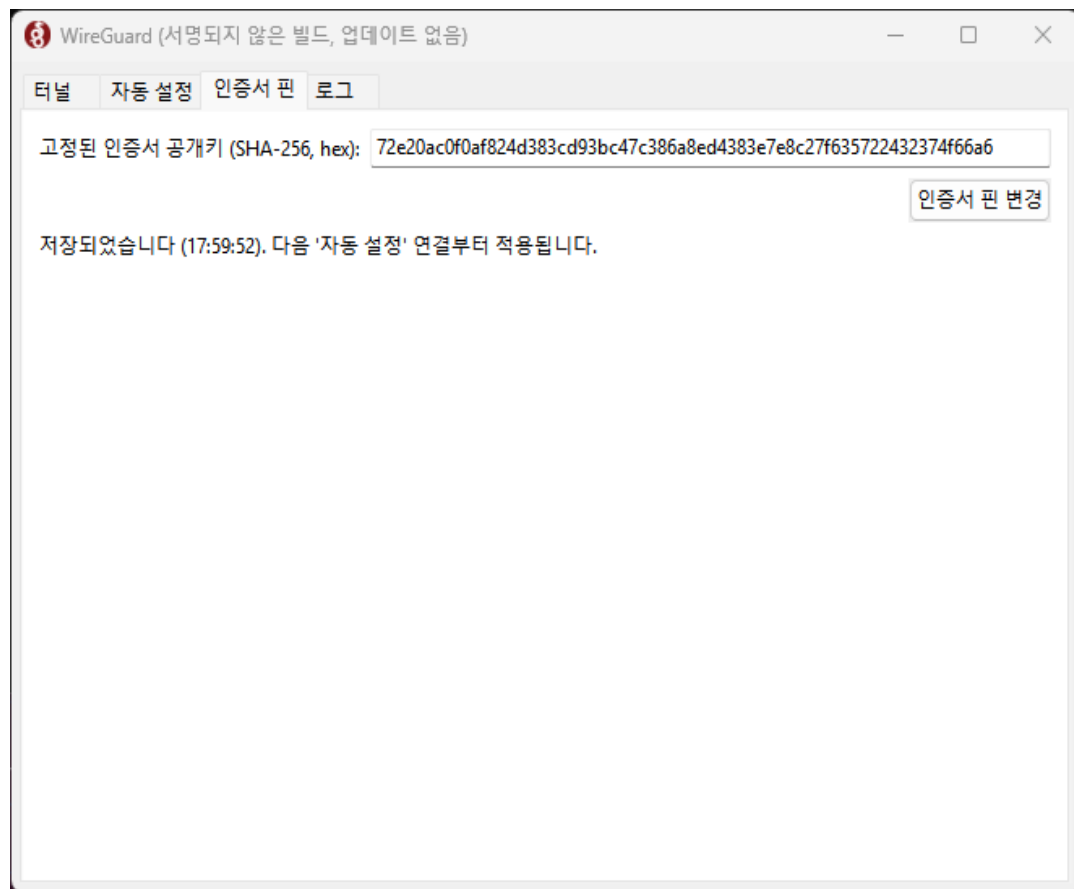
The image contains two side-by-side screenshots of the WireGuard application window. The window title is 'WireGuard (서명되지 않은 빌드, 업데이트 없음)'. The left screenshot shows the 'Direct Connect' tab. It has a dropdown menu for '상대 Peer:' with 'SN-2026-GI-iNet' selected, and a 'Direct Connect' button. Below this, it says '활성 Direct 링크:' and '필터링: 커짐 (ICMP, TCP/22, TCP/443, TCP/80, TCP/5432, TCP/5000, TCP/8080)'. A table shows the peer 'SN-2026-GI-iNet' with VPN IP '10.1.1.4', status '연결됨', and message '직접 연결되었습니다.'. The right screenshot shows the ' wg-auto0 ' interface selected. It displays configuration details: '공개 키: sSBT+CdlVbjz0n9f3fY2VP2UxWnM7fJlQjBGEbKpAE=', '사전 공유 키: 활성화됨', '허용된 IP: 10.1.1.0/24', '엔드포인트: 192.168.8.114:51820', '지속적 연결 유지: 15', '최근 핸드셰이크: 30 초 전에', and '전송: 44.42 MiB 받음, 2.82 MiB 보내기'.

- Direct Connect Peer 확인

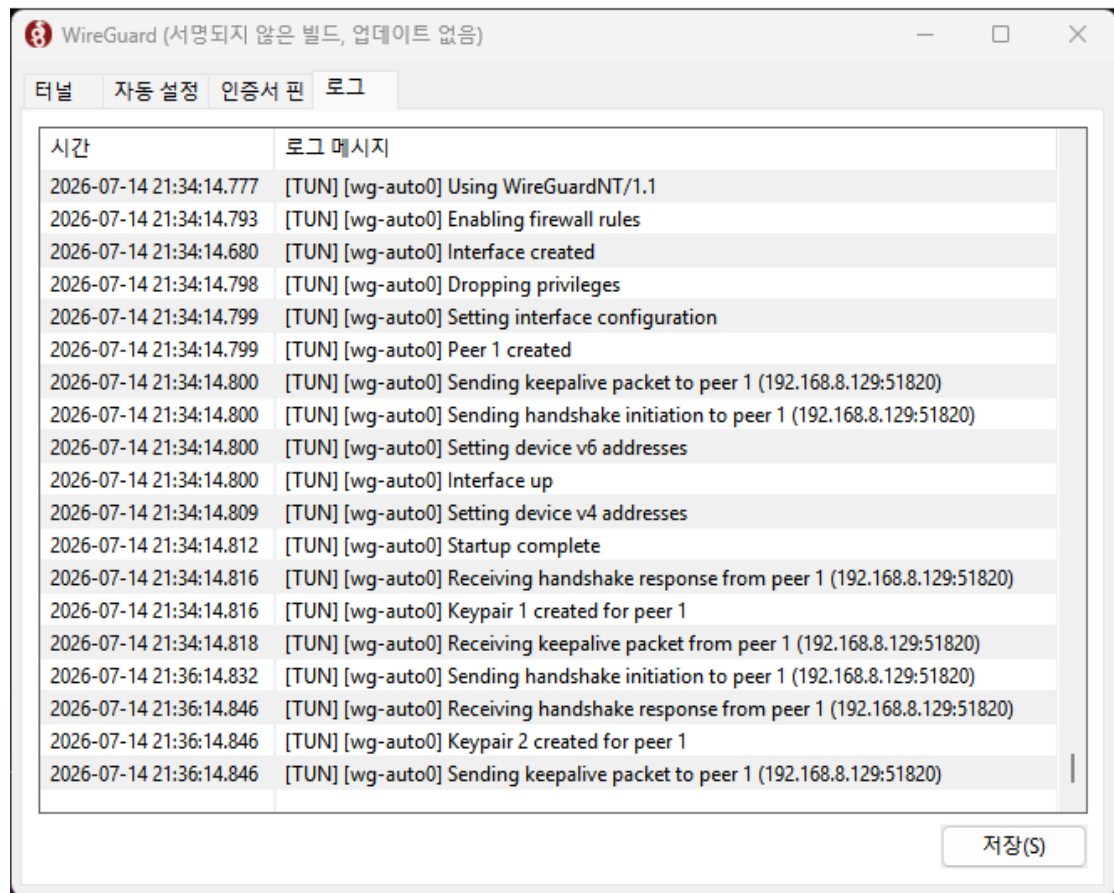


5.3 MightyConnect Windows(1) – User Interface(4)

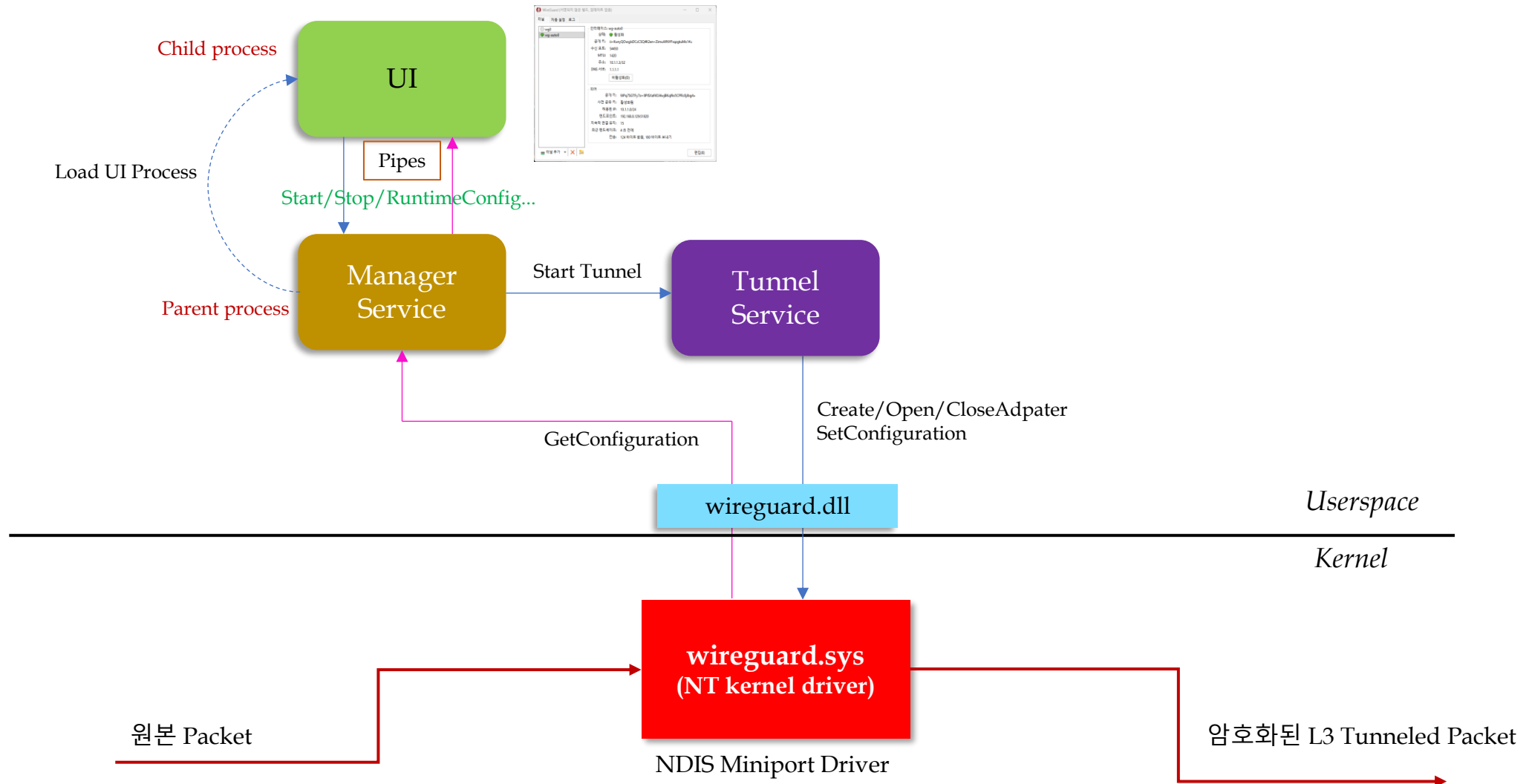
- X.509 인증서 Pin 값 변경 Page



- Log Page



5.3 MightyConnect Windows(2) – Architecture

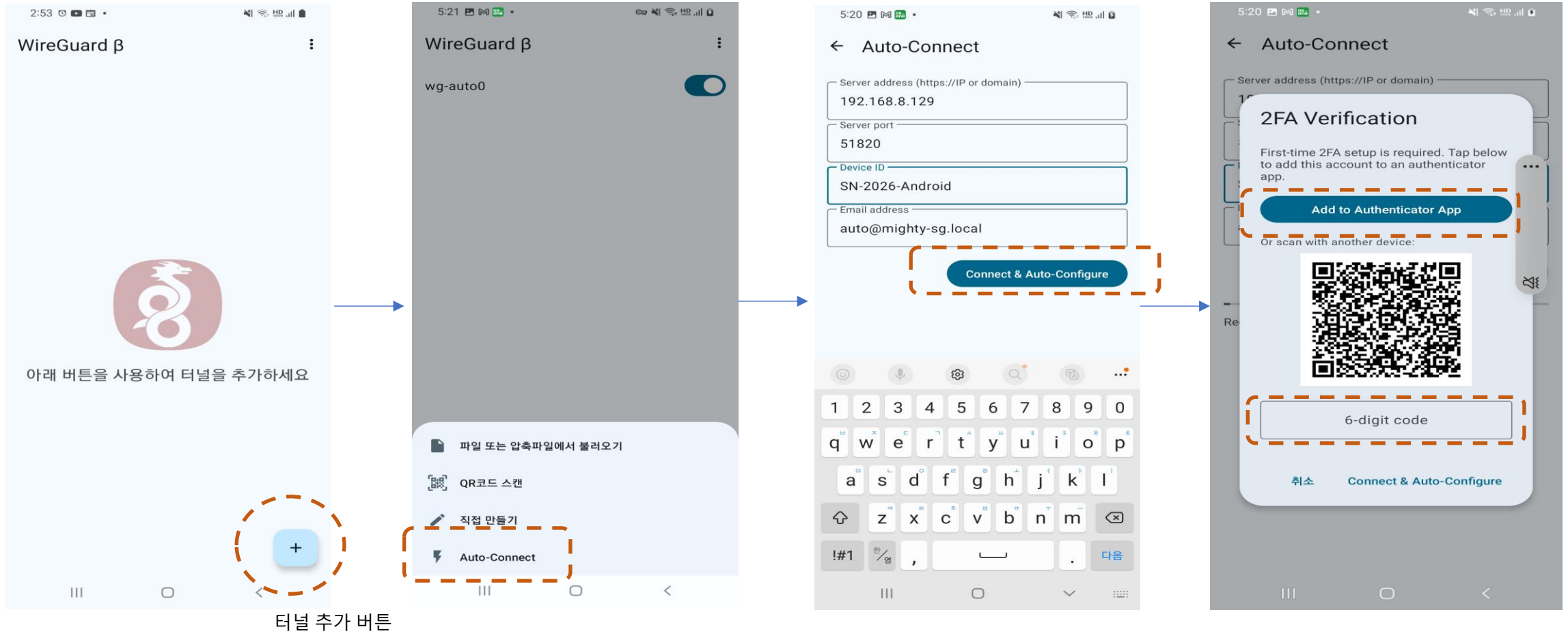


5.4 MightyConnect Android

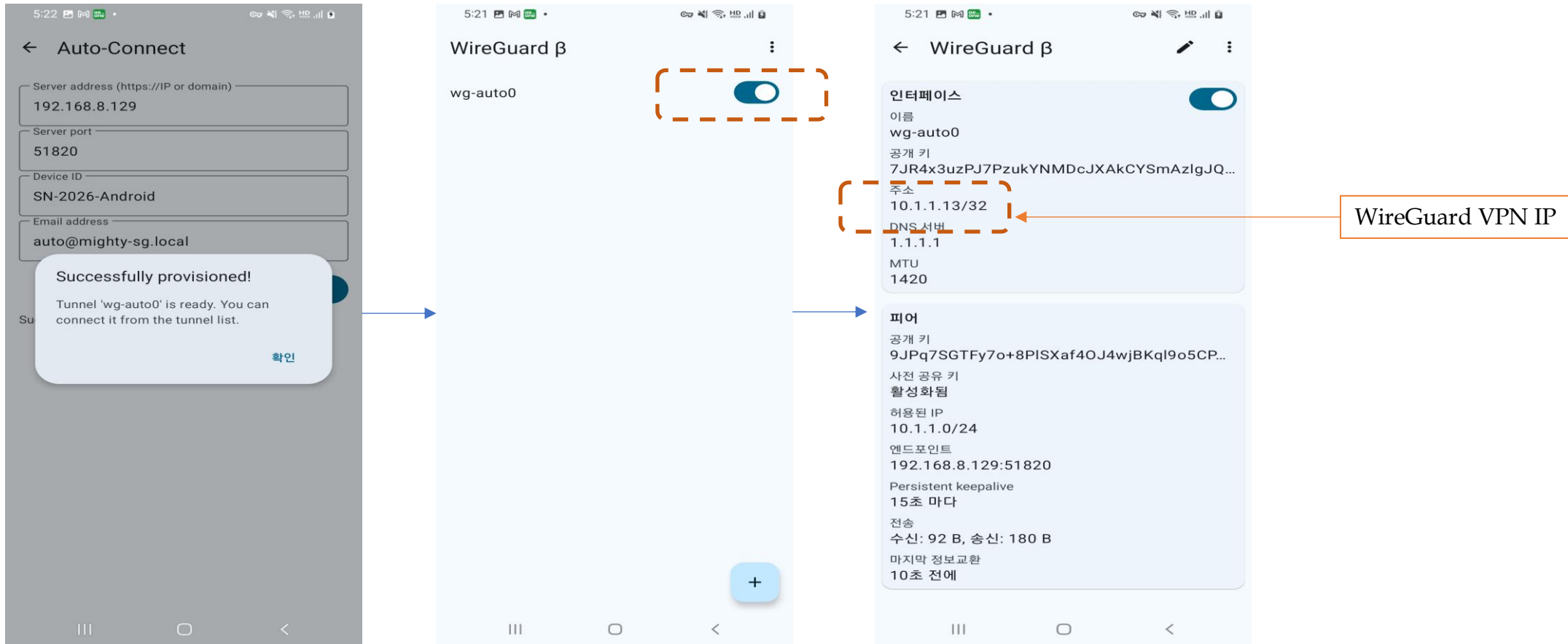
Android App



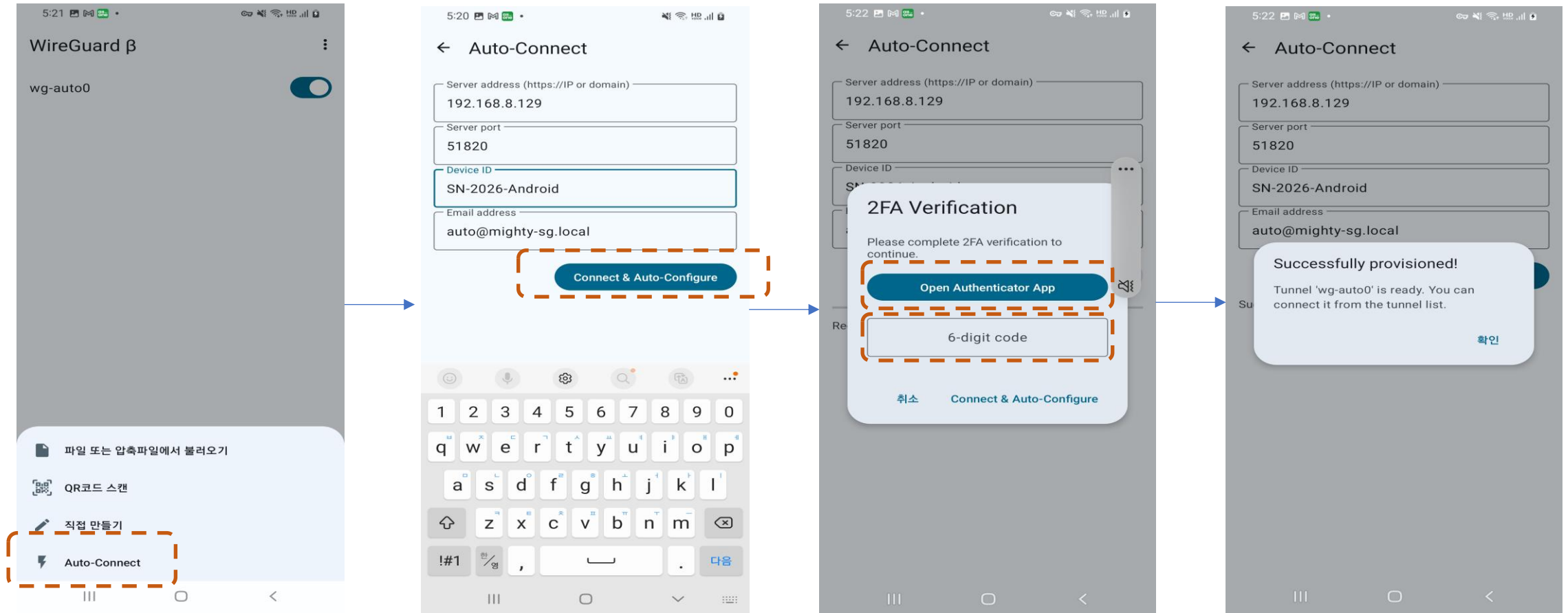
5.4 MightyConnect Android(1) – Auto Connect(1)



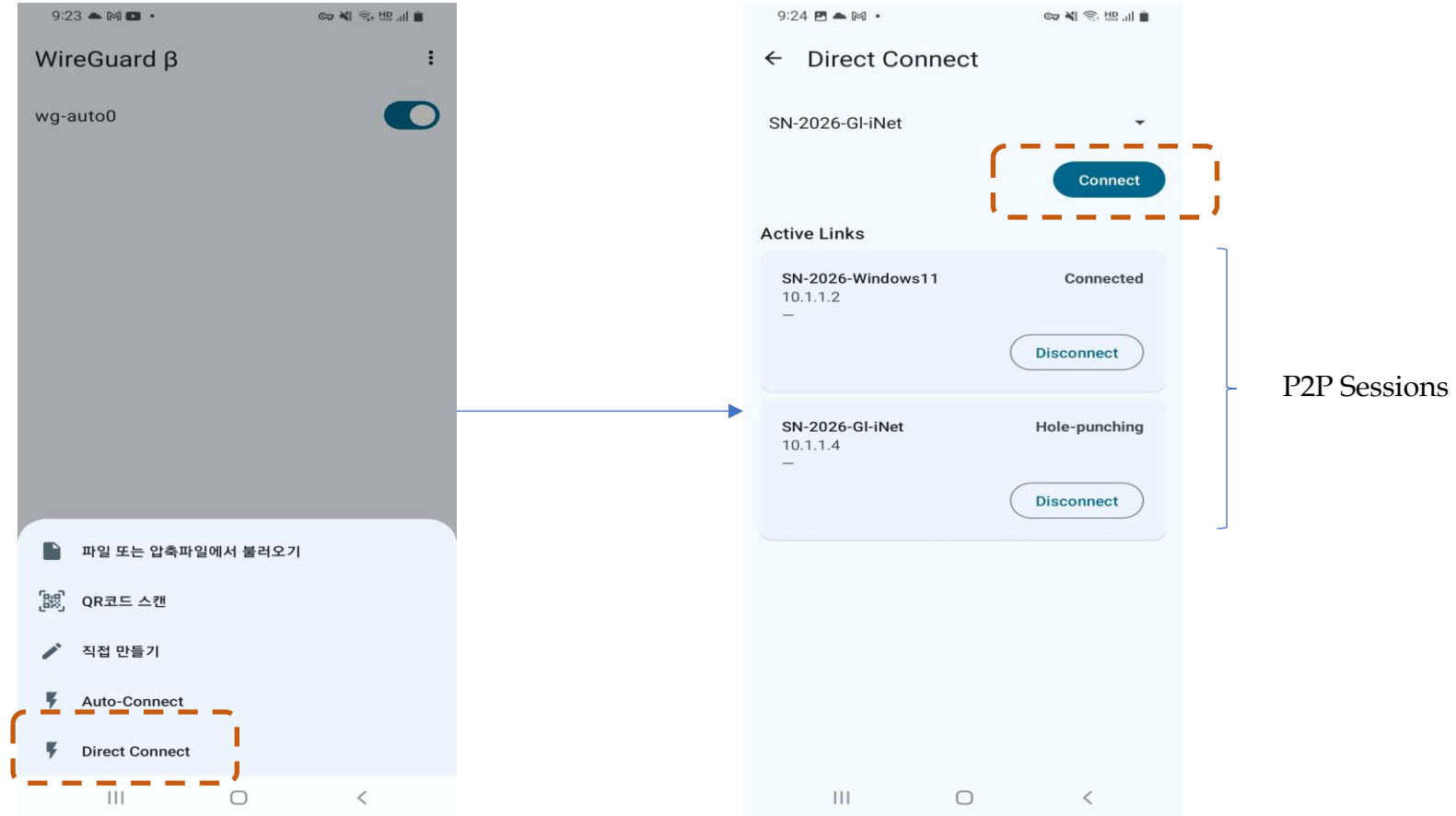
5.4 MightyConnect Android(1) – Auto Connect(2)



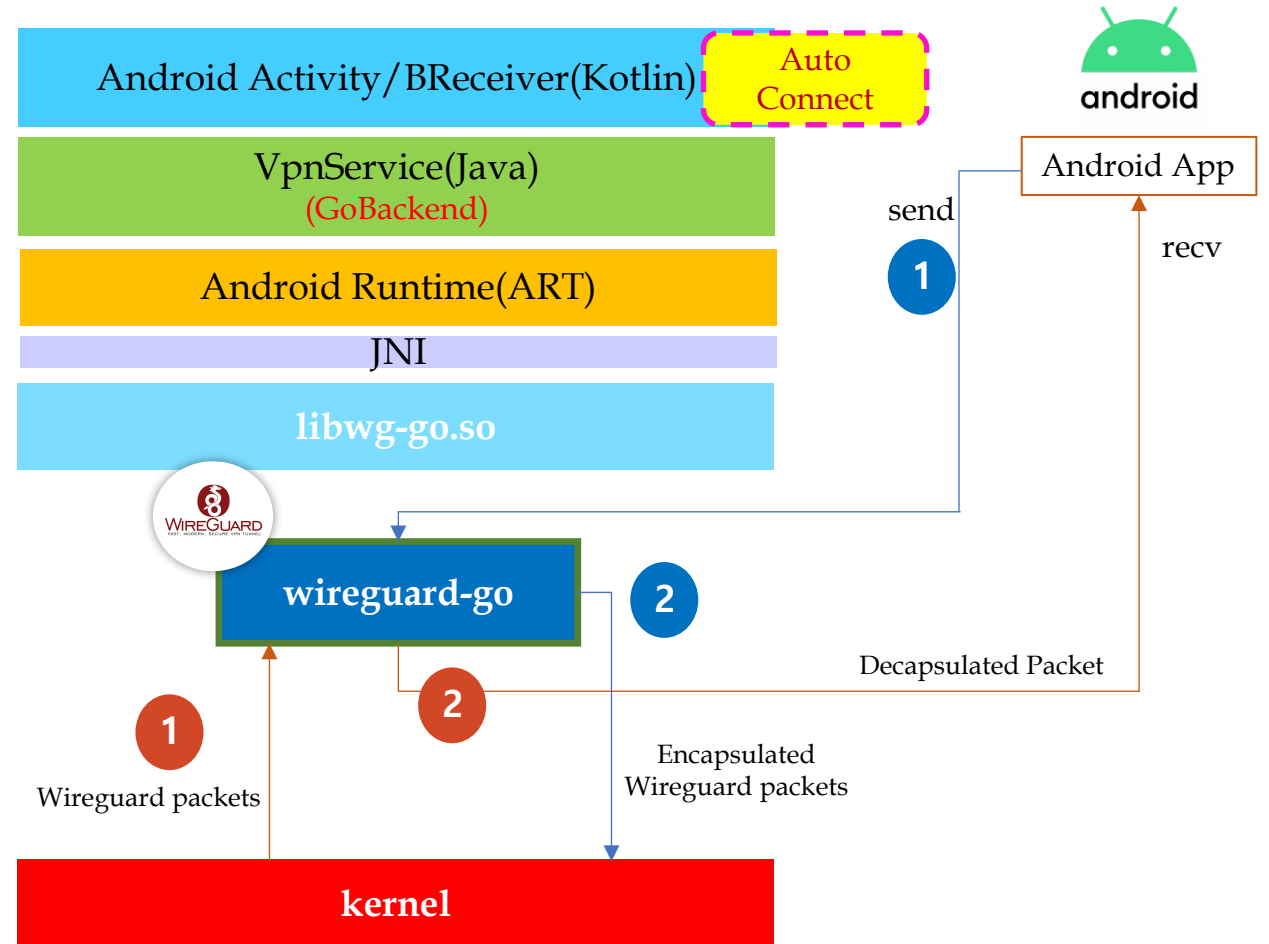
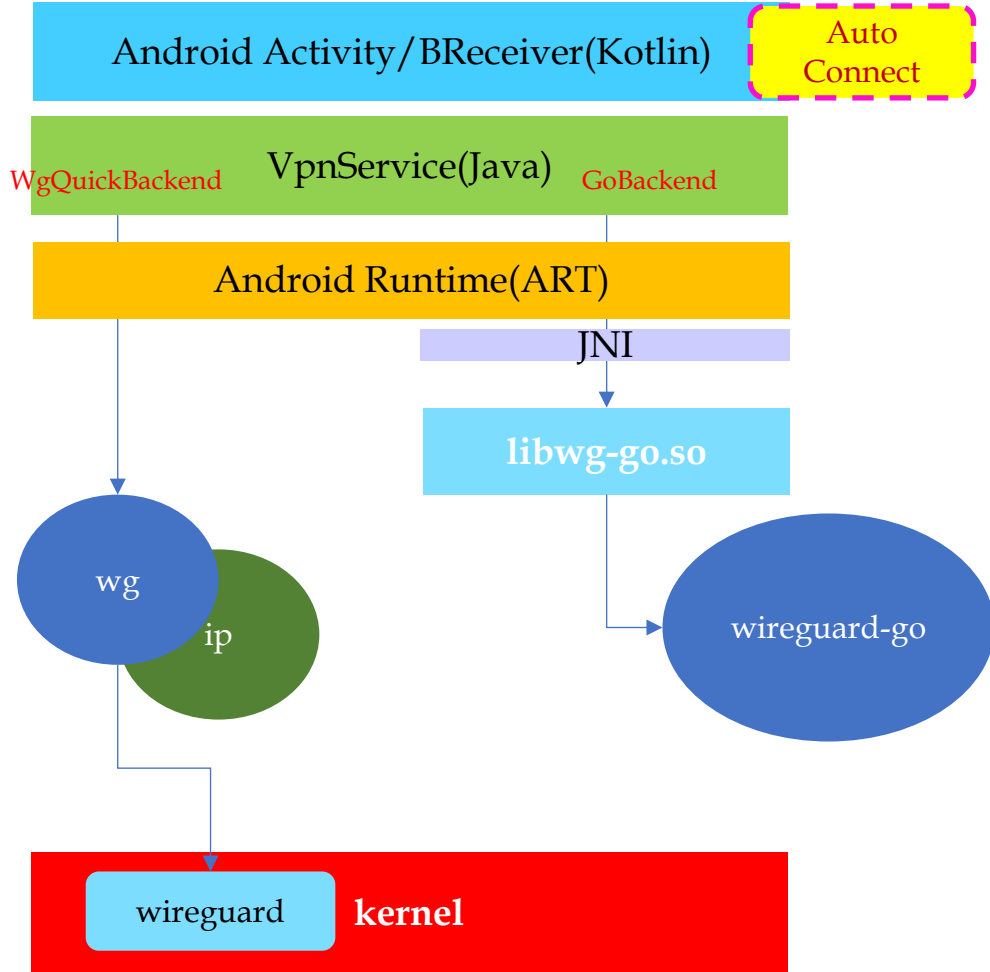
5.4 MightyConnect Android(1) – Auto Connect(3)



5.4 MightyConnect Android(2) – Direct Connect



5.4 MightyConnect Android(3) – Architecture

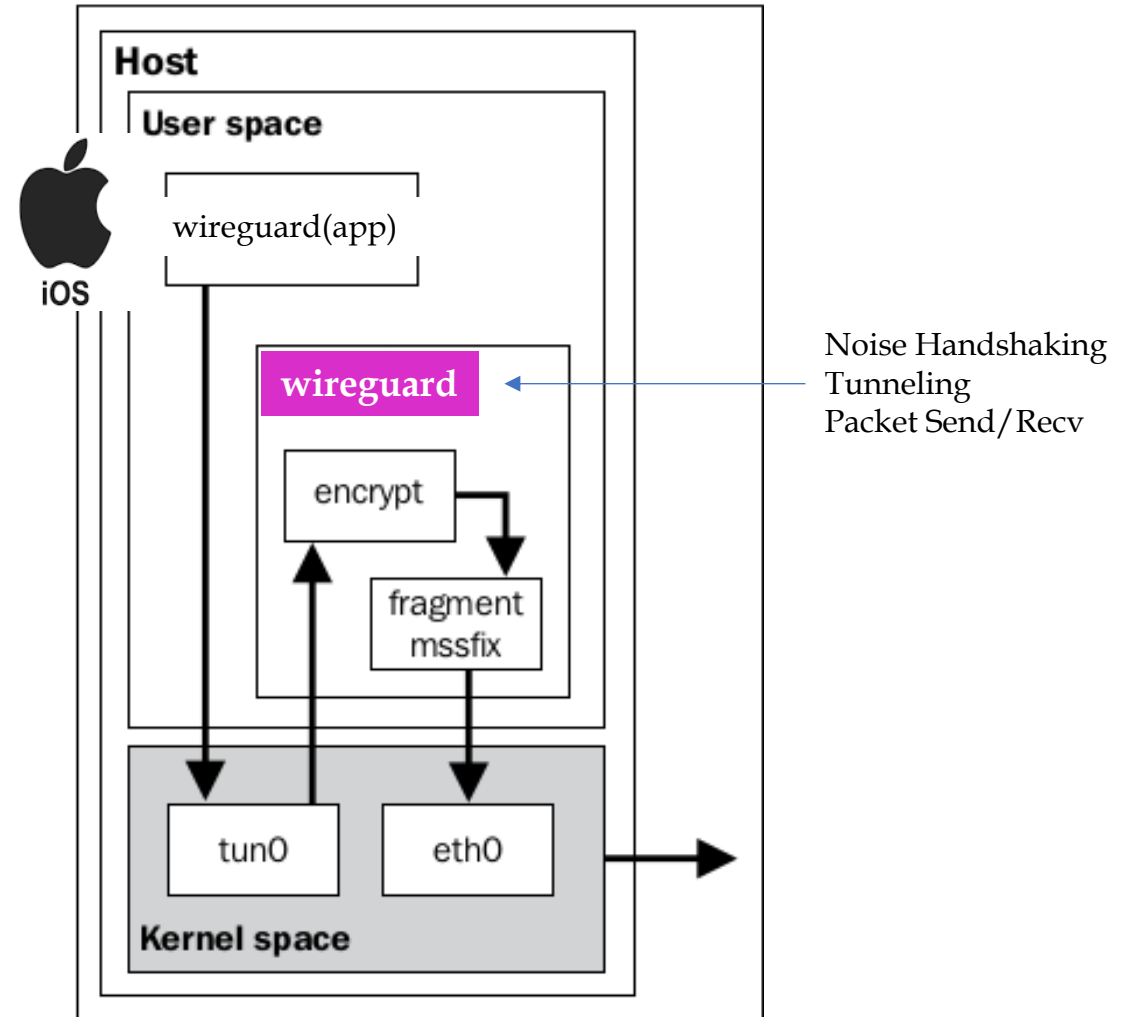
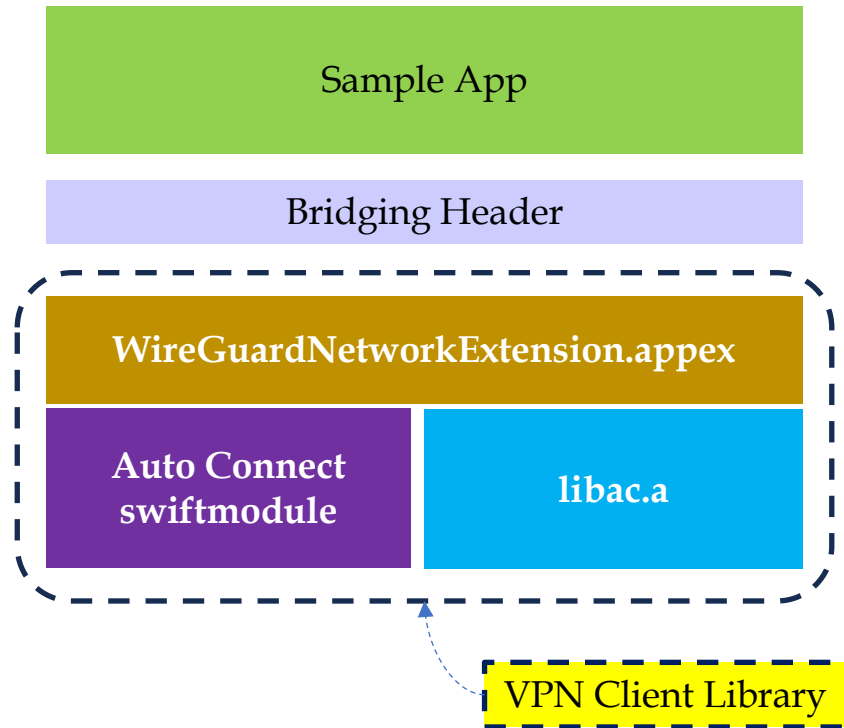


5.5 MightyConnect macOS/iOS

MightyConnect macOS/iOS Agent



5.5 MightyConnect macOS/iOS – Architecture



6. References

6. References

- [1] <https://www.wireguard.com>
- [2] <https://fd.io/>
- [3] <https://www.gl-inet.com/collections/all-products>
- [4] <https://www.friendlyelec.com/>
- [5] <https://ferrumgate.com/>
- [6] <https://tailscale.com/blog/more-throughput>
- [7] https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf
- [8] Performance Comparison of VPN Solutions, Lukas Osswald
- [9] Impressive Packet Processing Performance Enables Greater Workload Consolidation
- [10] <https://www.netgate.com/resources/articles-vector-packet-processing>
- [11] Windows is a registered trademark of Microsoft.
- [12] Android is a registered trademark of Google.
- [13] macOS & iOS is a registered trademark of Apple.
- [14] Ubuntu is a registered trademark of Canonical Ltd.
- [15] And Google~

We Secure the AI & IoT with MightyConnect™ !

©copyright 2026 MightyLink Lab., All rights reserved.

